

# Feynman Equation Results of Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

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## A The Feynman Equations

We consider learning the following 5 Feynman equations [Udrescu and Tegmark, 2020], with details in Table 1. The data corresponding to each Feynman equation is acquired from the Penn Machine Learning Benchmark (PMLB; <https://github.com/EpistasisLab/pmlb>) [Olson et al., 2017], containing  $10^5$  observations.

$$\begin{aligned}
 \text{I\_12\_2 : } F &= \frac{q_1 q_2}{4\pi\epsilon r^2}, & p &= 4 & (\text{CL}) \\
 \text{I\_12\_11 : } F &= q(E_f + vB \sin \theta), & p &= 5 & (\text{FCE}) \\
 \text{I\_24\_6 : } E_n &= \frac{1}{4}m(\omega^2 + \omega_0^2)x^2, & p &= 4 & (\text{EHFO}) \\
 \text{I\_50\_26 : } x &= x_1[\cos(\omega t) + \alpha \cos^2(\omega t)], & p &= 4 & (\text{HOQN}) \\
 \text{II\_36\_38 : } f &= \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T}M, & p &= 8 & (\text{FMMM})
 \end{aligned}$$

where, I\_12\_2 corresponds to Coulomb’s law (CL); I\_12\_11 represents the force on a charged particle in an electromagnetic field (FCE); I\_24\_6 describes the energy of a harmonic oscillator (EHFO); I\_50\_26 represents harmonic oscillation with a quadratic nonlinear correction (HOQN); and II\_36\_38 describes the frictional force on a moving magnetic moment (FMMM).

Table 1: Details on Feynman equations.

| Equation       | Output variable                    | Input variables                                                                                                                                              | Scientific interpretation                                                                                                                                                                                                                                                                        |
|----------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I_12_2 (CL)    | $F$ : electrostatic force          | electric charges ( $q_1, q_2$ ); permittivity ( $\epsilon$ ); distance between charges ( $r$ )                                                               | Coulomb’s law describing the magnitude of the electrostatic force between two point charges. The force is proportional to the product of the charges and inversely proportional to the squared separation distance, with the permittivity controlling the strength of interaction in the medium. |
| I_12_11 (FCE)  | $F$ : Lorentz force                | particle charge ( $q$ ); electric field ( $E_f$ ); magnetic field ( $B$ ); particle velocity ( $v$ ); angle between velocity and magnetic field ( $\theta$ ) | Force acting on a charged particle moving through combined electric and magnetic fields. The term $qE_f$ corresponds to the electric contribution, while $qvB \sin \theta$ captures the magnetic force perpendicular to the velocity–field plane.                                                |
| I_24_6 (EHFO)  | $E_n$ : harmonic oscillator energy | mass ( $m$ ); angular frequency ( $\omega$ ); natural angular frequency ( $\omega_0$ ); displacement ( $x$ )                                                 | Energy associated with a harmonic oscillator-like system. The energy scales with mass, squared displacement, and the combined squared frequencies, representing stored mechanical energy in oscillatory motion.                                                                                  |
| I_50_26 (HOQN) | $x$ : position or displacement     | amplitude/initial displacement scale ( $x_1$ ); angular frequency ( $\omega$ ); time ( $t$ ); nonlinear correction parameter ( $\alpha$ )                    | Oscillatory displacement with a nonlinear correction term. The first term represents standard cosine oscillation, while $\alpha \cos^2(\omega t)$ introduces a second-order harmonic correction to the motion.                                                                                   |

*Continued on next page*

Table 1: Details on Feynman equations (*continued*).

| Equation        | Output variable         | Input variables                                                                                                                                                                                                                      | Scientific interpretation                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IL_36_38 (FMMM) | $f$ : magnetic response | magnetic moment ( $\text{mom}$ ); magnetic field ( $H$ ); Boltzmann constant ( $k_b$ ); temperature ( $T$ ); polarizability parameter ( $\alpha$ ); permittivity ( $\epsilon$ ); speed of light ( $c$ ); mass-like parameter ( $M$ ) | Magnetic response governed by the balance between field-moment interaction and thermal energy. The first term captures the normalized contribution of the applied magnetic field, while the second term adds a correction involving polarizability, electromagnetic constants, and the material parameter $M$ . |

**Computational environment.** All Feynman equation experiments were implemented in **Python** and executed on a MacBook Air equipped with an Apple M2 processor and 8GB of unified memory.

## B Experimental Settings of Competitors

### B.1 DSR

We use the `dso.DeepSymbolicRegressor` implementation (<https://github.com/dso-org/deep-symbolic-optimization>) of Deep Symbolic Regression (DSR) [Petersen et al., 2021] for learning the 5 Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

DSR parameterizes a policy over symbolic expressions using an autoregressive recurrent neural network (RNN) which sequentially samples symbolic programs from a prescribed function set. Across all 5 Feynman equations, the operator set is configured as follows.

#### DSR operator set

```
1 function_set = [
2     "add",      # (x, y) -> x + y
3     "mul",      # (x, y) -> x*y
4     "neg",      # x -> -x
5     "inv",      # x -> 1/x
6     "sin",      # x -> sin(x)
7     "cos",      # x -> cos(x)
8     "exp",      # x -> exp(x)
9     "n2",       # x -> x^2
10    "n3",       # x -> x^3
11 ]
```

The regression task is run with protected operators enabled. For all equations, the the following DSR configuration is maintained.

#### DSR settings

```
1 dsr_settings = {
2     "n_samples": 200000, # total symbolic program sampling budget
3     "batch_size": 500,   # samples per RNN policy-gradient update
4     "max_length": 20,    # maximum length of each sampled expression
5 }
```

These settings correspond to approximately 400 RNN policy-gradient updates, unless early stopping occurs. During training, DSR retains the best-reward symbolic program encountered over the search. The final recovered expression is extracted from the fitted estimator through

**Extracting the final DSR expression**

```
1 estimator.program_
```

which stores the best symbolic program returned by `dso.DeepSymbolicRegressor`.

**B.2 QLattice**

We use the `feyn.QLattice` implementation (<https://docs.abzu.ai/>) of QLattice [Broløs et al., 2021] for learning the 5 Feynman equations in I.12.2 (CL)-II.36-38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed, with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

QLattice constructs a lattice-based search space over symbolic interactions among input variables and primitive functions, and iteratively generates and scores candidate symbolic models. Candidate formulas are generated through the `auto_run` routine, ranked according to a specified model selection criterion, and the top-ranked model returned by QLattice is used as the learned expression. The operator set across all 5 Feynman equations is configured using the primitive functions available in the `feyn` implementation as follows.

**QLattice operator set**

```
1 function_names = [
2     "inverse", # z -> 1/z
3     "linear",  # z -> a*z + b
4     "squared", # z -> z^2
5     "add",     # (z1, z2) -> z1 + z2
6     "multiply", # (z1, z2) -> z1 * z2
7     "exp",     # z -> exp(z)
8     "sin",     # z -> sin(z)
9     "cos",     # z -> cos(z)
10 ]
```

For all equations, the QLattice search is run with the following configuration.

**QLattice settings**

```
1 qlattice_settings = {
2     "n_epochs": 25,          # number of search epochs
3     "max_complexity": 20,    # complexity of symbolic models
4     "criterion": "bic",      # Bayesian information criterion
5 }
```

The final recovered expression is extracted from the top-ranked BIC model using Python's Sympy

package [Meurer et al., 2017] as

#### Extracting the final QLattice expression

```
1 # signif -> significant digits in fitted numerical coefficients
2
3 best_model = models[0] # top-ranked BIC model
4 expression = best_model.sympify(signif=6)
```

which converts the selected QLattice model into a symbolic expression.

### B.3 SISSO++

For sure independence screening and sparsifying operator (SISSO) [Ouyang et al., 2018], we use the SISSORegressor implementation (<https://sissoopp-developers.gitlab.io/sissoopp/>) from SISSO++ [Purcell et al., 2022] for learning the 5 Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

SISSO++ constructs a large symbolic descriptor space (complete enumeration) by recursively applying a prescribed set of mathematical operators to the primary input variables  $x_1, \dots, x_p$ . It then performs sure independence screening, SIS, followed by sparse descriptor selection through the sparsifying operator, S0.

For all equations, the SISSO++ regression task is run with the following common configuration.

#### SISSO++ settings

```
1 sisso_settings = {
2     "n_dim": 3,           # number of selected descriptors
3     "max_rung": 3,        # maximum recursive feature-construction depth
4     "n_sis_select": 50,   # candidate descriptors retained by SIS
5     "n_residual": 20,     # number of residual-based selections during S0
6 }
```

For SISSO++, we use problem-specific operator sets rather than the common collection of operators across all equations. This is because of the combinatorial growth of the SISSO++ descriptor space: as `max_rung` increases, recursively applying a large operator library to the primary variables can generate an extremely large number of candidate descriptors, making the SIS and S0 steps computationally expensive. Therefore, for each Feynman equation, we restrict the operator set to the algebraic and functional components required to represent the corresponding ground-truth expression. This provides a computationally feasible and structurally informed benchmark for SISSO++, while still allowing the method to search over a rich

nonlinear descriptor space generated from the specified operators.

#### SISSO++ operator set

```

1 # add: +, mult: *, div: /, sq: ^2, sin: sin(), cos: cos(),
2 siso_operator_sets = {
3     "I_12_2": ["add", "div", "sq"],          # CL
4     "I_12_11": ["add", "mult", "sin"],       # FCE
5     "I_24_6": ["add", "mult", "sq"],         # EHFO
6     "I_50_26": ["add", "mult", "cos", "sq"], # HOQN
7     "II_36_38": ["add", "mult", "div", "sq"], # FMMM
8 }
```

The final recovered expression is taken as the sparse linear combination of selected symbolic descriptors returned by `SISSORegressor`.

## B.4 gplearn

We use the `gplearn.genetic.SymbolicRegressor` (<https://gplearn.readthedocs.io/>) implementation of `gplearn` [Stephens, 2016] for learning the 5 Feynman equations in I\_12\_2 (CL)-II\_36\_38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

`gplearn` evolves symbolic expressions using genetic programming [Koza, 1992]. Starting from an initial population of randomly generated symbolic programs, it repeatedly applies evolutionary operations such as selection, crossover, mutation, and reproduction to improve predictive fit. In our experiments, symbolic programs are scored using RMSE, and a parsimony penalty is included to discourage overly complex expressions.

Across all 5 Feynman equations, the operator set is configured as follows.

#### gplearn operator set

```

1 function_set = [
2     "add",      # (x, y) -> x + y
3     "mul",      # (x, y) -> x*y
4     "neg",      # x -> -x
5     "inv",      # x -> 1/x
6     "sin",      # x -> sin(x)
7     "cos",      # x -> cos(x)
8     "exp",      # x -> exp(x)
9     "sq",       # x -> x^2
10    "cu",       # x -> x^3
11 ]
```



The unary functions `neg`, `inv`, `sin`, `cos`, `exp`, <sup>2</sup>, and <sup>3</sup> are supplied through custom wrapper `gplearn.functions.make_function`. The inverse operator is protected near zero, and the exponential operator is clipped for numerical stability.

For all equations, the following `gplearn` configurations are maintained.

#### gplearn settings

```

1  # metric: fitness metric to select symbolic expression;
2  # generations: number of evolutionary generations;
3  # population_size: number of symbolic programs maintained in each generation;
4  # tournament_size: tournament selection pressure used to choose parent programs;
5  # parsimony_coefficient: expression complexity during fitness evaluation;
6  # stopping_criteria: early stopping threshold for RMSE;
7  # p_crossover: subtree crossover probability;
8  # p_subtree_mutation: subtree mutation probability;
9  # p_hoist_mutation: hoist mutation probability;
10 # p_point_mutation: point mutation probability;
11 # max_samples: fraction of training samples per program.
12
13 gplearn_settings = {
14     "metric": "rmse",
15     "generations": 20,
16     "population_size": 2000,
17     "tournament_size": 20,
18     "parsimony_coefficient": 1e-4,
19
20     # gplearn default settings implemented below
21     "stopping_criteria": 0.00,
22     "p_crossover": 0.9,
23     "p_subtree_mutation": 0.01,
24     "p_hoist_mutation": 0.01,
25     "p_point_mutation": 0.01,
26     "max_samples": 1.0,
27 }
```

The final recovered expression is extracted from the fitted estimator through

#### Extracting the final gplearn expression

```
1 model._program
```

which stores the best symbolic program retained by `gplearn`.

## B.5 operon

We use the `pyoperon.sklearn.SymbolicRegressor` implementation (<https://github.com/heal-research/operon>) of `operon` [Burlacu et al., 2020] for learning the 5 Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected

into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

**operon** is a genetic programming-based SR method that evolves expression trees using evolutionary search. Starting from a population of candidate symbolic expressions, **operon** iteratively applies variation and selection operators to improve predictive fit while searching over a prescribed set of primitive symbols. In our implementation, the symbolic search is performed through the **SymbolicRegressor** interface from the Python package **pyoperon**.

Across all 5 Feynman equations, the primitive symbol set is configured as follows.

#### operon operator set

```
1 allowed_symbols = [
2     "add",      # (x, y) -> x + y
3     "mul",      # (x, y) -> x*y
4     "sub",      # (x, y) -> x - y
5     "div",      # (x, y) -> x/y
6     "sin",      # x -> sin(x)
7     "cos",      # x -> cos(x)
8     "exp",      # x -> exp(x)
9     "square",   # x -> x^2
10    "variable", # input variable leaf nodes
11    "constant", # numerical constant leaf nodes
12 ]
```

For all equations, **operon** is implemented with the following configuration.

#### operon settings

```
1 # generations: number of evolutionary generations;
2 # population_size: number of candidate expressions maintained in the population;
3 # max_length: maximum length of the evolved expressions
4
5 operon_settings = {
6     "generations": 2000,
7     "population_size": 1000,
8     "max_length": 20,
9 }
```

The **operon** estimator is initialized as follows.

#### operon module initialization

```
1 model = SymbolicRegressor(
2     allowed_symbols=allowed_symbols,
3     generations=2000,
4     population_size=1000,
5     max_length=20,
6 )
```

The final recovered expression is extracted from the fitted estimator using the available `operon` model-string interface viz.,

#### Extracting the final operon expression

```
1 model.get_model_string(model.model_)
```

whenever the fitted model is stored in `model_`; otherwise, the corresponding best available expression attribute returned by the fitted `SymbolicRegressor` is used.

## B.6 PySR

We use the `pysr.PySRRegressor` implementation (<https://github.com/MilesCranmer/PySR>) of PySR [Cranmer, 2023] for learning the 5 Feynman equations in I.12.2 (CL)-II.36-38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

PySR performs evolutionary SR by searching over expression trees composed of user-specified binary and unary operators. The search maintains multiple evolving populations of candidate expressions and optimizes them using mutation, crossover, selection, and constant optimization. In our experiments, the symbolic search is performed through the `PySRRegressor` interface, with squared-error loss and deterministic serial execution.

Across all 5 Feynman equations, the operator set is configured as follows.

#### PySR operator set

```
1 binary_operators = [
2     "+",          # (x, y) -> x + y
3     "*",          # (x, y) -> x*y
4 ]
5
6 unary_operators = [
7     "neg(x) = -x",      # x -> -x
8     "inv(x) = 1 / x",   # x -> 1/x
9     "sin",              # x -> sin(x)
10    "cos",              # x -> cos(x)
11    "exp",              # x -> exp(x)
12    "sq(x) = x * x",     # x -> x^2
13    "cu(x) = x * x * x", # x -> x^3
14 ]
```

The custom unary operators `neg`, `inv`, `square`, and `cube` are additionally mapped to their corresponding SymPy [Meurer et al., 2017] expressions to allow symbolic conversion of the final

recovered model.

For all equations, PySR is run with the following configuration.

#### PySR settings

```

1  # niterations: number of PySR evolutionary iterations;
2  # populations: number of populations evolved during the search;
3  # population_size: number of candidate expressions in each population;
4  # maxsize: maximum expression size.
5
6  pysr_settings = {
7      "niterations": 200,
8      "populations": 10,
9      "population_size": 50,
10     "maxsize": 20,
11     "elementwise_loss": "loss(x, y) = (x - y)^2", # squared-error loss
12     "model_selection": "best", # for symbolic expression selection
13     "deterministic": True, # to ensure reproducibility
14     "parallelism": "serial", # avoid variation owing to parallel execution
15 }
```

The custom SymPy mappings used for symbolic conversion are as follows.

#### PySR SymPy mappings for symbolic conversion

```

1  extra_sympy_mappings = {
2      "neg": lambda x: -x,
3      "inv": lambda x: 1 / x,
4      "sq": lambda x: x**2,
5      "cu": lambda x: x**3,
6  }
```

The final recovered expression is extracted from the best PySR model using

#### Extracting the best PySR model

```

1  model.get_best()["equation"]
```

whenever available; otherwise, the symbolic expression returned by

#### Extracting the final SymPy-supported PySR expression

```

1  model.sympy()
```

is used.

## B.7 BMS

We use the BMSRegressor implementation from the AutoRA module (<https://autoresearch.github.io/autora/user-guide/theorists/bms/>) [Musslick et al., 2024] of Bayesian machine

scientist (BMS) [Guimerà et al., 2020] for learning the 5 Feynman equations in I\_12.2 (CL)-II\_36.38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

Across all 5 Feynman equations, the BMS operator set is configured using the primitive operators supported by the BMSRegressor interface. The common operator configuration is as follows.

#### BMS operator set

```
1 bms_operator_set = [
2     "add",    # (x, y) -> x + y
3     "mul",    # (x, y) -> x*y
4     "neg",    # x -> -x
5     "div",    # (x, y) -> x/y
6     "sin",    # x -> sin(x)
7     "cos",    # x -> cos(x)
8     "exp",    # x -> exp(x)
9     "sq",     # x -> x^2
10    "cu",     # x -> x^3
11 ]
```

The aforementioned operators are passed through the ops dictionary and a uniform prior weight is assigned to each included operator.

#### BMS operator dictionary and weights

```
1 bms_ops = {
2     "+": 2,
3     "*": 2,
4     "/": 2,
5     "-": 1,
6     "sin": 1,
7     "cos": 1,
8     "exp": 1,
9     "pow2": 1,
10    "pow3": 1,
11 }
12
13 bms_prior = {
14     "Nopi_+": 1.0,
15     "Nopi_*": 1.0,
16     "Nopi_/": 1.0,
17     "Nopi_-": 1.0,
18     "Nopi_sin": 1.0,
19     "Nopi_cos": 1.0,
20     "Nopi_exp": 1.0,
21     "Nopi_pow2": 1.0,
```

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```

22     "Nopi_pow3": 1.0,
23 }

```

For all equations except II.36\_38 (FMMM), each BMS chain is run for `epochs = 2000` MCMC steps. For II.36\_38 (FMMM), which is the most structurally complex Feynman equation considered here, we use `epochs = 20000`. BMS is run using 5 independent parallel chains.

**BMS settings**

```

1 bms_settings = {
2     "n_parallel_chains": 5,
3     "epochs": 2000, # 20000 for II_36_38 (FMMM)
4     "parallel_chains": True,
5 }

```

After all chains are completed, the chain with the lowest in-sample RMSE (computed on a 90% train split) is selected, and its symbolic expression is used as the final recovered expression.

**Extracting the final BMS expression across parallel chains**

```

1 best_chain = min(
2     successful_chains,
3     key=lambda chain: chain["chain_train_rmse"],
4 )
5
6 final_expression = best_chain["expression"]

```

**B.8 BSR**

We use the `BSRRegressor` implementation from the `AutoRA` module (<https://autoresearch.github.io/autora/user-guide/theorists/bsr/>) of Bayesian symbolic regression (BSR) [Jin et al., 2020] for learning the 5 Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). For each equation, we subsampled  $n = 2000$  observations out of  $10^5$  observations. We consider the noiseless setting (with the original target) and noisy regimes, where independent zero-mean Gaussian noise with standard deviation  $\sigma$  is injected into the target. For each equation and noise level, 5 independent repetitions were performed with train/test RMSE and  $R^2$  computed on a 90/10 train/test split. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

Across all 5 Feynman equations, the BSR operator set is configured as follows.

**BSR operator set**

```

1 bsr_operator_set = [
2     "add",    # (x, y) -> x + y
3     "mul",    # (x, y) -> x*y

```

*continued on next page*

*continued from previous page*

```

4     "neg",    # x -> -x
5     "inv",    # x -> 1/x
6     "sin",    # x -> sin(x)
7     "cos",    # x -> cos(x)
8     "exp",    # x -> exp(x)
9     "sq",     # x -> x^2
10    "cu",     # x -> x^3
11 ]

```

The aforementioned operator set is converted to the corresponding BSR prior dictionary.

#### BSR operator prior dictionary

```

1 bsr_operator_prior = {
2     "+": 1.0,
3     "*": 1.0,
4     "neg": 1.0,
5     "inv": 1.0,
6     "sin": 1.0,
7     "cos": 1.0,
8     "exp": 1.0,
9     "pow2": 1.0,
10    "pow3": 1.0,
11 }

```

For all equations except II\_36\_38 (FMMM), we set `itr_num` = 2000, while for II\_36\_38 we set `itr_num` = 20000. We use 5 independent restarts for each BSR fit. The main BSR configuration is given below. Note that, the split probability parameters are maintained at their default specifications.

#### BSR settings

```

1 # split probability parameters maintained at default settings
2
3 bsr_settings = {
4     "n_restarts": 5,
5     "itr_num": 2000, # 20000 for II_36_38 (FMMM)
6     "prior_name": "restricted_ops",
7 }

```

The number of symbolic trees  $K$  is chosen to match the corresponding **HIERBOSSS** configuration described in [Section D.1](#). The equation-specific tree configurations are given below.

#### BSR $K$ configuration

```

1 bsr_tree_num = {
2     "I_12_2": 4,
3     "I_12_11": 3,
4     "I_24_6": 4,

```

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```

5     "I_50_26": 4,
6     "II_36_38": 4,
7 }

```

Thus, for a given equation, the BSR estimator is initialized as follow.

#### BSR module initialization

```

1 model = BSRRegressor(
2     tree_num=bsr_tree_num[equation_name],
3     itr_num=2000, # 20000 for II_36_38 (FMMM)
4 )

```

After all restarts are completed, the final recovered expression is taken from the successful restart with the lowest in-sample RMSE (computed on a 90% train set).

#### Extracting the final BSR expression across restarts

```

1 best_restart = min(
2     successful_restarts,
3     key=lambda restart: restart["chain_train_rmse"],
4 )
5
6 final_expression = best_restart["expression"]

```

The final expression is extracted as the additive symbolic model returned by `BSRRegressor`, together with the fitted coefficients of the selected symbolic trees.



## C Results of Competitors for Learning Feynman Equations

### C.1 Results for I\_12.2 (CL)

Table 2: DSR results for the Feynman equation I\_12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                  |            |           |             |            |             |
| 1                                              | $q_2 \{ \{ r(r+r)(\epsilon+\epsilon) \}^{-1} \}$                                                 | 0.038      | 0.044     | 0.808       | 0.760      | 62.7        |
| 2                                              | $\{ q_1^{-1} + (\epsilon r + r)r \}^{-1}$                                                        | 0.049      | 0.054     | 0.657       | 0.679      | 93.0        |
| 3                                              | $q_1 \{ r + r\epsilon \}^{-2}$                                                                   | 0.040      | 0.035     | 0.806       | 0.794      | 83.4        |
| 4                                              | $q_2 \{ \epsilon r + r \}^{-2}$                                                                  | 0.043      | 0.040     | 0.791       | 0.725      | 57.7        |
| 5                                              | $q_1 \{ q_1 + \epsilon \exp[r \exp\{\sin(q_2^{-2})\}] \}^{-1}$                                   | 0.028      | 0.023     | 0.898       | 0.923      | 85.3        |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                  |            |           |             |            |             |
| 1                                              | $\epsilon q_1 \exp\{- (\epsilon + r)\}$                                                          | 0.155      | 0.148     | 0.223       | 0.185      | 95.3        |
| 2                                              | $\sin\left[- \left\{ - \epsilon \exp(q_2^{-1})(r+r) + \epsilon \right\}^{-1}\right]$             | 0.159      | 0.149     | 0.160       | 0.116      | 112.0       |
| 3                                              | $q_1 \{ (r^3 + q_1)\epsilon \}^{-1}$                                                             | 0.162      | 0.156     | 0.185       | 0.276      | 109.8       |
| 4                                              | $\sin\left[\{ q_1 [r(\epsilon + q_1)]^{-1} \}^2\right]$                                          | 0.155      | 0.154     | 0.190       | 0.214      | 106.6       |
| 5                                              | $\sin\left[\{ q_2 (\epsilon + r + \exp r)^{-1} \}^2\right]$                                      | 0.156      | 0.145     | 0.163       | 0.237      | 113.3       |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                  |            |           |             |            |             |
| 1                                              | $\{ (\exp r + \epsilon) \cos(\epsilon^{-1}) \}^{-1}$                                             | 0.203      | 0.206     | 0.083       | 0.133      | 140.3       |
| 2                                              | $q_2 \sin\left[\{ r^2 + r\epsilon \}^{-2}\right]$                                                | 0.205      | 0.201     | 0.127       | 0.185      | 148.2       |
| 3                                              | $\sin\left[\left\{ \epsilon r^3 [q_2^2 \{ \exp(r) \}^{-1}]^{-1} + \epsilon \right\}^{-1}\right]$ | 0.205      | 0.208     | 0.116       | 0.220      | 124.7       |
| 4                                              | $\sin\left[\epsilon \{ r(\epsilon + q_2^{-1}) \}^{-2}\right]$                                    | 0.200      | 0.213     | 0.078       | 0.111      | 119.9       |
| 5                                              | $q_2 \left\{ \epsilon^{-1} + (r^3 + \exp \epsilon + \cos(q_1 + r))r \right\}^{-1}$               | 0.208      | 0.199     | 0.128       | 0.076      | 84.7        |

Table 3: QLattice results for the Feynman equation I\_12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                                                                                                                                          | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                                                                                             |            |           |             |            |             |
| 1                        | $ \begin{aligned} & -2.19 \times 10^{-4} q_2 + 2.03 \times 10^{-4} \\ & -0.100(-0.044 q_1 - 0.001) [0.065 r - 0.211 \\ & + (1.077 - 0.002 \epsilon)(-1.409 q_2 - 0.021)(0.155 r - 1.036)] \\ & \times \{ (0.264 - 0.464 r)(-0.347 \epsilon - 0.010) \}^{-1} \end{aligned} $ | 0.003      | 0.002     | 0.999       | 0.999      | 95.666      |

Continued on next page

Table 3: **QLattice** results for the Feynman equation I\_12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                                         | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 2                                              | $-8.94 \times 10^{-4}$ $- 2.298 \left[ -0.239\epsilon + (1.020 - 0.309q_2)(0.235\epsilon + 0.424) - 0.429 \right]$ $\times (0.015\epsilon + 0.417q_1 + 0.002r - 0.030)$ $\times \left\{ (0.003 - r)^2 (\epsilon + 0.567)^2 \right\}^{-1}$                                                                  | 0.001      | 0.001     | 1.000       | 1.000      | 95.883      |
| 3                                              | $-5.11 \times 10^{-5}$ $- 0.075(-1.224q_1 - 0.020)(0.067q_2 - 0.002)$ $\times [0.024\epsilon - 0.005q_2 + (1.588 - 0.213r)(2.815 - 0.355r) + 0.694]$ $\times \left\{ (0.024 - 0.817\epsilon)(0.428 - 0.808r) \right\}^{-1}$                                                                                | 0.002      | 0.002     | 1.000       | 1.000      | 91.454      |
| 4                                              | $-4.30 \times 10^{-5}q_1 + 3.97 \times 10^{-6}$ $+ 0.258(-0.249q_1 - 1.32 \times 10^{-4})(-0.071q_2 - 0.307)$ $\times (0.214q_2 + 8.66 \times 10^{-4})$ $\times \left\{ (0.088 - 0.585r)(-0.195\epsilon - 5.70 \times 10^{-4}) \right.$ $\left. \times (-0.340q_2 - 1.473)(-0.311r - 0.061) \right\}^{-1}$ | 0.000      | 0.000     | 1.000       | 1.000      | 92.951      |
| 5                                              | $-0.231(0.004 - 0.480q_1)(0.038\epsilon - 0.597)(0.484q_2 - 0.022)$ $\times \exp\{2.441 \exp(-0.563\epsilon)\}$ $\times \left[ -0.010q_2 + (1.341 - 1.314r)(1.189r + 1.129) - 1.544 \right]^{-1}$ $- 2.15 \times 10^{-4}$                                                                                  | 0.001      | 0.001     | 1.000       | 1.000      | 93.297      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                                                                                            |            |           |             |            |             |
| 1                                              | $0.373 \exp(-0.457\epsilon + 0.346q_1 + 0.261q_2 - 1.107r) + 0.002$                                                                                                                                                                                                                                        | 0.150      | 0.144     | 0.272       | 0.226      | 84.850      |
| 2                                              | $1.498 \exp[(0.424\epsilon - 0.244q_1 + 1.296)$ $\times (0.264q_2 - 1.283r - 0.635)] + 0.027$                                                                                                                                                                                                              | 0.153      | 0.148     | 0.223       | 0.120      | 83.578      |
| 3                                              | $1.895(0.233q_1 - 0.080)$ $\times \exp[0.356q_2 - 1.316r + (0.325\epsilon - 2.744)(0.343\epsilon - 0.123)]$ $+ 0.003$                                                                                                                                                                                      | 0.153      | 0.157     | 0.270       | 0.262      | 85.915      |
| 4                                              | $-2.544(0.090 - 0.207q_1) \exp(-0.492\epsilon + 0.345q_2 - 1.449r)$ $+ 0.011$                                                                                                                                                                                                                              | 0.152      | 0.148     | 0.230       | 0.271      | 81.465      |
| 5                                              | $0.408 \exp(-0.553\epsilon + 0.303q_1 + 0.370q_2 - 1.306r) + 0.013$                                                                                                                                                                                                                                        | 0.150      | 0.140     | 0.228       | 0.293      | 79.621      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                                                                                                                            |            |           |             |            |             |
| 1                                              | $0.465 \exp(-0.618\epsilon + 0.276q_1 + 0.282q_2 - 1.060r) + 0.014$                                                                                                                                                                                                                                        | 0.198      | 0.197     | 0.127       | 0.208      | 72.570      |
| 2                                              | $0.418 \exp(-0.468\epsilon + 0.335q_1 + 0.339q_2 - 1.382r) - 0.005$                                                                                                                                                                                                                                        | 0.201      | 0.195     | 0.164       | 0.233      | 78.285      |
| 3                                              | $0.458 \exp(-0.569\epsilon + 0.298q_1 + 0.371q_2 - 1.358r) + 0.013$                                                                                                                                                                                                                                        | 0.201      | 0.199     | 0.150       | 0.289      | 74.692      |
| 4                                              | $-0.277(1.244 - 0.209\epsilon)(0.193q_2 + 0.023)$ $\times \{0.180q_1 - 0.921r - 0.050\}^{-1} + 0.003$                                                                                                                                                                                                      | 0.197      | 0.208     | 0.109       | 0.152      | 66.408      |
| 5                                              | $0.385 \exp(-0.580\epsilon + 0.234q_1 + 0.421q_2 - 1.107r) + 0.006$                                                                                                                                                                                                                                        | 0.204      | 0.198     | 0.164       | 0.088      | 76.088      |

Table 4: **SISSO++** results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                              |            |           |             |            |             |
| 1                                              | $-4.642 \times 10^{-17} + 1.430 \times 10^{-22} \epsilon q_1 q_2^2 (q_2/r)^2$<br>$+ 2.206 \times 10^{-19} \frac{\epsilon q_2 (q_1/r)}{r^2 (r/q_1)} + 0.080 \frac{q_1 q_2}{\epsilon r^2}$     | 0.000      | 0.000     | 1.000       | 1.000      | 14.700      |
| 2                                              | $-5.029 \times 10^{-17} - 1.126 \times 10^{-18} \frac{q_2^2 (q_1/r)}{\epsilon^2}$<br>$- 3.676 \times 10^{-18} \frac{q_1/r}{(\epsilon/q_2)\epsilon^2} + 0.080 \frac{q_1 q_2}{\epsilon r^2}$   | 0.000      | 0.000     | 1.000       | 1.000      | 17.523      |
| 3                                              | $-4.654 \times 10^{-17} + 0.080 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 6.090 \times 10^{-20} \frac{q_2^2 (q_1 q_2)}{(r\epsilon)(r/q_2)} - 1.016 \times 10^{-18} \frac{(q_1/r)^2}{r^2 \epsilon}$ | 0.000      | 0.000     | 1.000       | 1.000      | 16.854      |
| 4                                              | $-4.955 \times 10^{-17} - 2.165 \times 10^{-17} \frac{q_1/r}{(r\epsilon)(r q_2)}$<br>$+ 0.080 \frac{q_1 q_2}{\epsilon r^2} - 6.040 \times 10^{-18} \frac{(q_1 q_2)/\epsilon^2}{(r/q_2)r^2}$  | 0.000      | 0.000     | 1.000       | 1.000      | 17.073      |
| 5                                              | $-4.998 \times 10^{-17} - 3.856 \times 10^{-19} \frac{\epsilon q_1 q_2}{r^2}$<br>$+ 0.080 \frac{q_1 q_2}{\epsilon r^2} - 6.617 \times 10^{-19} \left( \frac{q_1 q_2}{r\epsilon} \right)^2$   | 0.000      | 0.000     | 1.000       | 1.000      | 17.043      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                              |            |           |             |            |             |
| 1                                              | $-0.004 + c_1 \frac{r^5}{\epsilon q_2^2} - 0.009 \frac{\epsilon r}{q_1^4}$<br>$+ 0.079 \frac{q_1 q_2}{\epsilon r^2}$                                                                         | 0.147      | 0.148     | 0.258       | 0.238      | 17.756      |
| 2                                              | $0.007 - 0.020 \frac{q_1^2}{\epsilon^3 q_2^4} + 0.071 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.009 \frac{q_1 q_2}{\epsilon^2 r}$                                                                | 0.148      | 0.147     | 0.263       | 0.273      | 17.300      |
| 3                                              | $0.005 + c_1 \frac{r^7}{\epsilon} + c_2 \frac{q_1^4 q_2}{r^3}$<br>$+ 0.073 \frac{q_1 q_2}{\epsilon r^2}$                                                                                     | 0.147      | 0.150     | 0.276       | 0.201      | 17.438      |
| 4                                              | $-0.003 + c_1 \epsilon^5 q_1 q_2 / r + c_2 q_1 q_2^6 / r$<br>$+ 0.084 \frac{q_1 q_2}{\epsilon r^2}$                                                                                          | 0.150      | 0.146     | 0.258       | 0.261      | 17.967      |
| 5                                              | $-0.002 + 0.011 \frac{q_2^4}{\epsilon^2 q_1^4} + 0.095 \frac{q_1 q_2}{\epsilon r^2}$<br>$- 0.066 \frac{q_2}{\epsilon r^3}$                                                                   | 0.151      | 0.150     | 0.254       | 0.271      | 17.298      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                              |            |           |             |            |             |
| 1                                              | $0.003 + c_1 \frac{r^4 q_1}{\epsilon q_2^2} + 0.019 \frac{q_1 q_2}{r^3}$<br>$+ 0.077 \frac{q_1 q_2}{\epsilon^2 r^2}$                                                                         | 0.195      | 0.198     | 0.164       | 0.144      | 17.098      |
| 2                                              | $0.009 - 0.026 \frac{q_1^2}{\epsilon^3 q_2^4} + 0.068 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.012 \frac{q_1 q_2}{\epsilon^2 r}$                                                                | 0.198      | 0.196     | 0.168       | 0.180      | 17.476      |
| 3                                              | $0.006 + c_1 \frac{r^7}{\epsilon} + c_2 \frac{q_1^4 q_2}{r^3}$<br>$+ 0.071 \frac{q_1 q_2}{\epsilon r^2}$                                                                                     | 0.196      | 0.200     | 0.176       | 0.120      | 17.432      |

Continued on next page

Table 4: **SISSO++** results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi \epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|-----------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4    | $-0.005 + c_1 \epsilon^5 q_1 q_2 / r + c_2 q_1 q_2^6 / r$<br>$+ 0.086 \frac{q_1 q_2}{\epsilon r^2}$                               | 0.200      | 0.195     | 0.165       | 0.166      | 17.477      |
| 5    | $0.002 + 0.016 \frac{q_2^2}{\epsilon^2 q_1^4} + 0.038 \frac{q_1^3 q_2}{\epsilon r^2}$<br>$- 0.004 \frac{q_1^3 q_2}{\epsilon r^2}$ | 0.202      | 0.201     | 0.163       | 0.166      | 18.443      |

Table 5: **gplearn** results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi \epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                      | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $\frac{q_1 q_2^2}{\epsilon \{q_1 q_2^2 r^2 + q_1 (r^3 + 1) + q_2\}}$                                                                                    | 0.037      | 0.045     | 0.818       | 0.746      | 16.505      |
| 2                                              | $q_1^2 q_2 \left\{ q_1 [\epsilon \{r + \sin(q_1^{-1}) - 0.509\} + r + \sin(e^{-\epsilon}) - 1.018] + 3 \right\}^{-2}$                                   | 0.013      | 0.013     | 0.977       | 0.982      | 17.639      |
| 3                                              | $\left\{ r + [\epsilon + r + \sin(q_1) + \sin(q_2)] - \cos \left\{ \frac{r}{r + \{\epsilon + r + 2 \sin(q_1) + \sin(q_2)\}^2} \right\}^2 \right\}^{-1}$ | 0.031      | 0.029     | 0.886       | 0.857      | 17.541      |
| 4                                              | $e^{3q_2} \left\{ \epsilon [\{r + \sin(q_1^{-9})\} e^{q_2} + 1]^3 \right\}^{-1}$                                                                        | 0.048      | 0.043     | 0.738       | 0.692      | 15.868      |
| 5                                              | $\left[ \epsilon r^2 + \sin(q_1) + \{q_2 - \sin(q_1^{-1})\}^{-2} + (q_2 - q_2^{-1})^{-2} + q_2^{-1} + q_2^{-2} \right]^{-1}$                            | 0.030      | 0.031     | 0.882       | 0.860      | 19.191      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $q_1^4 \{\epsilon(\epsilon^2 + q_1^4 r^2 + q_1^2 r^2 + e^r)\}^{-1}$                                                                                     | 0.154      | 0.151     | 0.232       | 0.148      | 15.586      |
| 2                                              | $\{r(\epsilon r + 1)\}^{-1}$                                                                                                                            | 0.161      | 0.151     | 0.137       | 0.092      | 14.700      |
| 3                                              | $q_1 q_2 (\epsilon r + e^r)^{-2}$                                                                                                                       | 0.154      | 0.156     | 0.263       | 0.274      | 15.547      |
| 4                                              | $q_1^3 q_2 \left\{ r [\epsilon^4 r^3 + q_1^3 (q_2 r + 1) + q_1^2 (q_2 r + 1)] \right\}^{-1}$                                                            | 0.153      | 0.153     | 0.213       | 0.224      | 16.498      |
| 5                                              | $\exp \left[ \frac{r \{-\epsilon q_2^2 - q_1 (q_2 + 2)\}}{q_1 q_2^2} \right]$                                                                           | 0.152      | 0.142     | 0.204       | 0.270      | 16.168      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $q_1^2 \{\epsilon(q_1^2 r^2 + q_1 r^2 + 1)\}^{-1}$                                                                                                      | 0.200      | 0.202     | 0.106       | 0.173      | 16.173      |
| 2                                              | $\sin \left[ \sin \left\{ \sin \left( \sin \{ \epsilon r^3 \}^{-1} \right) \right\} \right]$                                                            | 0.207      | 0.208     | 0.112       | 0.121      | 15.123      |
| 3                                              | $\exp(-0.686/q_2)(\epsilon r^2)^{-1}$                                                                                                                   | 0.206      | 0.210     | 0.109       | 0.205      | 15.370      |
| 4                                              | $\{r^2(\epsilon + 1)\}^{-1}$                                                                                                                            | 0.201      | 0.216     | 0.066       | 0.092      | 14.981      |
| 5                                              | $q_2 \{\epsilon r (q_2 r + 2)\}^{-1}$                                                                                                                   | 0.209      | 0.195     | 0.125       | 0.111      | 16.827      |

Table 6: **operon** results for the Feynman equation I\_12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                                                             | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                                                |            |           |             |            |             |
| 1                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$                                                                                                                                                                                            | 0.000      | 0.000     | 1.000       | 1.000      | 18.776      |
| 2                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$                                                                                                                                                                                            | 0.000      | 0.000     | 1.000       | 1.000      | 18.709      |
| 3                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$                                                                                                                                                                                            | 0.000      | 0.000     | 1.000       | 1.000      | 20.110      |
| 4                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$                                                                                                                                                                                            | 0.000      | 0.000     | 1.000       | 1.000      | 20.938      |
| 5                                              | 0                                                                                                                                                                                                                              | 0.000      | 0.000     | 1.000       | 1.000      | 19.030      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                |            |           |             |            |             |
| 1                                              | -0.007                                                                                                                                                                                                                         | 0.148      | 0.145     | 0.288       | 0.222      | 26.672      |
| 2                                              | $-0.057(\epsilon r^2)^{-1} [0.035\epsilon r^2 - 0.245q_1^2 r \sin^2\{1.606q_1/r\} - 0.042q_1 q_2 r \sin^2\{1.606q_1/r\} - q_1 q_2 - 0.171q_2^2]$                                                                               | 0.152      | 0.146     | 0.229       | 0.144      | 23.136      |
| 3                                              | $0.061(\epsilon r^3)^{-1} [0.132\epsilon r^3 + 0.327q_1 q_2^2 + 0.432q_1 q_2 r + q_2 r \sin\{1.248q_2/r\}]$                                                                                                                    | 0.152      | 0.156     | 0.280       | 0.271      | 23.952      |
| 4                                              | $0.195\{\epsilon q_1 r(r - 0.335)\}^{-1} [0.015\epsilon q_1 r^2 - 0.005\epsilon q_1 r + 0.308q_1^2 q_2 - 0.553q_1 \cos(0.956\epsilon^2) + \cos(2.539q_1)]$                                                                     | 0.151      | 0.152     | 0.238       | 0.237      | 24.718      |
| 5                                              | 0.014                                                                                                                                                                                                                          | 0.149      | 0.139     | 0.235       | 0.300      | 22.389      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                                                |            |           |             |            |             |
| 1                                              | $-0.074e^{-\sin(16.325\epsilon/r)}(\epsilon r^2)^{-1} [0.284\epsilon r^2 e^{\sin(16.325\epsilon/r)} \sin(8.519q_1 r) - 0.501\epsilon r^2 e^{\sin(16.325\epsilon/r)} + 0.284\epsilon r^2 - q_1 q_2 e^{\sin(16.325\epsilon/r)}]$ | 0.196      | 0.199     | 0.143       | 0.195      | 26.691      |
| 2                                              | $0.111(\epsilon r^2)^{-1} [-0.073\epsilon q_1^2 - 0.117\epsilon q_1 r + 0.265\epsilon q_1 + 0.202\epsilon r^2 - 0.355\epsilon r + 0.172\epsilon + q_1 q_2 - 0.832q_2 r + 0.561q_2]$                                            | 0.200      | 0.194     | 0.170       | 0.235      | 20.375      |
| 3                                              | $-0.075q_2(\epsilon r^2)^{-1} [-q_1 + 0.527r \sin\{\cos(2.385\epsilon + 2.675r)\} + 0.527r \cos(2.375\epsilon + 2.591q_2)]$                                                                                                    | 0.200      | 0.199     | 0.159       | 0.285      | 26.512      |
| 4                                              | $0.015\{\epsilon r \cos(1.285/r^2)\}^{-1} [\epsilon r \cos(1.285/r^2) + 0.968q_1 q_2 e^{\sin(0.572q_2/r)}]$                                                                                                                    | 0.196      | 0.211     | 0.114       | 0.127      | 25.906      |
| 5                                              | $0.050e^{-0.310\epsilon r} \{\epsilon r \sin(0.619r)\}^{-1} [\epsilon q_1 q_2 \sin(0.619r) + 0.099\epsilon r e^{0.310\epsilon r} \sin(0.619r) + 0.601q_2^2]$                                                                   | 0.203      | 0.196     | 0.170       | 0.106      | 25.677      |

Table 7: **PySR** results for the Feynman equation I\_12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                  | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|-------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                     |            |           |             |            |             |
| 1                        | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$ | 0.000      | 0.000     | 1.000       | 1.000      | 71.351      |

Continued on next page

Table 7: PySR results for the Feynman equation I.12.2 (CL)  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------|------------|-----------|-------------|------------|-------------|
| 2                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$  | 0.000      | 0.000     | 1.000       | 1.000      | 72.965      |
| 3                                              | $q_1 q_2 (12.566 \epsilon r^2)^{-1}$ | 0.000      | 0.000     | 1.000       | 1.000      | 73.835      |
| 4                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$  | 0.000      | 0.000     | 1.000       | 1.000      | 61.572      |
| 5                                              | $0.080 q_1 q_2 (\epsilon r^2)^{-1}$  | 0.000      | 0.000     | 1.000       | 1.000      | 70.044      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                      |            |           |             |            |             |
| 1                                              | $(\epsilon r^3)^{-1}$                | 0.160      | 0.151     | 0.173       | 0.156      | 68.526      |
| 2                                              | $\exp\{-1.337r\}$                    | 0.166      | 0.155     | 0.077       | 0.040      | 68.534      |
| 3                                              | $0.082 q_1 q_2 (\epsilon r^2)^{-1}$  | 0.152      | 0.156     | 0.276       | 0.274      | 69.974      |
| 4                                              | $\exp\{-1.307r\}$                    | 0.164      | 0.158     | 0.099       | 0.172      | 83.431      |
| 5                                              | $0.163 r^{-1}$                       | 0.163      | 0.156     | 0.079       | 0.118      | 73.418      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                      |            |           |             |            |             |
| 1                                              | $\exp\{-1.314r\}$                    | 0.206      | 0.210     | 0.052       | 0.103      | 102.550     |
| 2                                              | $\exp\{-1.385r\}$                    | 0.211      | 0.218     | 0.079       | 0.036      | 74.571      |
| 3                                              | $\exp\{-1.365r\}$                    | 0.212      | 0.230     | 0.057       | 0.045      | 68.500      |
| 4                                              | $0.178 r^{-1}$                       | 0.204      | 0.218     | 0.046       | 0.072      | 70.455      |
| 5                                              | $\exp\{-1.348r\}$                    | 0.216      | 0.196     | 0.061       | 0.108      | 92.480      |

Table 8: BMS results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS.

| Rep.                                           | Learned expression                                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                   |            |           |             |            |             |
| 1                                              | $-\frac{q_1 q_2}{a_0 \epsilon r^2} - \frac{q_2 \sin(a_0)}{a_0^2}$ | 0.000      | 0.000     | 1.000       | 1.000      | 231.897     |
| 2                                              | $\frac{2q_1 q_2}{\epsilon r^2(2a_0 + 1)}$                         | 0.000      | 0.000     | 1.000       | 1.000      | 198.388     |
| 3                                              | $\frac{2e^{2/e} q_1 q_2}{a_0 \epsilon r^2}$                       | 0.000      | 0.000     | 1.000       | 1.000      | 201.371     |
| 4                                              | $\frac{a_0 q_1 q_2}{\epsilon r^2 \{e^{3a_0} + 1\}}$               | 0.000      | 0.000     | 1.000       | 1.000      | 218.641     |
| 5                                              | $\frac{a_0 q_1 q_2}{\epsilon r^2(2a_0 + 1)}$                      | 0.000      | 0.000     | 1.000       | 1.000      | 204.226     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                   |            |           |             |            |             |
| 1                                              | $\sin\left\{\frac{q_1 q_2}{a_0 \epsilon r^2}\right\}$             | 0.149      | 0.144     | 0.281       | 0.232      | 349.022     |
| 2                                              | $-\frac{q_1 q_2}{\epsilon r^2(a_0 + r)}$                          | 0.153      | 0.146     | 0.221       | 0.144      | 365.649     |

*Continued on next page*

Table 8: BMS results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS (*continued*).

| Rep.                                           | Learned expression                                             | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | $\frac{a_0 q_1 (a_0 r + q_2)}{\epsilon r^2}$                   | 0.152      | 0.156     | 0.275       | 0.274      | 363.815     |
| 4                                              | $a_0 q_1 q_2 (\epsilon r^2)^{-1}$                              | 0.152      | 0.148     | 0.228       | 0.275      | 363.012     |
| 5                                              | $-\frac{q_1 q_2}{a_0 \epsilon r^2}$                            | 0.149      | 0.139     | 0.232       | 0.305      | 365.000     |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                |            |           |             |            |             |
| 1                                              | $\sin\{a_0\{q_1 q_2 + \cos(a_0 q_2)\}(\epsilon r^2)^{-1}\}$    | 0.197      | 0.196     | 0.132       | 0.221      | 406.794     |
| 2                                              | $a_0 q_1 q_2 (\epsilon r^3)^{-1}$                              | 0.201      | 0.195     | 0.164       | 0.229      | 391.551     |
| 3                                              | $a_0 q_1 q_2 \sin\{a_0/(2r)\}(\epsilon r)^{-1}$                | 0.201      | 0.199     | 0.148       | 0.286      | 421.309     |
| 4                                              | $\sin\{q_1 q_2 \cos^3(a_0)(\epsilon r^2)^{-1}\}$               | 0.197      | 0.211     | 0.104       | 0.131      | 393.521     |
| 5                                              | $q_1 q_2 \{a_0^2 \epsilon r [\epsilon + r(2a_0 + q_1)]\}^{-1}$ | 0.204      | 0.195     | 0.166       | 0.112      | 382.448     |

Table 9: BSR results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                      |            |           |             |            |             |
| 1                                              | $0.141 - 0.066 \exp\{\cos(q_1)\} + 0.001 \exp(q_1) - 0.002 \sin\{(\epsilon + q_2)^3\} - 0.007 q_1 (1 + \epsilon)$                                                    | 0.081      | 0.081     | 0.153       | 0.191      | 189.495     |
| 2                                              | $2.105 - 2.008 \cos\{\exp(-\epsilon)\} + 0.076 \cos(r) - 0.014 \cos(\epsilon) - 0.004 q_1 r$                                                                         | 0.069      | 0.079     | 0.332       | 0.325      | 188.710     |
| 3                                              | $-0.013 - 5.302 \times 10^{-5} \exp(q_1) + 4.900 \times 10^{-4} \exp(q_2) + 3.541 \times 10^{-5} \frac{\cos(q_1)}{\sin(q_1^2)} + 0.019 q_1$                          | 0.086      | 0.073     | 0.095       | 0.093      | 188.960     |
| 4                                              | $0.052 - 0.019\{\epsilon + \exp[\cos(\epsilon)]\} + 6.265 \times 10^{-4} \exp(q_1) + 0.019 q_2 + 0.001 \sin(q_1^2)$                                                  | 0.086      | 0.071     | 0.171       | 0.154      | 188.388     |
| 5                                              | $0.013 + 0.005\{\cos(q_2^{-1}) + q_2 + r\} - 0.026 q_2^{-1} + 0.077 r^{-1} + 8.526 \times 10^{-6} (-r q_2)^3$                                                        | 0.075      | 0.069     | 0.260       | 0.288      | 189.140     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                      |            |           |             |            |             |
| 1                                              | $9.919 \times 10^{-11} + 2.028 \times 10^{-10} r + 2.141 \times 10^{-10} q_1 r^{-1} + 7.839 \times 10^{-12} \sin(\epsilon^2) - 6.104 \times 10^{-11} \sin^{-2}(q_1)$ | 0.186      | 0.168     | -0.117      | -0.047     | 190.750     |
| 2                                              | $-0.308 - 0.009 \cos(\epsilon^2) - 0.435 \exp(-r) - 0.008 \cos\{\sin(r)\} + 0.271 \exp(r^{-1})$                                                                      | 0.165      | 0.153     | 0.088       | 0.062      | 188.704     |
| 3                                              | $0.051 + 0.030 \exp\{\sin(\epsilon)\} + 0.096 \cos(r) + 9.383 \times 10^{-4} \sin(\epsilon^2) - 0.027 \cos(q_1)$                                                     | 0.171      | 0.168     | 0.090       | 0.151      | 188.453     |
| 4                                              | $0.023 + 4.914 \times 10^{-5} r \exp(q_2) + 0.058 \sin\{\sin(r)\} - 6.124 \times 10^{-5} \sin^{-1}(q_1) + 6.006 \times 10^{-4} \exp(q_1)$                            | 0.167      | 0.170     | 0.063       | 0.046      | 189.518     |

*Continued on next page*

Table 9: BSR results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi \epsilon r^2}$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5                                              | <i>numerical overflow</i>                                                                                                                                                                            | —          | —         | —           | —          | —           |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                      |            |           |             |            |             |
| 1                                              | $9.680 \times 10^{-18} + 3.878 \times 10^{-12} \exp(q_2 r) \\ - 4.213 \times 10^{-14} (-\epsilon q_1)^9 + 2.551 \times 10^{-11} \exp(\epsilon q_2 / r) \\ - 2.554 \times 10^{-17} \{\cos(q_1) - r\}$ | 0.220      | 0.230     | -0.076      | -0.080     | 188.188     |
| 2                                              | $6.596 \times 10^{-13} - 2.350 \times 10^{-13} \sin(q_2) - 3.259 \times 10^{-13} \cos(q_1) \\ + 6.368 \times 10^{-13} \exp\{\exp(r)\} + 2.641 \times 10^{-12} \epsilon r$                            | 0.223      | 0.233     | -0.032      | -0.095     | 187.441     |
| 3                                              | $0.068 - 0.132 q_1^{-1} - 0.010 \sin(-\epsilon^3) \\ + 0.050 \cos\{\cos(q_1)\} + 0.005 \sin\{\cos(q_1^2)\}$                                                                                          | 0.217      | 0.234     | 0.014       | 0.014      | 188.789     |
| 4                                              | $0.114 - 0.034 \cos\{\cos(q_1)\} - 0.027 \cos^2\{\sin(q_2^2)\} \\ + 5.686 \times 10^{-4} q_1^3 - 0.057 \sin^2\{\cos(\epsilon) + q_2\}$                                                               | 0.206      | 0.225     | 0.019       | 0.010      | 190.897     |
| 5                                              | $0.042 + 0.288 r^{-1} - 0.026 \epsilon^2 + 3.220 \times 10^{-5} \exp\{\cos(r q_1) + r\} \\ + 0.004 \epsilon^3$                                                                                       | 0.213      | 0.197     | 0.092       | 0.095      | 194.460     |



## C.2 Results for I\_12\_11 (FCE)

Table 10: DSR results for the Feynman equation I\_12\_11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                              | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                 |            |           |             |            |             |
| 1                                              | $v\{-B - E_f \sin(\theta)\}$                                    | 16.714     | 15.133    | 0.590       | 0.596      | 48.5        |
| 2                                              | $v\{q^2 \sin(\theta) + B\} + \theta^{-3}$                       | 14.464     | 11.720    | 0.681       | 0.788      | 46.3        |
| 3                                              | $q + \sin(\theta)v\{B[(\exp(\theta) + E_f)^{-1} + q] + B\} + B$ | 9.827      | 9.812     | 0.852       | 0.829      | 44.4        |
| 4                                              | $Bvq \sin^3(\theta)$                                            | 12.989     | 12.120    | 0.729       | 0.799      | 62.4        |
| 5                                              | $\{q + E_f \sin(\theta)\}^2$                                    | 16.952     | 20.242    | 0.566       | 0.443      | 42.0        |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                 |            |           |             |            |             |
| 1                                              | $B \sin(\theta)(q + v)$                                         | 15.523     | 14.612    | 0.642       | 0.667      | 42.6        |
| 2                                              | $Bqv[\{\exp(B)v + \theta\}^{-1} + \sin^3(\theta)] + \theta$     | 11.446     | 11.847    | 0.814       | 0.800      | 42.1        |
| 3                                              | $v[E_f B \sin(\theta) + \sin\{-\exp(-E_f)\}^3 + E_f]$           | 14.015     | 16.898    | 0.709       | 0.600      | 44.1        |
| 4                                              | $q^2 B \sin(\theta) + v$                                        | 15.082     | 13.882    | 0.644       | 0.764      | 41.7        |
| 5                                              | $B \sin(\theta) \exp\{\exp[\cos(v^{-1})]\}$                     | 15.955     | 15.260    | 0.620       | 0.603      | 42.5        |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                 |            |           |             |            |             |
| 1                                              | $vq \sin(\theta)\{B - \cos(-\theta)\}$                          | 11.291     | 11.289    | 0.804       | 0.806      | 42.6        |
| 2                                              | $v\{E_f + Bq \sin(\theta)\}$                                    | 5.422      | 5.420     | 0.955       | 0.951      | 42.0        |
| 3                                              | $(q + v) \sin(\theta)\{q - \sin(v)\}$                           | 14.773     | 15.797    | 0.681       | 0.668      | 42.7        |
| 4                                              | $\{v + qB \sin(\theta)\}q \cos(B^{-1})$                         | 13.950     | 15.722    | 0.723       | 0.604      | 42.6        |
| 5                                              | $Bvq \sin(\theta)$                                              | 10.228     | 10.272    | 0.840       | 0.843      | 42.6        |

Table 11: QLattice results for the Feynman equation I\_12\_11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                                                                                                                                                                                                        | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                                                                                                                                                           |            |           |             |            |             |
| 1                        | $ \begin{aligned} & -48.303(0.126q + 5.576 \times 10^{-4}) \\ & \times \left[ -0.163E_f + (0.165 - 0.393\theta)(1.908 - 0.326\theta) \right. \\ & \quad \times (4.395 - 1.400\theta)(-0.625B - 0.002) \\ & \quad \times (-4.409 \times 10^{-5}E_f - 1.210)(0.156v + 0.002) \\ & \quad \left. - 0.062 \right] \end{aligned} $              | 0.531      | 0.522     | 1.000       | 1.000      | 96.089      |
| 2                        | $ \begin{aligned} & -38.383(1.643 - 0.294\theta)(0.060q + 1.766 \times 10^{-5}) \\ & \times (-0.072\theta - 0.758) \left[ 0.602E_f + (2.118 - 0.672\theta)(0.107E_f + 3.200) \right. \\ & \quad \times (0.241\theta - 0.086)(0.459v - 0.170) \\ & \quad \left. \times (0.936B - 0.215E_f + 0.276) - 0.106 \right] + 0.867 \end{aligned} $ | 1.091      | 1.096     | 0.998       | 0.998      | 94.457      |

Continued on next page

Table 11: **QLattice** results for the Feynman equation I.12.11 (**FCE**):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | $19.478q + 42.662(0.536 - 0.027E_f)(-0.354q - 0.002)$<br>$\times [0.018\theta + (0.554 - 0.177\theta)(0.838B - 0.044)]$<br>$\times (0.451v - 0.029)\{-0.088E_f + (0.530\theta - 0.548)$<br>$\times (0.559\theta - 2.919) - 0.602\} + 2.340] + 0.143$             | 0.550      | 0.447     | 1.000       | 1.000      | 90.534      |
| 4                                              | $2.996q - 18.907(0.006 - 0.290B)(2.827 - 0.901\theta)$<br>$\times (3.622 - 0.621\theta)(0.242q + 0.009)$<br>$\times (0.629\theta - 0.263)(0.282v - 0.001)$<br>$- 29.826(1 - 0.060E_f)^2 + 19.714$                                                                | 1.433      | 1.548     | 0.997       | 0.997      | 91.767      |
| 5                                              | $20.585(4.180 - 1.332\theta)(2.972 \times 10^{-4}E_f - 0.351)$<br>$\times (0.060q + 0.001)[0.317B + (0.450 - 2.152B)(1.003 - 0.171\theta)$<br>$\times (1.197\theta - 0.474)(0.533v + 0.031) - 0.828]$<br>$+ 1.113(-0.568E_f - 0.559q + 1)^2 + 1.825$             | 0.899      | 0.996     | 0.999       | 0.999      | 92.527      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $-37.575(0.002 - 0.258B)(1.145q + 0.005)$<br>$\times (0.178\theta - 0.558)(0.243\theta - 1.422)$<br>$\times (-0.326v - 0.001)(0.003B - 0.843\theta + 0.339)$<br>$- 37.575(0.053 - 1.318q)(0.156 - 4.545 \times 10^{-4}v)$<br>$\times (0.132E_f - 0.009) + 0.246$ | 0.429      | 0.441     | 1.000       | 1.000      | 91.071      |
| 2                                              | $20.395(0.025q - 0.003)[2.120E_f + (0.017 - 1.983v)(0.757 - 0.130\theta)$<br>$\times (1.015 - 0.001E_f)(1.442 - 0.015q)$<br>$\times (2.870 - 0.911\theta)(-1.397B - 0.004)$<br>$\times (0.586\theta - 0.244) - 0.782] + 0.684$                                   | 0.635      | 0.657     | 0.999       | 0.999      | 93.601      |
| 3                                              | $-0.021E_f + 9.681q - 17.240(-0.219q - 0.002)$<br>$\times [0.266E_f + (0.138 - 0.326\theta)(3.221 - 0.552\theta)$<br>$\times (3.269 - 1.042\theta)(0.385B + 5.972 \times 10^{-4})$<br>$\times (-0.485v - 0.001) - 2.566] - 0.096$                                | 0.462      | 0.502     | 1.000       | 1.000      | 91.798      |
| 4                                              | $34.556(0.013 - 0.048\theta)(-0.007E_f - 3.883)$<br>$\times (0.318q - 0.007)[0.653\theta + (2.967 - 0.508\theta)$<br>$\times \{0.144E_f + (-0.328B - 0.063)(0.201\theta - 0.625)$<br>$\times (1.753v + 0.146) + 1.108\} - 3.621] + 0.273$                        | 1.070      | 1.086     | 0.998       | 0.999      | 91.377      |
| 5                                              | $38.879[-0.048E_f + (0.392 - 0.950\theta)(1.024 - 0.326\theta)$<br>$\times (0.382B + 0.001)(0.102v - 8.045 \times 10^{-4})$<br>$\times (-0.525\theta + 7.335 \times 10^{-4}v + 3.070) - 6.423 \times 10^{-4}]$<br>$\times (-0.532q + 0.002v - 0.009) - 0.018$    | 0.451      | 0.374     | 1.000       | 1.000      | 92.678      |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $77.446(0.084q - 8.236 \times 10^{-4})$<br>$\times [0.156E_f - 0.001q + (0.183 - 0.438\theta)(2.442 - 0.778\theta)$<br>$\times (0.389B - 0.002)(0.776v + 0.005)$<br>$\times (-0.001q + 0.200\theta - 1.169) - 0.005] + 0.143$                                    | 1.233      | 1.253     | 0.998       | 0.998      | 95.471      |
| 2                                              | $22.135(0.105 - 1.077q)[-0.043E_f + (0.078 - 0.193\theta)(3.925 - 0.669\theta)$<br>$\times (0.289\theta - 0.909)(0.141v + 0.002)$<br>$\times (-1.071B + 0.043q - 0.150) + 0.004] + 0.471$                                                                        | 1.132      | 1.113     | 0.998       | 0.998      | 92.821      |

Continued on next page

Table 11: **QLattice** results for the Feynman equation I.12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                                                                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | $2.899E_f + 25.179(0.144q + 0.002)$ $\times [1.058(0.037 - 0.091\theta)(2.247 - 0.385\theta)$ $\times (-1.059B - 0.025)\{-0.649\theta + (3.902 - 1.244\theta)$ $\times (0.739v - 0.523) + 2.026\} + 0.828] - 8.668$                              | 1.686      | 1.682     | 0.996       | 0.996      | 96.179      |
| 4    | $-29.274(3.813 \times 10^{-4} - 0.085q)$ $\times [0.406E_f + (9.843 \times 10^{-4} - 0.419B)(0.281 - 0.683\theta)$ $\times (0.951 - 0.303\theta)(1.904 - 0.325\theta)$ $\times (1.872v - 0.002) - 0.009] - 0.152$                                | 1.129      | 1.165     | 0.998       | 0.998      | 96.701      |
| 5    | $-21.119(2.281 \times 10^{-4} - 0.033q)$ $\times [1.391E_f - 0.525\theta + (0.099 - 0.589v)(0.130 - 0.819B)$ $\times (0.527 - 1.152\theta)(2.975 - 0.514\theta)$ $\times (5.584 \times 10^{-4} q + 0.835)(0.845\theta - 2.657) + 1.597] - 0.143$ | 1.238      | 1.140     | 0.998       | 0.998      | 86.482      |

Table 12: **SISSO++** results for the Feynman equation I.12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                              |            |           |             |            |             |
| 1                                              | $-9.001 \times 10^{-16} - 1.354 \times 10^{-16}(v + B + \sin B)(\sin v + v)$ $+ E_f q + Bv q \sin(\theta)$   | 0.000      | 0.000     | 1.000       | 1.000      | 42.554      |
| 2                                              | $8.925 \times 10^{-16} + \{\sin(v) + E_f q + E_f q + v\}$ $- \{E_f q + v + \sin(v)\} + Bv q \sin(\theta)$    | 0.000      | 0.000     | 1.000       | 1.000      | 44.682      |
| 3                                              | $-8.076 \times 10^{-17} - 2.549 \times 10^{-16}\{\sin(E_f) + \sin(q) + E_f\}$ $+ E_f q + Bv q \sin(\theta)$  | 0.000      | 0.000     | 1.000       | 1.000      | 45.950      |
| 4                                              | $4.641 \times 10^{-16} - 1.679 \times 10^{-16}(\theta + v + vB)q \sin(\theta)$ $+ E_f q + Bv q \sin(\theta)$ | 0.000      | 0.000     | 1.000       | 1.000      | 41.782      |
| 5                                              | $4.508 \times 10^{-16} + 2.420 \times 10^{-16}\{\sin(q) + E_f + \sin(E_f)\}$ $+ E_f q + Bv q \sin(\theta)$   | 0.000      | 0.000     | 1.000       | 1.000      | 43.372      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                              |            |           |             |            |             |
| 1                                              | $0.019 + 0.998 E_f q + 0.994 Bv q \sin(\theta)$ $+ 0.004 q \sin(\theta)(vB + v)$                             | 0.245      | 0.248     | 1.000       | 1.000      | 42.689      |
| 2                                              | $0.011 + 1.000 E_f q + 0.996 Bv q \sin(\theta)$ $+ 0.004 v \sin(\theta)\{\sin(B) + Bq\}$                     | 0.248      | 0.246     | 1.000       | 1.000      | 42.115      |
| 3                                              | $0.094 + 1.026 E_f q - 0.026\{\sin(B) + E_f q + B\}$ $+ 1.000 Bv q \sin(\theta)$                             | 0.245      | 0.249     | 1.000       | 1.000      | 42.134      |
| 4                                              | $0.031 - 0.006\{\sin(B) + \sin(E_f) + B + q + \sin(E_f)\}$ $+ 1.001 E_f q + 1.000 Bv q \sin(\theta)$         | 0.250      | 0.244     | 1.000       | 1.000      | 43.311      |
| 5                                              | $0.012 + 0.003(v + B + \sin q) \sin(E_f) \sin(q)$ $+ 0.999 E_f q + 1.000 Bv q \sin(\theta)$                  | 0.252      | 0.251     | 1.000       | 1.000      | 43.518      |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                                              |            |           |             |            |             |

Continued on next page

Table 12: **SISSO** ++ results for the Feynman equation I.12.11 (**FCE**):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                      | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|---------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 1    | $0.076 + 0.991 E_f q + 0.987 Bv q \sin(\theta) + 0.008 v \sin(\theta)(Bq + E_f + q)$                    | 0.978      | 0.994     | 0.998       | 0.998      | 43.563      |
| 2    | $0.043 + 0.999 E_f q + 0.982 Bv q \sin(\theta) + 0.017 v \sin(\theta)\{\sin(B) + Bq\}$                  | 0.992      | 0.985     | 0.999       | 0.999      | 42.256      |
| 3    | $0.375 + 1.102\{\sin(B) + E_f q + E_f q + B\} - 1.205\{\sin(B) + E_f q + B\} + 1.001 Bv q \sin(\theta)$ | 0.981      | 0.995     | 0.999       | 0.998      | 46.512      |
| 4    | $0.124 - 0.024\{\sin(B) + \sin(E_f) + B + q + \sin(E_f)\} + 1.002 E_f q + 0.999 Bv q \sin(\theta)$      | 1.001      | 0.975     | 0.999       | 0.998      | 42.650      |
| 5    | $0.003 + 0.912 E_f q + 0.044\{\sin(B) + E_f q + \sin(E_f) + E_f q\} + 1.001 Bv q \sin(\theta)$          | 1.010      | 1.005     | 0.998       | 0.998      | 43.966      |

Table 13: **gplearn** results for the Feynman equation I.12.11 (**FCE**):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                      |            |           |             |            |             |
| 1                                              | $q\{E_f + Bv \sin(\theta)\}$                                                                         | 0.000      | 0.000     | 1.000       | 1.000      | 15.034      |
| 2                                              | $qE_f + q^2 \sin(\theta) + \sin^3(\theta) + v^2 \sin\{\sin[\sin(\theta)]\} + B^2 \sin(\theta)$       | 8.008      | 7.412     | 0.902       | 0.915      | 17.532      |
| 3                                              | $q\{E_f + Bv \sin(\theta)\}$                                                                         | 0.000      | 0.000     | 1.000       | 1.000      | 15.353      |
| 4                                              | $q\{E_f + Bv \sin(\theta)\}$                                                                         | 0.000      | 0.000     | 1.000       | 1.000      | 17.214      |
| 5                                              | $qv\{1 + B \sin(\theta)\}$                                                                           | 5.166      | 5.210     | 0.960       | 0.963      | 16.575      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                      |            |           |             |            |             |
| 1                                              | $qE_f \exp\{\sin[\sin\{\sin(\theta)\}]\} + 2Bv \sin(\theta) + qB \sin\{\sin[\sin\{\sin(\theta)\}]\}$ | 8.225      | 7.592     | 0.899       | 0.910      | 17.947      |
| 2                                              | $q + E_f + q \sin(\theta)\{q + 2v - \theta\} + B^2 \sin(\theta) + \exp(0.374^3)$                     | 7.488      | 7.239     | 0.921       | 0.925      | 16.998      |
| 3                                              | $q\{E_f + Bv \sin(\theta)\}$                                                                         | 0.254      | 0.260     | 1.000       | 1.000      | 15.842      |
| 4                                              | $(E_f + q)\{0.571^{-1} + \sin(\theta)\} + 0.877 qBv \sin\{\sin(\theta)\}$                            | 4.251      | 4.668     | 0.972       | 0.973      | 17.093      |
| 5                                              | $qE_f + qBv \sin\{\sin(\theta)\} + v \sin(\theta)$                                                   | 1.896      | 1.733     | 0.995       | 0.995      | 16.505      |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                                      |            |           |             |            |             |
| 1                                              | $q\{E_f + Bv \sin(\theta)\}$                                                                         | 0.988      | 0.978     | 0.999       | 0.999      | 17.098      |
| 2                                              | $2q + E_f + B^2 \sin(\theta) + \sin\{q - \exp[B^2 \sin(\theta)]\} - 0.384 + 2qv \sin(\theta)$        | 6.973      | 7.579     | 0.925       | 0.905      | 19.384      |
| 3                                              | $qE_f + \sin(\theta)\left[q^2 + B^2 + v^2 + \sin(\theta) - 0.239 - 2 \exp\{\sin(\theta)\}\right]$    | 7.699      | 8.219     | 0.913       | 0.910      | 15.741      |

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Table 13: **gplearn** results for the Feynman equation I\_12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                          | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|-------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4    | $qE_f + v[qB \sin(\theta) + \exp\{-(0.569^{-1})^3\} + \exp\{-\exp(0.569^{-1})\} + \exp\{-(0.569^{-1})^3\}]$ | 0.985      | 1.052     | 0.999       | 0.998      | 16.375      |
| 5    | $q\{E_f + Bv \sin(\theta)\}$                                                                                | 1.021      | 0.973     | 0.998       | 0.999      | 15.157      |

Table 14: **operon** results for the Feynman equation I\_12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                   |            |           |             |            |             |
| 1                                              | $-0.324 + 1.497[2.035q + \{-0.548qv \sin(1.002\theta) - 0.023E_f\}(-1.218B) - e^{0.711} 0.459q \sin(1.021E_f)]$                   | 0.816      | 0.864     | 0.999       | 0.999      | 27.316      |
| 2                                              | $4.572 + 7216.561[0.462q \sin^2(0.022E_f) + 0.022B \sin(0.996\theta)] \times 0.022q\{0.273v + 0.022E_f \sin[\sin(0.871\theta)]\}$ | 2.459      | 2.396     | 0.991       | 0.991      | 29.131      |
| 3                                              | $9.072 + 3.994(-0.481E_f)[\cos(0.521q) - \{0.520qv \sin(\theta) - 0.177qB^{-1} \sin(1.029E_f)\}]$                                 | 1.535      | 1.552     | 0.996       | 0.996      | 26.635      |
| 4                                              | $-8.787 - 0.866[-3.317E_f - 3.563q + 2.948q \sin(1.001\theta)\{-1.297v + \exp(-3.417E_f + 0.678q \sin(0.976\theta))\} 0.302B]$    | 1.274      | 1.362     | 0.997       | 0.997      | 27.056      |
| 5                                              | $0.028 + 0.496[2.020qE_f + 1.822B \sin(\theta)\{1.115qv\}]$                                                                       | 0.049      | 0.046     | 1.000       | 1.000      | 22.940      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                   |            |           |             |            |             |
| 1                                              | $6.992 - 13.761[(0.256q/1.414E_f)^2 + \cos(-0.129E_f) - 0.122E_f + 0.093qBv \sin(-1.001\theta) - 0.256q]$                         | 1.025      | 1.023     | 0.998       | 0.998      | 25.396      |
| 2                                              | $2.510 - 0.303[-1.782q + 2.586B - 0.535qE_f^2 - 2.303B\{0.359q + 1.430qv \sin(1.002\theta)\}]$                                    | 0.753      | 0.816     | 0.999       | 0.999      | 25.028      |
| 3                                              | $0.178 + 2.419[\{0.179qv \sin(1.007\theta) + c_1 qE_f B^{-1}\} 2.303B]$                                                           | 0.448      | 0.443     | 1.000       | 1.000      | 26.952      |
| 4                                              | $-8.880 + 3.419[1.214 qBv \sin(1.002\theta) + 0.874E_f + 0.878q]$                                                                 | 1.344      | 1.311     | 0.997       | 0.998      | 22.984      |
| 5                                              | $-0.002 + 0.279[1.790qE_f + 1.790qE_f + 0.012E_f + 3.598qBv \sin(\theta)]$                                                        | 0.249      | 0.231     | 1.000       | 1.000      | 23.109      |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                                                                   |            |           |             |            |             |
| 1                                              | $0.056 - 180.230(0.806q)[2.276(0.073B)(0.073v) \sin(-0.995\theta) - 0.007E_f - (0.073B)(0.073v) \sin(-0.988\theta)]$              | 0.986      | 0.985     | 0.999       | 0.999      | 24.154      |
| 2                                              | $-0.155 - 0.011[\{(0.327v/-1.160)[(-0.972 - 1.873B) \sin(-0.993\theta) + \sin(-0.964\theta)] - 0.525E_f\}\{0.693q/(-0.063)^2\}]$  | 1.013      | 0.984     | 0.998       | 0.998      | 23.321      |

Continued on next page

Table 14: **operon** results for the Feynman equation I.12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                     | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|----------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | $3.797 + 0.085 \left[ \{3.382B \sin(1.036\theta) + 0.442E_f - \sin[\sin(1.409\theta)]\} 1.533v - 1.230v + 0.807qE_f \right]$           | 1.641      | 1.707     | 0.996       | 0.996      | 26.583      |
| 4    | $-1.497 + 1.487 \left[ 0.369 \{0.660E_f + \cos[\sin(1.027\theta)]\} + 0.837q \{0.800Bv \sin(1.003\theta) + 0.720E_f + 0.305\} \right]$ | 1.003      | 1.075     | 0.999       | 0.998      | 25.101      |
| 5    | $-5.545 + 8.714 \left[ -0.373q \exp(-0.434E_f) - 0.115qBv \sin(-1.003\theta) - 0.216E_f + 0.459q \right]$                              | 1.319      | 1.266     | 0.997       | 0.998      | 24.734      |

Table 15: **PySR** results for the Feynman equation I.12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                              |            |           |             |            |             |
| 1                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.000      | 0.000     | 1.000       | 1.000      | 136.217     |
| 2                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.000      | 0.000     | 1.000       | 1.000      | 135.984     |
| 3                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.000      | 0.000     | 1.000       | 1.000      | 122.059     |
| 4                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.000      | 0.000     | 1.000       | 1.000      | 115.471     |
| 5                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.000      | 0.000     | 1.000       | 1.000      | 126.500     |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                              |            |           |             |            |             |
| 1                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.248      | 0.237     | 1.000       | 1.000      | 119.976     |
| 2                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.255      | 0.244     | 1.000       | 1.000      | 129.114     |
| 3                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.254      | 0.260     | 1.000       | 1.000      | 128.247     |
| 4                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.253      | 0.247     | 1.000       | 1.000      | 124.288     |
| 5                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.249      | 0.231     | 1.000       | 1.000      | 105.516     |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                              |            |           |             |            |             |
| 1                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.988      | 0.978     | 0.999       | 0.999      | 124.517     |
| 2                                              | $q\{E_f + Bv \sin(\theta)\}$ | 1.006      | 0.959     | 0.998       | 0.998      | 115.130     |
| 3                                              | $q\{E_f + Bv \sin(\theta)\}$ | 1.007      | 0.996     | 0.999       | 0.999      | 112.441     |
| 4                                              | $q\{E_f + Bv \sin(\theta)\}$ | 0.987      | 1.056     | 0.999       | 0.998      | 117.745     |
| 5                                              | $q\{E_f + Bv \sin(\theta)\}$ | 1.021      | 0.973     | 0.998       | 0.999      | 146.730     |

Table 16: BMS results for the Feynman equation I\_12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS.

| Rep.                                           | Learned expression                                                             | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                |            |           |             |            |             |
| 1                                              | $qE_f + Bqv \sin(\theta)$                                                      | 0.000      | 0.000     | 1.000       | 1.000      | 267.921     |
| 2                                              | $q\{E_f + Bv \sin(\theta)\}$                                                   | 0.000      | 0.000     | 1.000       | 1.000      | 257.230     |
| 3                                              | $qE_f + a_0 Bqv \sin(\theta)$                                                  | 0.000      | 0.000     | 1.000       | 1.000      | 318.374     |
| 4                                              | $q\{E_f + Bv \sin(\theta)\} + \sin^3(a_0)$                                     | 0.000      | 0.000     | 1.000       | 1.000      | 282.034     |
| 5                                              | $qE_f - a_0 Bqv \sin(\theta)$                                                  | 0.000      | 0.000     | 1.000       | 1.000      | 235.786     |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                |            |           |             |            |             |
| 1                                              | $qE_f + Bqv \sin(\theta)$                                                      | 0.248      | 0.237     | 1.000       | 1.000      | 294.344     |
| 2                                              | $qE_f + B\{qv + 2a_0\} \sin(\theta)$                                           | 0.255      | 0.244     | 1.000       | 1.000      | 314.851     |
| 3                                              | $qE_f - Bqv \sin\{-(\theta + 2a_0)\}$                                          | 0.254      | 0.260     | 1.000       | 1.000      | 281.084     |
| 4                                              | $qE_f + Bqv \sin(\theta)$                                                      | 0.253      | 0.247     | 1.000       | 1.000      | 303.930     |
| 5                                              | $qE_f + Bq(v + a_0\theta) \sin(\theta)$                                        | 0.249      | 0.232     | 1.000       | 1.000      | 272.353     |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                |            |           |             |            |             |
| 1                                              | $qE_f + Bqv \sin(\theta)$                                                      | 0.988      | 0.978     | 0.999       | 0.999      | 285.738     |
| 2                                              | $qE_f + Bv\{q + a_0 e^\theta\} \sin(\theta)$                                   | 1.003      | 0.962     | 0.998       | 0.998      | 302.691     |
| 3                                              | $q\left[E_f - Bv \sin(\theta)\left\{-\exp\{a_0(B + \theta)\}\right\}^9\right]$ | 1.007      | 0.996     | 0.999       | 0.999      | 344.995     |
| 4                                              | <i>numerical overflow</i>                                                      | —          | —         | —           | —          | —           |
| 5                                              | $qE_f + Bqv \sin(\theta)$                                                      | 1.021      | 0.973     | 0.998       | 0.999      | 246.953     |

Table 17: BSR results for the Feynman equation I\_12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                   |            |           |             |            |             |
| 1                                              | $-25.320 - 1.303 \sin(v) + 0.022(B^3 - v) + 87.165 \theta^{-1}$                   | 19.583     | 17.165    | 0.437       | 0.481      | 164.419     |
| 2                                              | $33.811 - 0.067(2\theta)^3 - 1.468B + 0.055e^B$                                   | 16.695     | 17.040    | 0.575       | 0.551      | 167.938     |
| 3                                              | $20.823 + 2.685E_f - 0.464\{e^\theta + \cos(\theta) + v\} + 0.536 \theta \sin(v)$ | 17.821     | 16.277    | 0.512       | 0.531      | 165.756     |
| 4                                              | $17.839 + 0.142\{B + e^{\cos(B)}\} - 0.447e^\theta + 2.890E_f$                    | 18.087     | 19.149    | 0.474       | 0.499      | 165.655     |
| 5                                              | $-29.316 + 0.775\{v + \cos(e^{q\theta})\} + 1.235B + 87.299 \theta^{-1}$          | 18.864     | 20.446    | 0.462       | 0.431      | 164.489     |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                   |            |           |             |            |             |

Continued on next page

Table 17: BSR results for the Feynman equation I\_12\_11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 1                                              | $19.906 - 0.970(v + q\theta) + 19.892 \sin\{\sin(e^v) + \theta\} - 1.615 \cos(E_f)$                                                                | 18.838     | 19.449    | 0.472       | 0.410      | 167.556     |
| 2                                              | $3.495 \times 10^{-10} + 2.548 \times 10^{-10} e^{-\cos^2(B)} + 4.434 \times 10^{-9} E_f^2 - 8.757 \times 10^{-9} e^{v\theta}$                     | 28.094     | 28.733    | -0.119      | -0.176     | 165.173     |
| 3                                              | $6.308 + 0.105 E_f^3 - 0.069 B \cos(B) - 1.104 \sin(B)$                                                                                            | 25.714     | 26.146    | 0.022       | 0.041      | 165.598     |
| 4                                              | $25.442 - 0.395 B^2 \theta - 0.144 \sin(v) - 5.693 v^{-1}$                                                                                         | 23.227     | 25.609    | 0.157       | 0.195      | 165.248     |
| 5                                              | $6.565 + 0.088 E_f^3 + 0.039(B^3 + B) - 5.808 \sin(q)$                                                                                             | 25.314     | 24.442    | 0.044       | -0.018     | 164.711     |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                                                                                    |            |           |             |            |             |
| 1                                              | $9.736 + 0.775 \cos^2(E_f) - 9.109 q^{-2} + 0.061 e^q$                                                                                             | 25.246     | 25.885    | 0.022       | -0.020     | 167.802     |
| 2                                              | $11.635 + 0.048 v^3 + 0.525 \sin\{E_f e^{1/q}\} \sin(B) - 3.670 \cos\{\cos(q)\}$                                                                   | 25.429     | 24.696    | 0.005       | -0.012     | 164.314     |
| 3                                              | $-5.198 + 2.083(B + \theta) + 29.568 \sin(\theta) + 3.449 \sin(B)$                                                                                 | 15.892     | 17.252    | 0.631       | 0.604      | 164.415     |
| 4                                              | $1.150 \times 10^{-7} - 2.940 \times 10^{-4} \theta^6 + 5.812 \times 10^{-8} \sin^2\{\cos(B) + E_f\} + 1.265 \times 10^{-7} \{v + \exp(B + v)\}^2$ | 28.476     | 28.072    | -0.156      | -0.264     | 165.759     |
| 5                                              | $32.862 - 2.512 q \theta - 0.009 \{\sin(E_f B^2)\}^{-1} + 0.054 B^2$                                                                               | 22.195     | 22.518    | 0.247       | 0.246      | 164.758     |



### C.3 Results for I\_24.6 (EHFO)

Table 18: DSR results for the Feynman equation I\_24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                   |            |           |             |            |             |
| 1                                              | $\omega_0^2 + m\omega x^2$                        | 5.183      | 4.936     | 0.863       | 0.882      | 31.6        |
| 2                                              | $x\{x + m\omega\omega_0\}$                        | 4.917      | 4.782     | 0.886       | 0.889      | 31.2        |
| 3                                              | $x\{x + m\omega\omega_0\}$                        | 4.697      | 4.721     | 0.889       | 0.899      | 30.6        |
| 4                                              | $mx^2\{\omega + \cos(\omega_0 + x)\}$             | 4.701      | 4.793     | 0.885       | 0.890      | 33.6        |
| 5                                              | $x\{m\omega\omega_0 + x\}$                        | 4.733      | 4.727     | 0.889       | 0.899      | 32.0        |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                   |            |           |             |            |             |
| 1                                              | $x^2 + m\omega\omega_0 x^2$                       | 4.752      | 4.784     | 0.893       | 0.885      | 33.2        |
| 2                                              | $\omega + \{-\omega_0 + \omega + m\omega_0\}x^2$  | 4.245      | 4.385     | 0.914       | 0.904      | 32.1        |
| 3                                              | $x\{m\omega\omega_0 + x\}$                        | 4.867      | 4.877     | 0.889       | 0.884      | 34.9        |
| 4                                              | $x^2 + m\omega\omega_0 x^2$                       | 4.709      | 4.732     | 0.897       | 0.886      | 31.2        |
| 5                                              | $m\{-\omega\omega_0^2 \cos(x) + \omega + x + m\}$ | 4.773      | 4.405     | 0.894       | 0.913      | 32.2        |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                   |            |           |             |            |             |
| 1                                              | $x\{m\omega\omega_0 + x\}$                        | 4.845      | 5.300     | 0.888       | 0.869      | 34.7        |
| 2                                              | $m\omega\omega_0 x + x^2$                         | 4.862      | 4.900     | 0.890       | 0.864      | 33.2        |
| 3                                              | $m\omega\omega_0 x + x^2$                         | 4.936      | 4.922     | 0.883       | 0.886      | 30.3        |
| 4                                              | $x\{x + m\omega\omega_0\}$                        | 4.892      | 4.577     | 0.888       | 0.894      | 29.8        |
| 5                                              | $m\omega\omega_0 x + x^2$                         | 5.088      | 4.995     | 0.888       | 0.889      | 30.2        |

Table 19: QLattice results for the Feynman equation I\_24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                                                                                                                    | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                                                                       |            |           |             |            |             |
| 1                        | $-8.917(-0.036m - 2.068 \times 10^{-4})(0.177x + 0.454)(1.553x - 0.855)$ $\times [1.874(0.050 - \omega)^2 + (0.193 - 0.006m)(2.987 - 0.980\omega)$ $+ (0.679\omega_0 - 0.346)\{1.481x + (0.062m + 2.235)(1.097\omega_0 + 0.301)$ $- 2.960\}] + 0.417$ | 0.292      | 0.285     | 1.000       | 1.000      | 95.182      |
| 2                        | $-13.136(-0.408m - 0.004)[-0.009\omega + (2.563 - 2.036x)$ $\times (-0.565x - 0.729) + 1.886] \exp\{(2.029 - 0.474\omega_0)(0.462\omega - 1.980)\}$ $- 0.061$                                                                                         | 0.284      | 0.280     | 1.000       | 1.000      | 99.424      |

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Table 19: **QLattice** results for the Feynman equation I\_24.6 (**EHFO**):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                                                    | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | $\begin{aligned} & -0.051\omega_0 - 17.463(-0.802m - 0.022)(0.794x + 0.099) \\ & \times \exp\left[-(0.784 - 0.121x)(-0.004\omega - 0.356)(2.746x - 5.284) \right. \\ & \quad \left. - (1.255 - 0.295\omega_0)(3.185 - 0.745\omega)\right] - 0.139 \end{aligned}$                                                      | 0.286      | 0.300     | 1.000       | 1.000      | 93.490      |
| 4                                              | $\begin{aligned} & 0.039m - 11.667(0.969 - 2.099x) \\ & \times \left[0.250(0.469m + 0.018)(-0.667x - 0.544) \right. \\ & \times \{-0.475\omega_0 + 0.565(2.908 - 0.676\omega_0)(-0.080\omega - 0.296) \\ & \quad \left. \times (0.710\omega - 2.618)e^{0.449\omega_0} - 1.202\} - 0.018\right] + 0.288 \end{aligned}$ | 0.253      | 0.271     | 1.000       | 1.000      | 95.968      |
| 5                                              | $\begin{aligned} & -10.519(1.361 - 2.571x) \\ & \times \left[-0.008m + (0.130m + 7.335 \times 10^{-4})\{0.698(-0.363x - 1)^2 \right. \\ & \quad \left. + 0.809\}\{1.834 \exp[(1.871 - 0.419\omega_0)(0.441\omega - 1.976)] - 0.033\} \right. \\ & \quad \left. - 0.005\right] + 0.300 \end{aligned}$                  | 0.275      | 0.270     | 1.000       | 1.000      | 94.381      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                                                                                                       |            |           |             |            |             |
| 1                                              | $\begin{aligned} & 2.459(-6.498 \times 10^{-5}m - 1)^2(0.122m + 1.847 \times 10^{-4}) \\ & \times (-0.008\omega + 1.825x - 0.955) \\ & \times \left[1.256\omega_0 + 0.720(\omega - 0.068)^2 - 2.947 + (0.547 - 0.123\omega_0)^{-1}\right] \\ & \quad \times e^{0.295x} - 0.017 \end{aligned}$                         | 0.251      | 0.265     | 1.000       | 1.000      | 94.493      |
| 2                                              | $\begin{aligned} & -26.030(0.638m + 0.005)(1.145x - 0.241) \\ & \times \left[-0.324x - 0.786 + (1.472 - 0.009\omega_0)^{-1}\right] \\ & \times \exp\{(1.745 - 0.405\omega)(0.538\omega_0 - 2.303)\} - 0.107 \end{aligned}$                                                                                            | 0.318      | 0.310     | 1.000       | 1.000      | 92.749      |
| 3                                              | $\begin{aligned} & 0.497(0.002 - 0.197m)(1.393x - 0.658) \\ & \times \left[3.234(0.045 - \omega)^2 + 2.792(\omega_0 - 0.060)^2(-0.025x - 1)^2\right] \\ & \quad \times (0.032\omega - 0.520x - 0.617) + 0.166 \end{aligned}$                                                                                          | 0.263      | 0.262     | 1.000       | 1.000      | 97.141      |
| 4                                              | $\begin{aligned} & 1.316(0.001 - 0.674m)(0.466 - 1.261x)(0.426x + 0.440) \\ & \times \exp\left[0.532\omega_0 + 0.052x + (1.300 - 0.695\omega)(0.309\omega_0 - 1.341)\right] \\ & \quad - 0.081 \end{aligned}$                                                                                                         | 0.362      | 0.350     | 0.999       | 0.999      | 94.907      |
| 5                                              | $\begin{aligned} & -18.166(0.271 - 1.498x)(-0.570m - 0.003)(-0.460x - 0.112) \\ & \times \exp\left[-0.033\omega + (1.093 - 0.246\omega_0)(0.888\omega - 3.807)\right] - 0.077 \end{aligned}$                                                                                                                          | 0.322      | 0.295     | 1.000       | 1.000      | 94.036      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                                                                                                                                                                                       |            |           |             |            |             |
| 1                                              | $\begin{aligned} & -33.528(0.760 - 1.956x)(0.058x + 0.037)(1.545m + 0.017\omega_0 - 0.059) \\ & \times \exp\left[(1.933 - 0.456\omega)(0.003m + 1.022)(0.473\omega_0 - 2.023)\right] + 0.078 \end{aligned}$                                                                                                           | 0.393      | 0.445     | 0.999       | 0.999      | 98.325      |
| 2                                              | $\begin{aligned} & -5.035(0.748 - 1.850x)(0.158m - 0.002)(0.188x + 0.120) \\ & \quad \times \exp\left[0.523\omega + 0.439\omega_0 \right. \\ & \quad \left. + (0.145\omega - 0.335)(0.343\omega_0 - 2.334)(1.201\omega_0 - 2.243) \right. \\ & \quad \left. \times (0.004x + 0.836)\right] + 0.028 \end{aligned}$     | 0.357      | 0.333     | 0.999       | 0.999      | 93.343      |
| 3                                              | $\begin{aligned} & -8.165(0.114 - 0.484x) \\ & \times \left[(0.694 - 0.530m)(0.339x + 0.122) \right. \\ & \quad \left. + (-0.197x - 0.061)\{-0.906m + (0.012 - 3.019m) \right. \\ & \quad \left. \times [0.103(\omega + 0.025)^2 + 0.103(-\omega_0 - 0.027)^2] + 1.223\}\right] - 0.015 \end{aligned}$                | 0.285      | 0.295     | 1.000       | 1.000      | 94.746      |
| 4                                              | $\begin{aligned} & 20.732(0.382 - 1.114x)(0.386m - 0.001)(-0.394x - 0.199) \\ & \times \left[(-0.191\omega - 0.095)\{-0.375(-0.002m - 0.554)(-0.027\omega - 2.015) \right. \\ & \quad \left. \times (0.809\omega - 1.322) - 0.441\} + 0.069(-\omega_0 - 0.015)^2\right] + 0.108 \end{aligned}$                        | 0.277      | 0.294     | 1.000       | 1.000      | 95.126      |

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Table 19: **QLattice** results for the Feynman equation I.24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                                                                                                                           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5    | $19.375 \left[ 0.575m + (5.514 \times 10^{-4} - 0.106m)(4.037 - 1.571x) \right. \\ \left. \times (0.185\omega + 0.947) + 5.200 \times 10^{-5} \right] (0.359\omega + 1.133x - 0.748) \\ \times \exp\{(2.362 - 0.544\omega)(0.397\omega_0 - 1.491)\} - 0.003$ | 0.412      | 0.393     | 0.999       | 0.999      | 93.102      |

Table 20: **SISSO++** results for the Feynman equation I.24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                                     | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                        |            |           |             |            |             |
| 1                                              | $-5.394 \times 10^{-16} - 7.822 \times 10^{-18} (x\omega)(xm)(\omega_0 + \omega)(\omega_0 m) \\ - 5.118 \times 10^{-18} (x\omega)\omega^2(x + \omega + x^2) + 0.250 mx^2(\omega_0^2 + \omega^2)$       | 0.000      | 0.000     | 1.000       | 1.000      | 20.426      |
| 2                                              | $3.502 \times 10^{-16} + \omega^4 x^2 + 0.250 mx^2(\omega_0^2 + \omega^2) \\ - 6.855 \times 10^{-18} (x + \omega_0 + \omega_0^2)(x\omega)(xm)$                                                         | 0.000      | 0.000     | 1.000       | 1.000      | 19.014      |
| 3                                              | $3.471 \times 10^{-16} + 6.981 \times 10^{-18} x^2(\omega_0 m)(\omega^2 + x) \\ + 0.250 mx^2(\omega_0^2 + \omega^2) + 1.516 \times 10^{-17} (x + \omega_0)^2(xm)\omega$                                | 0.000      | 0.000     | 1.000       | 1.000      | 19.793      |
| 4                                              | $2.494 \times 10^{-16} - 1.995 \times 10^{-17} (x + m)\omega_0^2 x^2 \\ - 6.534 \times 10^{-17} (x\omega + \omega_0)(xm)(\omega_0 + \omega) \\ + 0.250 mx^2(\omega_0^2 + \omega^2)$                    | 0.000      | 0.000     | 1.000       | 1.000      | 19.437      |
| 5                                              | $4.628 \times 10^{-16} - 7.202 \times 10^{-18} (x\omega_0 + \omega + m)(x\omega)(\omega m) \\ - 4.890 \times 10^{-17} (x\omega_0 + xm)(x\omega + \omega_0 + m) \\ + 0.250 mx^2(\omega_0^2 + \omega^2)$ | 0.000      | 0.000     | 1.000       | 1.000      | 19.006      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                        |            |           |             |            |             |
| 1                                              | $0.017 + 0.250 mx^2(\omega_0^2 + \omega^2) + 0.001(x\omega_0)m(x + \omega)^2 \\ - 0.003(x\omega_0)m(x\omega + x + \omega)$                                                                             | 0.146      | 0.149     | 1.000       | 1.000      | 19.172      |
| 2                                              | $0.014 - 1.842 \times 10^{-5} (\omega_0 \omega)^2 (\omega m)^2 \\ + 0.249 mx^2(\omega_0^2 + \omega^2) + 2.906 \times 10^{-4} x^2(\omega m)(\omega_0^2 + \omega + m)$                                   | 0.149      | 0.148     | 1.000       | 1.000      | 18.933      |
| 3                                              | $0.007 + 0.250 mx^2(\omega_0^2 + \omega^2) \\ + 6.386 \times 10^{-4} (x + \omega_0)(x\omega)(xm + \omega_0 m) \\ - 6.527 \times 10^{-4} (x^2 + x + m)(xm)(\omega_0 \omega)$                            | 0.147      | 0.150     | 1.000       | 1.000      | 18.841      |
| 4                                              | $-0.001 + 7.489 \times 10^{-4} (\omega_0 \omega)\omega_0^2(\omega_0 + \omega + \omega_0^2) \\ - 7.301 \times 10^{-4} (\omega_0 + \omega + \omega^2)\omega_0^4 \\ + 0.250 mx^2(\omega_0^2 + \omega^2)$  | 0.150      | 0.146     | 1.000       | 1.000      | 19.038      |
| 5                                              | $-0.003 + 8.834 \times 10^{-6} \omega_0^4 (\omega_0 m)^2 \\ - 7.401 \times 10^{-5} (x\omega_0)\omega_0^2(x^2 + \omega_0) \\ + 0.250 mx^2(\omega_0^2 + \omega^2)$                                       | 0.151      | 0.150     | 1.000       | 1.000      | 19.092      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                                                                        |            |           |             |            |             |
| 1                                              | $-0.012 - 2.034 \times 10^{-4} x^4 \omega^4 + 3.247 \times 10^{-4} (x\omega)(x + \omega)x^2 \omega^2 \\ + 0.250 mx^2(\omega_0^2 + \omega^2)$                                                           | 0.244      | 0.249     | 1.000       | 1.000      | 19.302      |
| 2                                              | $0.023 - 3.070 \times 10^{-5} (\omega_0 \omega)^2 (\omega m)^2 \\ + 0.249 mx^2(\omega_0^2 + \omega^2) + 4.844 \times 10^{-4} x^2(\omega m)(\omega_0^2 + \omega + m)$                                   | 0.248      | 0.246     | 1.000       | 1.000      | 18.769      |

Continued on next page

Table 20: **SISSO** ++ results for the Feynman equation I.24.6 (**EHFO**):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                                  | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | $0.011 + 0.250 mx^2(\omega_0^2 + \omega^2)$<br>$+ 0.001(x + \omega_0)(x\omega)(x + \omega_0)m - 0.001(x\omega)(\omega_0 m)(x^2 + x + m)$                            | 0.245      | 0.250     | 1.000       | 1.000      | 18.868      |
| 4    | $-0.001 + 0.001(\omega_0\omega)\omega_0^2(\omega_0 + \omega + \omega_0^2)$<br>$- 0.001(\omega_0 + \omega + \omega^2)\omega_0^4 + 0.250 mx^2(\omega_0^2 + \omega^2)$ | 0.250      | 0.243     | 1.000       | 1.000      | 19.545      |
| 5    | $-0.005 + 1.472 \times 10^{-5}(\omega_0^2)(\omega_0 m)^2$<br>$- 1.233 \times 10^{-4}(x\omega_0)\omega_0^2(x^2 + \omega_0)$<br>$+ 0.250 mx^2(\omega_0^2 + \omega^2)$ | 0.252      | 0.251     | 1.000       | 1.000      | 19.332      |

Table 21: **gplearn** results for the Feynman equation I.24.6 (**EHFO**):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                        | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                           |            |           |             |            |             |
| 1                                              | $m\omega_0\{x^2 + \cos[\cos(x) + \omega x]\}$<br>$+ \cos\{x \cos(\omega - 0.615)\} + \cos(\omega + x) + \omega(\omega - 0.615)$                           | 2.953      | 3.633     | 0.956       | 0.936      | 16.194      |
| 2                                              | $m\omega x^2 + x^4 \omega_0^8 \sin^8(0.286) + m^2 x^2 \omega_0^6 \sin^6(0.286)$                                                                           | 3.393      | 3.114     | 0.946       | 0.953      | 21.492      |
| 3                                              | $e^{-1/\omega_0} [e^{-2/\omega_0} \{m x^2 (\omega + 1) e^{-1/\omega_0} + x(0.842 + m\omega x)\}$<br>$+ x(\omega_0^{-1} + m\omega x)]$                     | 2.182      | 2.154     | 0.976       | 0.979      | 17.269      |
| 4                                              | $m\omega\omega_0 x + x^2$                                                                                                                                 | 4.729      | 4.574     | 0.884       | 0.900      | 14.323      |
| 5                                              | $m\omega\omega_0 + m + x^3 - 0.081$<br>$+ \cos(m\omega) + \cos(x\omega) + \cos(m\omega_0)$                                                                | 6.596      | 7.069     | 0.784       | 0.775      | 14.708      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                           |            |           |             |            |             |
| 1                                              | $x(\omega_0^2 - \omega_0) - x + m\omega x^2$                                                                                                              | 3.220      | 3.448     | 0.951       | 0.940      | 14.527      |
| 2                                              | $\omega_0^2 + m\omega x^2$                                                                                                                                | 5.192      | 4.815     | 0.872       | 0.885      | 14.160      |
| 3                                              | $m\omega\omega_0 x + x^2 + \cos(mx) + 4 \cos(2x)$                                                                                                         | 3.289      | 3.232     | 0.949       | 0.949      | 14.774      |
| 4                                              | $m\omega\omega_0 x + x^2$                                                                                                                                 | 4.709      | 4.732     | 0.897       | 0.886      | 14.162      |
| 5                                              | $\sin(0.678^2) \cos(\sin \omega) [m\omega_0 x^2 \cos\{\cos[\sin(m \sin \omega \cos(\sin \omega))]\} + \omega x^2]$<br>$+ m\omega_0 x^2 \cos(\sin \omega)$ | 2.464      | 2.428     | 0.972       | 0.974      | 16.442      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                           |            |           |             |            |             |
| 1                                              | $x^2 + \omega\omega_0(mx - 2x^{-3})$                                                                                                                      | 4.011      | 4.487     | 0.923       | 0.906      | 15.331      |
| 2                                              | $m\omega x^2 + 2\omega_0^2 - \sin^3(\omega_0) - \sin^3(m) - 6.454$                                                                                        | 3.390      | 3.271     | 0.946       | 0.939      | 15.304      |
| 3                                              | $\frac{\omega^2 + (x - 0.123)^2 + m\omega_0(x - 0.123)^2}{m e^x} - \frac{m e^x}{\omega_0 \omega^2}$                                                       | 3.449      | 3.497     | 0.943       | 0.943      | 16.284      |
| 4                                              | $x^2 + m\omega\omega_0 x$                                                                                                                                 | 4.892      | 4.577     | 0.888       | 0.894      | 14.208      |
| 5                                              | $\cos(x + \omega) + 2\omega^2 + m\omega_0 x^2 + e^{-e^x}$<br>$+ 2\{\sin(-0.286)\}^{-1} + \exp\{\cos(mx)\}$                                                | 3.168      | 3.523     | 0.957       | 0.945      | 17.446      |

Table 22: operon results for the Feynman equation I\_24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                        | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                           |            |           |             |            |             |
| 1                                              | $-3.546 + 0.531[0.770x \cdot 0.606\omega\{(0.569x)^2 + 1.004\omega + (0.756\omega_0)(-1.298x)(-1.391m)\} + (0.756\omega_0)(-1.298x)(-1.633m)]$                            | 1.699      | 1.991     | 0.985       | 0.981      | 23.375      |
| 2                                              | $-0.541 + 1.069\left[\frac{2.718x}{\sin\{\sin(0.749\omega_0)\}} \frac{0.833m}{\sin(0.749\omega)}\right] \times \sin\{2.416\omega_0 \exp(-0.993)(-0.164\omega)\}(-1.230x)$ | 1.544      | 1.533     | 0.989       | 0.989      | 29.189      |
| 3                                              | $0.607 - 3.798[\sin(-1.594\omega_0)(0.318m) + 0.318m(0.650\omega_0)(-1.010x)] \times [\exp(-1.344\omega) + \sin(-1.483\omega_0) + (-0.663\omega)(-1.010x)]$               | 0.842      | 0.853     | 0.996       | 0.997      | 28.206      |
| 4                                              | $-0.633 - 0.510[(-0.442\omega)(1.043x)\{(-2.308m)(-0.442\omega)\} + (0.320\omega)(2.116\omega_0)(1.045x)\{\cos(-1.177\omega) + 1.750x\}(-0.846m)]$                        | 1.515      | 1.557     | 0.988       | 0.988      | 24.353      |
| 5                                              | $0.485 - 2.089[e^{1.002\omega}(0.707)(-0.290x)e^{0.300\omega_0} + \sin\{e^{0.477\omega_0}\}] \times \{(0.434m)(2.718x)e^{-0.581\omega}\}$                                 | 0.672      | 0.704     | 0.998       | 0.998      | 29.436      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                           |            |           |             |            |             |
| 1                                              | $0.011 - 0.089[-0.066\omega_0 + \{(-1.043\omega_0 - 1.101\omega)(-1.911x)\}^2 + \{(-1.047\omega_0 + 0.990\omega)(-2.012x)\}^2](-0.333m)$                                  | 0.149      | 0.143     | 1.000       | 1.000      | 22.110      |
| 2                                              | $-0.002 - 0.027(1.805m)[(3.824\omega_0)(0.509x)(-1.326x)(1.997\omega_0) - \{(1.649\omega)(-1.377x)\}^2]$                                                                  | 0.153      | 0.146     | 1.000       | 1.000      | 20.319      |
| 3                                              | $-1.079 + 0.201(0.693\omega)[(1.777x)^2 + (-0.123\omega_0)\{(-1.754x)^2 - 0.693\omega\}(0.424\omega)] \times (0.060\omega + 0.961m)e^{0.559\omega_0}$                     | 1.632      | 1.553     | 0.987       | 0.988      | 23.922      |
| 4                                              | $-0.068 + 0.737[\{-2.053\omega(0.122x) - (-0.270\omega_0)^4 + 0.136\omega - (0.307x)^2\} \times (-0.874m)\{(1.210x)(1.305\omega_0) + 2.053\omega\}]$                      | 1.099      | 1.097     | 0.994       | 0.994      | 22.964      |
| 5                                              | $-0.077 + 22.227[(2.456 - 1.226\omega_0 - 1.949\omega)e^{(-0.193\omega_0)^2}(0.506x) \times (-0.148m)][0.506x \cos\{\cos(e^{-0.887\omega})\}]$                            | 0.853      | 0.812     | 0.997       | 0.997      | 28.447      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                                           |            |           |             |            |             |
| 1                                              | $-2.085 - 0.156[\frac{1.014\omega}{1.123\omega_0} - 0.693\omega_0 - (-0.877x)(1.287\omega_0)] \times [-0.958x + 1.661m\{(-0.958x)(1.446\omega) - 1.414\omega_0\}]$        | 1.002      | 1.192     | 0.995       | 0.993      | 23.710      |
| 2                                              | $-0.341 + 7.613[(0.493x)(0.313m)(0.493x) \times \{0.619\omega_0 - \sin(1.106\omega_0) - [(0.493x - 1.712\omega) + (0.592x - 0.493x)(-2.037\omega_0)]\}]$                  | 0.879      | 0.946     | 0.996       | 0.995      | 25.662      |
| 3                                              | $-1.330 + 1.419[\frac{(-2.871x)(-0.505\omega)}{(1.145\omega_0)^2} - \{(0.188\omega_0)(1.425) + 1.218\omega\}(0.336x)(-2.572x)(0.858m)(0.327\omega_0)]$                    | 1.490      | 1.698     | 0.989       | 0.986      | 22.994      |
| 4                                              | $0.248 - 0.281[\{(0.485x - 1.914x) - 3.145\omega\}(0.485x)(0.707m)(1.875\omega) - \{(-0.809\omega_0)(0.784x)\}^2(2.190m) + 2.074\omega(2.190m)]$                          | 0.737      | 0.838     | 0.997       | 0.996      | 24.293      |
| 5                                              | $-0.216 - 0.260[\{(-3.487x)(0.711\omega_0) - (0.116x)(1.614\omega)(5.721\omega)\} \times (0.116x)\{0.456 + 0.483 + (-0.238\omega_0)^2\}(6.283m)]$                         | 0.778      | 0.724     | 0.997       | 0.998      | 22.522      |

Table 23: PySR results for the Feynman equation I.24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                              | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                 |            |           |             |            |             |
| 1                                              | $0.250 mx^2(\omega^2 + \omega_0^2)$                                             | 0.000      | 0.000     | 1.000       | 1.000      | 66.755      |
| 2                                              | $mx^2\{\omega + \omega_0 - 1.095 + \sin[\cos(\omega + 1.120)]\}$                | 0.792      | 0.791     | 0.997       | 0.997      | 85.093      |
| 3                                              | $mx^2\{0.328 - 0.529(\omega + \omega_0 - 0.164\omega\omega_0)\}^2$              | 0.545      | 0.615     | 0.999       | 0.998      | 63.298      |
| 4                                              | $mx^2\{\omega_0 + \sin(-0.780\omega_0) + \omega + \sin(-0.684\omega - 0.202)\}$ | 0.176      | 0.170     | 1.000       | 1.000      | 74.764      |
| 5                                              | $mx^2\{\omega + 0.252\omega_0^2 - 0.925\}$                                      | 0.759      | 0.836     | 0.997       | 0.997      | 59.788      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                 |            |           |             |            |             |
| 1                                              | $0.250 mx^2(\omega_0^2 + \omega^2)$                                             | 0.149      | 0.143     | 1.000       | 1.000      | 63.275      |
| 2                                              | $m[x^2\{\omega - 0.946 + 0.251\omega_0^2\} + \cos\{\omega \sin(\omega)\}]$      | 0.440      | 0.401     | 0.999       | 0.999      | 73.962      |
| 3                                              | $x[mx\{e^{0.422\omega} + \omega_0 - 2.257\} + \cos\{\omega_0 \sin(\omega_0)\}]$ | 0.383      | 0.373     | 0.999       | 0.999      | 77.065      |
| 4                                              | $mx^2\{-1.246 \sin(\omega_0 + 0.687) + \omega + 0.628\}$                        | 0.826      | 0.796     | 0.997       | 0.997      | 86.585      |
| 5                                              | $mx\{x - 0.387 \sin(\omega_0) \sin(\omega)\}\{\omega + \omega_0 - 1.648\}$      | 0.688      | 0.684     | 0.998       | 0.998      | 82.477      |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                 |            |           |             |            |             |
| 1                                              | $\sin(\omega^2) + mx^2\{0.250\omega_0^2 + \omega - 0.936\}$                     | 0.657      | 0.767     | 0.998       | 0.997      | 70.417      |
| 2                                              | $m[(\omega_0 - 2.003)^2 + x^2(\omega_0 + \omega - 1.897)]$                      | 0.939      | 0.987     | 0.996       | 0.994      | 64.486      |
| 3                                              | $mx^2\{\omega - 57.814 \cos(-0.093\omega_0) + 56.898\}$                         | 0.772      | 0.728     | 0.997       | 0.998      | 81.822      |
| 4                                              | $mx^2\{\omega_0 + 0.261(\omega - 1.998)^2 + \omega - 1.918\}$                   | 0.798      | 0.768     | 0.997       | 0.997      | 65.787      |
| 5                                              | $0.250 mx^2(\omega^2 + \omega_0^2)$                                             | 0.255      | 0.243     | 1.000       | 1.000      | 57.047      |

Table 24: BMS results for the Feynman equation I.24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS.

| Rep.                                           | Learned expression                                                                                                                           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                              |            |           |             |            |             |
| 1                                              | $m[a_0^2(a_0 + 1)^2 x^2(a_0 + \omega_0 + \omega) - x]$                                                                                       | 1.286      | 1.268     | 0.992       | 0.992      | 202.400     |
| 2                                              | $a_0^4 mx^2(\omega + a_0)\{\exp[-\sin(\omega_0) \sin(\omega)] + \omega_0\}$                                                                  | 1.238      | 1.160     | 0.993       | 0.993      | 245.711     |
| 3                                              | $mx^2\{2a_0 + \omega_0 + \omega\}$                                                                                                           | 1.103      | 1.143     | 0.994       | 0.994      | 241.044     |
| 4                                              | $mx^2\{\omega + \omega_0 + 2a_0\}$                                                                                                           | 1.083      | 1.171     | 0.994       | 0.993      | 241.607     |
| 5                                              | $\left[\omega_0 + x\{2\omega + \omega_0 a_0 \cos^2(a_0 \omega)\}\right] \times x a_0^6 m(\omega_0 + a_0) + \cos(\omega + x) + \cos(a_0 + m)$ | 1.286      | 1.355     | 0.992       | 0.992      | 248.992     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                              |            |           |             |            |             |
| 1                                              | $mx^2(a_0 + \omega_0 + \omega) + (\omega_0 + a_0)^2$                                                                                         | 0.946      | 0.950     | 0.996       | 0.995      | 208.908     |

Continued on next page

Table 24: BMS results for the Feynman equation I.24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS (*continued*).

| Rep.                                           | Learned expression                                                                                                        | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 2                                              | $\{(\omega + \omega_0)^2 + a_0 m \omega_0\} m \sin(a_0) x^2 + \{a_0(a_0 + \omega^2) + \cos(\omega_0)\}^2$                 | 0.819      | 0.566     | 0.997       | 0.998      | 253.143     |
| 3                                              | $x\{a_0 + (a_0 + a_0^3 + \omega_0)m x + a_0^2\} + m \omega x^2$                                                           | 1.103      | 1.221     | 0.994       | 0.993      | 247.547     |
| 4                                              | $m x^2 \{\cos(a_0 + \omega_0) + \omega\} + x$                                                                             | 1.502      | 1.437     | 0.989       | 0.989      | 229.507     |
| 5                                              | $m x^2(a_0 + \omega_0 + \omega) + \exp(a_0)$                                                                              | 1.106      | 1.027     | 0.994       | 0.995      | 224.604     |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                           |            |           |             |            |             |
| 1                                              | $x[mx(\omega + a_0)a_0(a_0 + \omega_0) + \sin\{\omega_0 + \omega_0 + a_0 + \sin(a_0\omega_0 + \omega)\}] + a_0^2\omega_0$ | 1.229      | 1.433     | 0.993       | 0.990      | 251.926     |
| 2                                              | $m\omega_0 x^2 + \omega(\omega + a_0) - \{mx \cos(\omega) + \sin(x)\omega + \omega + x^2\}$                               | 1.707      | 1.667     | 0.986       | 0.984      | 233.893     |
| 3                                              | $a_0[\{a_0 x(\exp[\sin(\sin a_0)])^3 + m\}(\omega + a_0 + \omega_0)]^2 + x \cos\{\omega_0 + \omega_0 \sin(\omega)\}$      | 0.970      | 0.901     | 0.995       | 0.996      | 246.667     |
| 4                                              | $m\{\omega x + \omega_0\}x^2 a_0^2 \omega_0 e^{a_0} + mx(a_0 + \omega)$                                                   | 1.563      | 1.585     | 0.989       | 0.987      | 239.007     |
| 5                                              | $-a_0 m x(\omega + \omega_0)\{a_0[a_0 \omega \omega_0]^2 - x\}x(\omega + \omega_0) + a_0$                                 | 0.983      | 0.906     | 0.996       | 0.996      | 241.769     |

Table 25: BSR results for the Feynman equation I.24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                         | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                            |            |           |             |            |             |
| 1                                              | $-8.483 + 3.470e^{\cos(m)} - 0.249e^m + 0.514 \sin\{(\omega_0 + \omega)^3\} + 4.413(m + \omega_0 m)$       | 11.979     | 12.930    | 0.268       | 0.191      | 190.583     |
| 2                                              | $77.459 - 88.506 \cos(e^{-\omega_0}) - 8.088 \cos(x) - 12.942 \cos(\omega_0) + 5.242mx$                    | 7.071      | 6.964     | 0.763       | 0.764      | 195.174     |
| 3                                              | $-5.975 + 0.200e^m + 1.005e^\omega + 0.009 \frac{\cos(m)}{\sin(m^2)} + 7.105m$                             | 12.327     | 12.784    | 0.238       | 0.263      | 200.450     |
| 4                                              | $-59.296 + 18.947\{\omega_0 + e^{\cos(\omega_0)}\} + 1.090e^m + 8.011\omega - 1.337 \sin(m^2)$             | 11.193     | 10.677    | 0.349       | 0.453      | 200.248     |
| 5                                              | $-21.076 + 6.629\{\cos(\omega^{-1}) + \omega + x\} + 16.613\omega^{-1} - 11.968x^{-1} - 0.050(-x\omega)^3$ | 8.496      | 9.116     | 0.641       | 0.625      | 195.475     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                            |            |           |             |            |             |
| 1                                              | $-43.096 + 24.557x + 11.088mx^{-1} - 0.697 \sin(\omega_0^2) + 0.111 \sin^{-2}(m)$                          | 9.273      | 9.513     | 0.591       | 0.546      | 198.356     |
| 2                                              | $27.426 + 2.462 \cos(\omega_0^2) + 67.775e^{-x} + 49.295 \cos\{\sin(x)\} - 30.300e^{x^{-1}}$               | 10.210     | 10.241    | 0.505       | 0.478      | 191.194     |
| 3                                              | $29.326 - 9.605e^{\sin(\omega_0)} - 21.107 \cos(x) - 1.870 \sin(\omega_0^2) - 11.162 \cos(m)$              | 7.887      | 7.834     | 0.707       | 0.702      | 191.329     |
| 4                                              | $18.336 + 0.667xe^\omega - 30.475 \sin\{\sin(x)\} - 2.175 \sin(m)^{-1} + 1.496e^m$                         | 7.578      | 7.591     | 0.732       | 0.707      | 191.405     |

Continued on next page

Table 25: BSR results for the Feynman equation I\_24.6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5                                              | $-30.878 + 0.729 \cos(\omega) - 5.018m^{-1} - 6.903(\omega + \omega_0) + 8.028(x + m + \omega\omega_0 + x)$                                                                                      | 6.226      | 5.786     | 0.820       | 0.850      | 190.559     |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $2.171 \times 10^{-7} + 1.295 \times 10^{-4} e^{\omega x} - 3.129 \times 10^{-7} \{(-\omega_0 m)^3\}^3 + 1.016 \times 10^{-6} e^{x^{-1} \omega_0 \omega} - 6.119 \times 10^{-7} \{\cos(m) - x\}$ | 22.016     | 22.501    | -1.321      | -1.363     | 190.690     |
| 2                                              | $1.211 \times 10^{-7} + 8.496 \times 10^{-8} \sin(\omega) - 6.173 \times 10^{-8} \cos(m) + 1.594 \times 10^{-7} e^{e^x} + 5.889 \times 10^{-7} \omega_0 x$                                       | 22.210     | 21.097    | -1.300      | -1.519     | 190.801     |
| 3                                              | $45.128 - 16.608m^{-1} + 1.523 \sin(-\omega_0^3) - 20.747 \cos\{\cos(m)\} - 2.033 \sin\{\cos(m^2)\}$                                                                                             | 13.339     | 13.665    | 0.146       | 0.122      | 192.065     |
| 4                                              | $-9.015 + 19.457 \cos\{\cos(m)\} + 1.670 \cos^2\{\sin(\omega^2)\} + 1.134m^3 - 0.678 \sin^2\{\cos(\omega_0) + \omega\}$                                                                          | 13.416     | 13.236    | 0.155       | 0.117      | 205.891     |
| 5                                              | $21.838 - 32.250x^{-1} + 1.481\omega_0^2 + 0.577 \exp\{\cos(xm) + x\} + 0.269\omega_0^3$                                                                                                         | 9.159      | 8.730     | 0.638       | 0.660      | 191.737     |



## C.4 Results for I\_50\_26 (HOQN)

Table 26: DSR results for the Feynman equation I\_50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                     | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                        |            |           |             |            |             |
| 1                                              | $\{\exp[\cos(t\omega + \exp(x_1)^{-1})]\}^2$           | 1.566      | 1.378     | 0.395       | 0.525      | 44.8        |
| 2                                              | $\{\exp[\cos(\omega t)] - (t^{-1})^3 \alpha^{-1}\}^2$  | 1.455      | 1.454     | 0.506       | 0.461      | 46.5        |
| 3                                              | $x_1 \exp[\cos(\omega t)]$                             | 1.430      | 1.514     | 0.471       | 0.400      | 43.8        |
| 4                                              | $\exp[\cos(\omega t)] + (\omega t)^{-1}$               | 1.654      | 1.778     | 0.311       | 0.209      | 45.0        |
| 5                                              | $\alpha \exp[\cos\{-[-(\omega t)^{-1}]^{-1}\}]$        | 1.494      | 1.518     | 0.472       | 0.354      | 44.3        |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                        |            |           |             |            |             |
| 1                                              | $x_1 \exp[\omega \sin^2(\omega) \cos(\omega + t)]$     | 1.607      | 1.659     | 0.300       | 0.357      | 47.1        |
| 2                                              | $\{\exp[\cos(t - \omega)]\}^2$                         | 1.496      | 1.348     | 0.456       | 0.437      | 44.7        |
| 3                                              | $\{\exp[\cos(-\omega - t)]\}^2$                        | 1.481      | 1.393     | 0.457       | 0.612      | 44.8        |
| 4                                              | $\{\sin[\exp\{\exp(t)^{-1}\}] + \cos(\omega t)\}^3$    | 1.462      | 1.526     | 0.449       | 0.290      | 46.4        |
| 5                                              | $x_1 \exp[\cos(-t\omega)]$                             | 1.494      | 1.449     | 0.436       | 0.474      | 45.8        |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                        |            |           |             |            |             |
| 1                                              | $\{\exp[\cos(-\omega t)]\}^2$                          | 1.477      | 1.370     | 0.458       | 0.461      | 44.9        |
| 2                                              | $t\alpha \cos^2(t\omega)$                              | 1.485      | 1.568     | 0.438       | 0.463      | 47.7        |
| 3                                              | $\exp[\cos\{t((\exp(x_1)t)^{-3} + \omega)\}]$          | 1.664      | 1.825     | 0.283       | 0.292      | 47.1        |
| 4                                              | $\alpha \exp[\{\exp[\exp(x_1^3)] + \cos(t\omega)\}^3]$ | 1.577      | 1.497     | 0.333       | 0.518      | 45.3        |
| 5                                              | $x_1 \exp[\cos(t\omega)]$                              | 1.447      | 1.482     | 0.431       | 0.429      | 45.0        |

Table 27: QLattice results for the Feynman equation I\_50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Row-specific constants are introduced inside the learned-expression column for compactness.

| Rep.                     | Learned expression                                                                                                                                                                                                         | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                                            |            |           |             |            |             |
| 1                        | $0.530 + \frac{1.358}{A_1},$ $A_1 = -0.700\omega - 0.708t - 0.143x_1$ $+ 88.824(1 - 0.203\omega - 0.201t)^2 + 4.059$                                                                                                       | 1.186      | 1.041     | 0.653       | 0.729      | 85.414      |
| 2                        | $-0.330t + 0.313x_1 + 1.256(0.105 - 0.211x_1)$ $\times (3.152 - 2.507\alpha) - 0.186 + \frac{1.256}{A_2},$ $A_2 = 43.822B_2^2 + 0.199,$ $B_2 = 0.353\omega + 0.162t - 0.404(0.505\omega + 0.743)$ $\times e^{-0.403t} - 1$ | 1.043      | 1.058     | 0.747       | 0.714      | 85.057      |

Continued on next page

Table 27: **QLattice** results for the Feynman equation L50\_26 (**HOQN**):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Row-specific constants are introduced inside the learned-expression column for compactness (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                              | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | $-5.665(-0.118\alpha - 0.237) \exp(A_3 B_3) + 0.518,$<br>$A_3 = -0.239\alpha - 0.966x_1 + (2.127 - 1.290\omega)$<br>$\times (2.701 - 2.085t) - 1.151$<br>$B_3 = -0.042\alpha - 1.596\omega - 0.264(t + 0.491)^2 + 6.067$                                                                        | 1.281      | 1.367     | 0.576       | 0.511      | 87.361      |
| 4                                              | $-0.114(3.120 - 4.436\alpha)(0.776x_1 + 1)^2 e^{-0.126\omega} A_4^{-1}$<br>$+ 0.463,$<br>$A_4 = (0.697\alpha + 0.174) B_4^2 + 0.357e^{-0.046\omega},$<br>$B_4 = 134.737\{-0.197t + (0.209\omega - 1)^2 + 0.269\}$                                                                               | 1.058      | 1.221     | 0.718       | 0.627      | 84.035      |
| 5                                              | $2.335 \exp(A_5 B_5 C_5) + 0.817,$<br>$A_5 = 0.278\omega + 1.233,$<br>$B_5 = 0.889(1 - 0.945\omega)^2 + (0.722t - 2.922)$<br>$\times (0.229x_1 + 2.571) + 0.944,$<br>$C_5 = -0.202\alpha - 2.093\omega - 2.157t - 0.230x_1 + 11.011$                                                            | 1.113      | 1.140     | 0.707       | 0.636      | 84.498      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                                                                                 |            |           |             |            |             |
| 1                                              | $0.449(1.220t - 1.359) A_1 + 0.978,$<br>$A_1 = 1.354\alpha + (3.986 - 2.332t)(-0.168x_1 - 0.238)e^{B_1}$<br>$- 3.123,$<br>$B_1 = (2.781\omega - 5.056)\{-0.671(2.312t - 4.288)C_1$<br>$+ 1.066\},$<br>$C_1 = 1.269\omega + 0.738t - 0.206x_1 - 4.366$                                           | 1.176      | 1.103     | 0.625       | 0.715      | 83.788      |
| 2                                              | $2.071(0.678t - 1.130) A_2 + 0.703,$<br>$A_2 = 0.311x_1 + e^{B_2 C_2 D_2} - 1.348,$<br>$B_2 = 1.960 - 1.127\omega, C_2 = 1.994\omega - 2.320,$<br>$D_2 = \{1.173t + (0.916 - 0.013\alpha)(1.327\omega - 3.304)$<br>$- 3.591\}\{1.537t + (2.852 - 0.607x_1)(0.139\alpha - 0.772)$<br>$- 1.898\}$ | 1.173      | 1.090     | 0.665       | 0.632      | 89.515      |
| 3                                              | $1.224 \exp(A_3) + 0.767,$<br>$A_3 = 0.738x_1 + 20.096(3.050 - 0.741\omega)(-0.192\omega$<br>$- 0.063) B_3^2 e^{-0.070\omega},$<br>$B_3 = -0.005\alpha - 0.457(1.221\omega - 1.627)$<br>$\times (1.351t - 1.883) + 1$                                                                           | 1.138      | 1.207     | 0.679       | 0.709      | 97.835      |
| 4                                              | $0.990\alpha + 5.697 \exp(A_4) - 1.152,$<br>$A_4 = 42.077(2.792 - 0.532t)(-0.026\omega - 2.553)$<br>$\times (-0.553t - 1.258)(0.452x_1 - 1.699)(1 - 0.201\omega$<br>$- 0.197t)^2$                                                                                                               | 1.020      | 1.004     | 0.732       | 0.693      | 80.437      |
| 5                                              | $4.614(0.098\alpha - 1)^2 A_5^2 + 0.023,$<br>$A_5 = -0.386\alpha - 0.265x_1 + 0.849 - e^{-45.106 B_5^2},$<br>$B_5 = 0.418\omega + 0.268(1.747 - 0.661t)(-0.178\alpha$<br>$- 2.005) - 1$                                                                                                         | 0.944      | 0.956     | 0.775       | 0.771      | 79.372      |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                                                                                                                                                                                                                 |            |           |             |            |             |
| 1                                              | $1.029\alpha + 5.267 \exp(A_1 B_1 C_1) - 1.342,$<br>$A_1 = -0.314\alpha - 0.862,$<br>$B_1 = -0.024\alpha + (3.639 - 2.339t)(1.788\omega - 3.272)$<br>$+ 2.433,$<br>$C_1 = -1.586\omega - 2.031t + 9.235$                                                                                        | 1.129      | 0.998     | 0.683       | 0.714      | 80.967      |

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Table 27: **QLattice** results for the Feynman equation I.50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Row-specific constants are introduced inside the learned-expression column for compactness (*continued*).

| Rep. | Learned expression                                                                                                                                                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 2    | $1.031\alpha - 2.827(0.227\omega - 2.374)\exp(A_2 B_2 C_2) - 1.257,$<br>$A_2 = 0.461 - 2.930t, B_2 = 0.842\omega + 1.024t - 4.755,$<br>$C_2 = 1.203t + (0.127 - 0.594t)(2.409 - 1.268\omega) - 3.949$                                | 1.120      | 1.182     | 0.681       | 0.694      | 83.521      |
| 3    | $0.331 - \frac{5.052}{(1.130x_1 - 4.640)A_3},$<br>$A_3 = (-1.929\omega - 0.248t + 4.478)B_3$<br>$+ 101.679e^{-1.687t},$<br>$B_3 = 0.556\omega + 1.051t + 0.691(1 - 0.713t)^2$<br>$\times (\omega - 0.893)^2(0.147t - 2.259) - 0.897$ | 1.189      | 1.207     | 0.634       | 0.691      | 83.611      |
| 4    | $0.994\alpha - 0.044t + 0.728x_1 + 1.085e^{A_4 B_4} - 2.569,$<br>$A_4 = -0.269\omega - 2.001,$<br>$B_4 = 0.076\omega + 100.427(1 - 0.200\omega - 0.196t)^2 - 0.810$                                                                  | 1.018      | 0.980     | 0.723       | 0.793      | 95.180      |
| 5    | $3.635(0.264 - 0.269t)(1.068x_1 - 0.401)A_5$<br>$+ 0.368,$<br>$A_5 = -0.429\alpha - 0.742\omega - 1.490e^{B_5} + 3.223,$<br>$B_5 = 1.619(1 - 0.702\omega)^2(2.258 - 1.258t)(5.321 - 2.166\omega)$<br>$\times (6.294 - 2.535t)$       | 1.234      | 1.300     | 0.586       | 0.561      | 83.509      |

Table 28: **SISSO++** results for the Feynman equation I.50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                            | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                               |            |           |             |            |             |
| 1                                              | $-1.698 + 0.034(\alpha + t + \omega + x_1)(\alpha x_1 + \alpha)$<br>$+ 3.251 \cos^2(t\omega) + 0.196 \cos(t\omega)(tx_1\omega)$               | 0.585      | 0.585     | 0.914       | 0.903      | 26.002      |
| 2                                              | $-1.476 + 0.044\{(\alpha + t)(\alpha x_1) + (\alpha + \omega)(\alpha x_1)\}$<br>$+ 3.232 \cos^2(t\omega) + 0.196 \cos(t\omega)(\omega x_1 t)$ | 0.589      | 0.595     | 0.910       | 0.910      | 27.093      |
| 3                                              | $-2.374 + 0.306\{\alpha x_1 + \alpha + x_1 + \cos(t + \omega)\}$<br>$+ 3.338 \cos^2(t\omega) + 0.193 \cos(t\omega)(\omega x_1 t)$             | 0.585      | 0.566     | 0.904       | 0.910      | 25.880      |
| 4                                              | $-1.755 + 0.035(\alpha + t + \alpha + \omega)(\alpha x_1 + x_1)$<br>$+ 3.268 \cos^2(t\omega) + 0.191 \cos(t\omega)(tx_1\omega)$               | 0.596      | 0.576     | 0.905       | 0.910      | 26.835      |
| 5                                              | $-2.511 + 0.211\{\alpha + t + \alpha x_1 + \alpha x_1 + \cos(t)\}$<br>$+ 3.347 \cos^2(t\omega) + 0.206 \cos(t\omega)(tx_1\omega)$             | 0.593      | 0.604     | 0.910       | 0.907      | 26.050      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                               |            |           |             |            |             |
| 1                                              | $-1.704 + 0.034(\alpha + \omega + t + x_1)(\alpha x_1 + \alpha)$<br>$+ 3.249 \cos^2(t\omega) + 0.195 \cos(t\omega)(\omega x_1 t)$             | 0.606      | 0.608     | 0.908       | 0.897      | 25.956      |
| 2                                              | $-1.491 + 0.051(\alpha + t + t + \omega)(\alpha x_1 + \cos t)$<br>$+ 3.242 \cos^2(t\omega) + 0.200 \cos(t\omega)(t\omega x_1)$                | 0.604      | 0.600     | 0.906       | 0.908      | 27.228      |
| 3                                              | $-2.367 + 0.306\{\alpha x_1 + \alpha + x_1 + \cos(t + \omega)\}$<br>$+ 3.341 \cos^2(t\omega) + 0.192 \cos(t\omega)(\omega x_1 t)$             | 0.605      | 0.592     | 0.898       | 0.902      | 25.630      |

*Continued on next page*

Table 28: SISSO + + results for the Feynman equation I\_50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4                                              | $-1.759 + 0.035(\alpha + t + \alpha + \omega)(\alpha x_1 + x_1)$<br>$+ 3.267 \cos^2(t\omega) + 0.191 \cos(t\omega)(tx_1\omega)$    | 0.613      | 0.595     | 0.900       | 0.904      | 26.012      |
| 5                                              | $-2.518 + 0.213\{\alpha + t + \alpha x_1 + \alpha x_1 + \cos(t)\}$<br>$+ 3.331 \cos^2(t\omega) + 0.206 \cos(t\omega)(t\omega x_1)$ | 0.614      | 0.622     | 0.904       | 0.902      | 25.768      |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                                                    |            |           |             |            |             |
| 1                                              | $-1.705 + 0.034(\alpha x_1 + \alpha)(\alpha + x_1 + t + \omega)$<br>$+ 3.249 \cos^2(t\omega) + 0.195 \cos(t\omega)(\omega x_1 t)$  | 0.615      | 0.617     | 0.905       | 0.894      | 25.894      |
| 2                                              | $-1.491 + 0.051(\alpha + t + t + \omega)(\alpha x_1 + \cos t)$<br>$+ 3.242 \cos^2(t\omega) + 0.200 \cos(t\omega)(tx_1\omega)$      | 0.611      | 0.607     | 0.904       | 0.907      | 25.341      |
| 3                                              | $-2.365 + 0.306\{\alpha x_1 + \alpha + x_1 + \cos(t + \omega)\}$<br>$+ 3.341 \cos^2(t\omega) + 0.192 \cos(t\omega)(tx_1\omega)$    | 0.613      | 0.602     | 0.895       | 0.899      | 25.675      |
| 4                                              | $-1.760 + 0.035(\alpha + t + \alpha + \omega)(\alpha x_1 + x_1)$<br>$+ 3.267 \cos^2(t\omega) + 0.191 \cos(t\omega)(\omega x_1 t)$  | 0.621      | 0.603     | 0.898       | 0.902      | 25.893      |
| 5                                              | $-2.519 + 0.213\{\alpha + t + \alpha x_1 + \alpha x_1 + \cos(t)\}$<br>$+ 3.328 \cos^2(t\omega) + 0.206 \cos(t\omega)(tx_1\omega)$  | 0.622      | 0.630     | 0.901       | 0.900      | 27.191      |

Table 29: gplearn results for the Feynman equation I\_50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                                                                                                                       | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                                                                                                                                                                                                          |            |           |             |            |             |
| 1                                              | $\{\alpha \cos^2(\omega t) + x_1[0.145\alpha x_1 \cos^2(\omega t)$<br>$+ 0.145\omega\{0.145t + 0.145 \cos[0.145\omega t\{\alpha x_1 \cos^2(\omega t) + \sin(\omega t)\}]\}$<br>$\times \{x_1 \cos^2(\omega t) + \cos^2(t)\} \cos(\omega t) + 0.145t + 0.021x_1$<br>$+ 0.290 \cos(\omega t)\} \cos\{\sin(\omega t)\} + \cos(\omega t)$                                                    | 0.350      | 0.326     | 0.970       | 0.973      | 20.960      |
| 2                                              | $\{0.125\alpha + 0.125\omega^2(0.488t - 0.488 \sin \omega)$<br>$\times \cos[\omega(0.488t - 0.488 \sin \omega) - \cos\{\omega \cos(\sin t)\} + \cos(\sin t)]$<br>$\times \cos[-t + \sin \omega + \cos\{\omega \cos(\sin t)\} + \cos\{-t + \sin \omega + \cos x_1$<br>$+ \cos[\omega(0.488t - 0.488 \sin \omega) \cos\{-t + \sin \omega + \cos x_1$<br>$+ \cos(\omega \cos(\sin t))\}]\}$ | 1.172      | 1.269     | 0.680       | 0.589      | 20.866      |
| 3                                              | $\alpha x_1 \cos^2(\omega t) + 2 \cos(\omega t)$                                                                                                                                                                                                                                                                                                                                         | 0.413      | 0.419     | 0.956       | 0.954      | 16.811      |
| 4                                              | $\alpha\{e^{\sin(e^{\cos[\sin(e^{\cos \alpha})] \cos(\omega t)})} + \cos(\sin x_1)\} \cos^2(\omega t)$                                                                                                                                                                                                                                                                                   | 0.730      | 0.719     | 0.866       | 0.870      | 19.996      |
| 5                                              | $0.281\alpha\omega tx_1 \cos^2(\omega t)$<br>$\times \cos\{\sin[\cos(\sin\{\cos[\sin(\omega t) \cos(\sin\{\sin[\sin(0.281\alpha\omega tx_1$<br>$\times \cos^2(\omega t) \cos(\sin(\omega t)) \cos(\sin(\cos(\omega t))\})\}) + \sin(\cos(\omega t))\})]\}$<br>$+ \sin(\cos(\omega t))$                                                                                                   | 0.411      | 0.389     | 0.960       | 0.958      | 17.816      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                                                                                                                                                                          |            |           |             |            |             |

*Continued on next page*

Table 29: **gplearn** results for the Feynman equation I\_50.26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 1                                              | $0.391x_1\{2\alpha \cos^2(\omega t) - \sin(0.868x_1 - \cos(\omega t)) + \cos^4(\omega t) + 3 \cos(\omega t) + \cos(\cos x_1)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.319      | 0.324     | 0.972       | 0.976      | 19.991      |
| 2                                              | $\alpha + \sin\{(\omega + t) \cos(\omega + t)\} + \cos\{\omega + t + \sin[(\alpha + \omega + \cos\{\alpha + \sin(\omega + t)\}) \cos(\omega + t)]\} + \cos\{t + x_1 + \sin[(\alpha + \omega) \cos(\omega + t)]\} + \cos\{\sin(\omega t)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.889      | 0.790     | 0.808       | 0.807      | 18.550      |
| 3                                              | $\alpha + 0.284x_1 + \sin^2(1.004/t) + \sin^2\{[\alpha - \sin(\cos \omega) + \cos^2 t - 0.615] \times \sin^2(\cos \omega)\} + \sin^2(\cos \omega) + \cos^2 t - 0.615 - \frac{1.004}{x_1} - \frac{2.004}{t}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1.593      | 1.864     | 0.371       | 0.306      | 18.179      |
| 4                                              | $2\alpha \cos^2(\omega t) + 2 \cos(\omega t)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.725      | 0.707     | 0.865       | 0.847      | 16.677      |
| 5                                              | $-0.395\{-0.395\omega \cos \omega + [-0.395\omega \cos \omega + t + \{-0.395\omega \cos \omega + (0.823\alpha + t - 0.665e^{1/t}) \cos(\sin x_1)\} \times \cos\{\sin(1.992t + \sin[e^{-e^{-(0.992t-0.665e^{1.040/\cos \omega} + 1.504e^{-1/t})^2} + \cos^2 \omega]\})\} - 0.395[-0.395\omega \cos \omega + \{0.823\alpha + t + [-0.395\omega \cos \omega + (0.823\alpha + 1.992t) \cos(\sin x_1)]\} \times \cos\{\sin(1.992t + \sin[e^{-e^{-(0.395\omega \cos \omega + 0.992t - 0.665e^{1/t})^2} + \cos^2 \omega]\})\} + 0.992 \cos \omega\} \cos(\sin x_1)\} \times \cos \omega \cos\{\sin(1.992t + \sin[e^{-e^{-0.992t}} + \cos^2 \omega]\}) \times \cos(\sin x_1) \cos \omega \cos\{\sin(1.992t + \sin[e^{-e^{-0.992t}} + \cos^2 \omega]\}) + (0.823\alpha + t - 0.665e^{1/t}) \cos(\sin x_1)\} \times \cos\{\sin[-0.395\omega \cos \omega + 1.992t + \cos^2 \omega]\} \times \cos\{\sin[-0.395\omega \cos \omega + 1.992t + \cos^2 \omega - e^{-e^{1/t}}/\{\alpha \cos(\sin x_1)\}]\}$ | 1.214      | 1.192     | 0.628       | 0.643      | 29.848      |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |           |             |            |             |
| 1                                              | $x_1\{\alpha + \sin[(\alpha + \cos \omega) \cos^3(\omega t) + \sin^3\{\sin^3[(\alpha + \cos(\omega + 0.528x_1)) \sin^2((\alpha + \cos(\omega + 0.528x_1)) \times \cos^2(\omega t) - \omega^{-3} - t^{-3}) + \cos^2 x_1 \cos(\omega t)]\}\} \cos^2(\omega t) + \cos(\omega t)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.524      | 0.533     | 0.932       | 0.919      | 20.623      |
| 2                                              | $\alpha + \cos(\omega t) + 2 \cos(2\omega t) + \cos\{\omega t \cos[e^{-\omega\{\alpha + t + 2 \cos(\omega t) + 2 \cos(2\omega t)\}}]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.829      | 0.904     | 0.825       | 0.821      | 18.325      |
| 3                                              | $0.434\alpha^2 \cos^2(\omega t) - \omega\{-0.032\alpha^2 x_1^2 \cos^2(\omega t) + 0.226\alpha - 0.259x_1 + 0.470 \cos(\omega^{-1}) \cos(\omega - 0.409) - 0.475 \cos(\omega t)\} \times \cos\{\sin[\sin[\omega \cos(\omega + t) \cos(\sin t) - \cos\{t + \{0.434\alpha^2 \cos^2(\omega t) - \omega(-0.249\alpha - 0.470 \cos(\omega - 0.409) \cos(\cos(\omega + t)) - 0.475 \cos(t + \cos(t + e^{-\omega-t}))\}) \times \cos(\sin[\sin\{\omega \cos(0.259\alpha x_1)\}])^{-1}\}]\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0.695      | 0.775     | 0.875       | 0.872      | 24.314      |
| 4                                              | $0.376\alpha\omega^2 e^{-\cos[t \cos\{\sin(\omega t)\}]} \sin(0.427x_1) \times \sin\{\sin[0.376\omega t \sin(e^{1/x_1} \sin[\sin[\cos(x_1^{-1})]])]\} \cos\{\sin(\omega t)\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.820      | 0.762     | 0.820       | 0.875      | 18.661      |

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Table 29: **gplearn** results for the Feynman equation I\_50\_26 (**HOQN**):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5    | $\alpha + \left[ \{0.263\omega + 0.263t + 0.263 \cos^2(0.263\omega + 0.263t + 0.263 \sin[\cos(0.572\omega^2)])\} \right.$ $+ 0.263 \cos\{\cos[\cos(0.572\omega^2 \cos t)]\} - 0.263x_1^{-1}\}^3$ $+ \sin(\omega^2 \cos t) - \cos t - \cos^2(0.572x_1) - \cos(0.572\omega^2 \cos t)$ $- \cos^2\{0.572 \cos[(\omega + \cos\{(0.263\omega + 0.263t + 0.263 \sin(\omega^2 \cos t) - 0.076)^3\})\}$ $\times (-0.150\omega^2 \cos t + 0.263t - 0.076) \cos(0.572\omega^2 \cos t)] + 0.572x_1^{-1}\}$ $\times \sin\{\sin(\omega^2 \cos t)\} - \sin x_1 - \cos(0.572\omega^2 \cos t)$ | 1.062      | 1.071     | 0.693       | 0.702      | 22.754      |

Table 30: **operon** results for the Feynman equation I\_50\_26 (**HOQN**):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                       | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                          |            |           |             |            |             |
| 1                                              | $0.232\omega t \left[ 0.726\alpha x_1 \cos\{\sin(2.580\omega)\} \right.$ $+ 1.011x_1 \sin(4.924t) \sin\{\sin(3.889\omega)\} \left. \right] - 0.500$                      | 1.117      | 1.076     | 0.692       | 0.710      | 30.061      |
| 2                                              | $-0.972 \left[ 0.248\omega t (-0.693\alpha - 0.910x_1) + 1.537 \right]$ $\times \left[ 0.224\alpha + 0.118x_1 - \sin(1.767\omega t) \right] + 0.372$                     | 0.520      | 0.580     | 0.937       | 0.914      | 22.804      |
| 3                                              | $0.003\omega^2 (2.752\alpha + 1.266tx_1)^2$ $\times \left[ 0.211t \sin(1.773\omega t) - \sin(0.512\omega + 0.729) \right]^2 - 0.253$                                     | 0.639      | 0.657     | 0.895       | 0.887      | 25.191      |
| 4                                              | $0.134\alpha x_1 (1.540 - 1.160\omega t)$ $\times \left[ -\sin\{1.094\omega (0.157 - 1.684t)\} - 0.994 \right] - 0.165$                                                  | 0.531      | 0.606     | 0.929       | 0.908      | 22.250      |
| 5                                              | $-0.049\alpha tx_1 \left[ 1.613\omega t \sin(1.773\omega t) - 5.275\omega + 0.792x_1 \right] - 0.527$                                                                    | 0.604      | 0.549     | 0.914       | 0.915      | 23.483      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                          |            |           |             |            |             |
| 1                                              | $-0.442\omega t \left[ -0.302\alpha x_1 - \sin(1.837\omega t - 28.722) \right] - 0.692$                                                                                  | 0.804      | 0.816     | 0.825       | 0.844      | 24.823      |
| 2                                              | $1.297 \left[ 0.653\alpha + \sin\{\sin(1.223\omega t)\} \right]$ $\times \left[ 0.277\omega x_1 + \sin\{\sin(2.304\omega t)\} \right] - 0.009$                           | 0.769      | 0.704     | 0.856       | 0.847      | 29.172      |
| 3                                              | $4.241 \left[ 0.236\alpha x_1 + \frac{0.233x_1}{\cos(1.013\omega t)} \right] \cos^2(1.013\omega t) - 0.007$                                                              | 0.208      | 0.205     | 0.989       | 0.992      | 24.984      |
| 4                                              | $2.148 \left[ 0.761\alpha \sin\{\sin[\sin(0.037\omega^2 t^2)]\} - 0.047x_1 \right]$ $\times \left[ -0.693x_1 \sin(1.748\omega t) + 0.698x_1 \right] - 0.224$             | 0.483      | 0.458     | 0.940       | 0.936      | 25.884      |
| 5                                              | $0.169x_1 \left[ 5.934\alpha + \frac{5.908}{\cos(1.006\omega t)} \right] \cos^2(1.003\omega t) - 0.011$                                                                  | 0.155      | 0.143     | 0.994       | 0.995      | 26.387      |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                                                                                          |            |           |             |            |             |
| 1                                              | $2.183x_1 \left[ 0.434\omega - e^{-0.681\alpha} \right] \left[ 0.441t - e^{-0.681\alpha} \right]$ $\times \left[ e^{-1.557\alpha} + \cos(2.018\omega t) \right] - 0.173$ | 0.633      | 0.591     | 0.901       | 0.900      | 29.021      |
| 2                                              | $4.712 \left[ 0.306\omega x_1 \cos(1.230t) + \sin(1.091t) \right]$ $\times \sin^2\{0.449t^2 \cos^2(0.979\omega)\} \cos(1.299t) + 0.853$                                  | 1.259      | 1.324     | 0.596       | 0.617      | 29.952      |

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Table 30: **operon** results for the Feynman equation I.50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | $0.576\alpha\omega x_1 [0.126t + \cos(1.016\omega t)] \times \sin(0.379t) \cos(1.016\omega t) - 0.075$                               | 0.511      | 0.503     | 0.932       | 0.946      | 28.517      |
| 4    | $0.120\alpha\omega t x_1 e^{-\sin\{\sin(1.757\omega t)\}} \sin(0.309\omega t) - 0.511$                                               | 0.625      | 0.631     | 0.895       | 0.914      | 28.719      |
| 5    | $-0.774x_1 \left[ 1.383\alpha - \frac{1.481\alpha}{\cos\{\cos(1.003\omega t)\}} + 0.354 \right] + 1.954 \cos(1.001\omega t) + 0.534$ | 0.458      | 0.447     | 0.943       | 0.948      | 25.782      |

Table 31: **PySR** results for the Feynman equation I.50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                    | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                       |            |           |             |            |             |
| 1                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.000      | 0.000     | 1.000       | 1.000      | 100.024     |
| 2                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.000      | 0.000     | 1.000       | 1.000      | 120.296     |
| 3                                              | $x_1 \cos(\omega t) \{\alpha \cos(\omega t) + 1.000\}$                                                | 0.000      | 0.000     | 1.000       | 1.000      | 127.832     |
| 4                                              | $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                    | 0.000      | 0.000     | 1.000       | 1.000      | 108.169     |
| 5                                              | $0.247 \left[ \{x_1 + \alpha\} \{\cos(\omega t) + 0.119\} + \alpha^{-1} \right]^2 - 0.310$            | 0.219      | 0.229     | 0.989       | 0.985      | 87.682      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                       |            |           |             |            |             |
| 1                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.149      | 0.142     | 0.994       | 0.995      | 99.897      |
| 2                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.153      | 0.146     | 0.994       | 0.993      | 102.262     |
| 3                                              | $x_1 \cos(\omega t) \{\alpha \cos(\omega t) + 0.999\}$                                                | 0.152      | 0.156     | 0.994       | 0.995      | 105.729     |
| 4                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.152      | 0.148     | 0.994       | 0.993      | 107.485     |
| 5                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.149      | 0.139     | 0.994       | 0.995      | 101.043     |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                       |            |           |             |            |             |
| 1                                              | $0.242 \left[ \{x_1 + \alpha\} \left\{ \frac{1}{2\alpha} + \cos(\omega t) \right\} \right]^2 - 0.283$ | 0.258      | 0.246     | 0.983       | 0.983      | 92.106      |
| 2                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.181      | 0.173     | 0.992       | 0.993      | 98.596      |
| 3                                              | $x_1 \cos(\omega t) \{\alpha \cos(\omega t) + 1.003\}$                                                | 0.181      | 0.179     | 0.991       | 0.993      | 119.619     |
| 4                                              | $x_1 \{\cos(\omega t) + \alpha \cos^2(\omega t)\}$                                                    | 0.178      | 0.190     | 0.992       | 0.992      | 99.745      |
| 5                                              | $x_1 \left[ \{\cos(\omega t) - 0.232\} (0.385\alpha + 0.628) + 0.687 \right]^2 - 0.139$               | 0.210      | 0.208     | 0.988       | 0.989      | 91.471      |

Table 32: BMS results for the Feynman equation I.50.26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS.

| Rep.                                           | Learned expression                                                                                                                                                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $-\omega^2 a_0^2 (x_1 + a_0 - x_1 t) \{ \alpha a_0 + \alpha \cos[\sin(\omega t)] \}$                                                                                                                             | 0.785      | 0.827     | 0.848       | 0.829      | 268.458     |
| 2                                              | $\{2\alpha + \omega t + \sin \alpha + x_1 + a_0^3\} \cos[\sin(\omega t)] + 2a_0$                                                                                                                                 | 0.999      | 1.053     | 0.767       | 0.717      | 280.043     |
| 3                                              | $\sin(\omega [a_0 t^2 + a_0 \sin(2a_0)(t + a_0)^3 (\alpha + \omega + x_1) \times \cos\{-2\omega t\}]) \{a_0 + \sin(\alpha + 2\omega) + \alpha x_1 + x_1\}$                                                       | 0.857      | 0.802     | 0.810       | 0.832      | 231.521     |
| 4                                              | $\frac{a_0^2 \sin(a_0) \omega t x_1}{\times \{-\cos \alpha + \sin(2\omega t + a_0 x_1 + a_0^2) + \sin(a_0^3) + a_0\}}$                                                                                           | 0.978      | 0.970     | 0.759       | 0.764      | 281.771     |
| 5                                              | $\exp[a_0^2 \{3\omega\} \cos(\omega t) \cos t \sin(\omega + t) \times \{x_1 + a_0 [\sin(\sin a_0) + a_0] [\alpha(\omega^2 + \alpha + a_0) + \omega]\}]^3$                                                        | 1.089      | 1.122     | 0.719       | 0.647      | 308.486     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $a_0(t + \omega) \left[ -a_0^2(t^2 + \omega x_1 \alpha + a_0 \omega x_1 \sin\{a_0 \alpha(t^2 - x_1)\}) \times \cos(\omega t) \right]^2$                                                                          | 0.739      | 0.701     | 0.852       | 0.885      | 286.765     |
| 2                                              | $\omega [tx_1 \{ \cos(a_0^3) + \sin(2a_0) \} \{ \cos(\omega t) + a_0 + \cos(2\omega t) + a_0^3 + \alpha \} + \alpha \exp\{-\exp[a_0 + \sin(a_0) + \exp(t + a_0)]\}]$                                             | 0.693      | 0.641     | 0.883       | 0.873      | 326.618     |
| 3                                              | $[\{t + \cos[\sin(\omega t)](\alpha + \omega + \alpha + t) + x_1 + \omega\} \sin a_0 + a_0]^2$                                                                                                                   | 0.819      | 0.893     | 0.834       | 0.841      | 302.188     |
| 4                                              | $\omega a_0(\alpha + t) \left[ \alpha + x_1 \{ \sin a_0 + \sin[\sin(a_0^3)] + 2\alpha \} + \omega e^t \{ a_0 \omega e^{x_1} \cos(\omega t) + a_0^2 e^{-a_0} \} \right]$                                          | 1.242      | 1.250     | 0.602       | 0.524      | 247.684     |
| 5                                              | $-\{ \omega + a_0 \omega (x_1 + \omega t) \{ a_0 + \cos(2\omega t) \} + a_0 \sin(-\alpha)(2\alpha) \} \times \cos[\sin\{\sin(\omega t)\}] \alpha (-a_0 x_1 t)$                                                   | 0.764      | 0.808     | 0.852       | 0.836      | 309.985     |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $a_0 x_1 (t + \alpha) \left[ a_0 + \omega(\omega + a_0)(\alpha - a_0 x_1) \right] \times [a_0 \{ a_0^3 \alpha a_0 - \omega \alpha t^3 \} \cos^2\{\cos(\omega t)\} + t]$                                          | 1.273      | 1.239     | 0.597       | 0.559      | 290.545     |
| 2                                              | $a_0 \alpha \omega t x_1 \{ a_0 + \cos(2\omega t) + \cos[a_0 + \sin(\omega + \cos(-x_1 \sin(a_0 + a_0 \cos a_0 + a_0 x_1 + t)))] \}$                                                                             | 0.727      | 0.657     | 0.865       | 0.906      | 331.377     |
| 3                                              | $\left[ a_0 \cos(e^{\sin t}) + \{ \cos(\omega t) + a_0 \} e^{\cos(a_0 x_1)} \right] \times x_1 (a_0 + a_0 \sin a_0) a_0 \{ [\cos(-\omega \sin a_0) + \cos a_0] \times [\sin x_1 + \alpha] + \omega \} + a_0 ]^2$ | 0.521      | 0.521     | 0.930       | 0.942      | 317.026     |
| 4                                              | $\left[ x_1 a_0 \{ a_0 e^{\cos[\omega(t + a_0 x_1)]} e^{\omega(\sin a_0 + a_0)(a_0 + e^t)} \}^3 + \alpha - a_0 \right] \times \omega t a_0 \{ (\cos t + x_1) \alpha + t \}$                                      | 1.139      | 0.956     | 0.652       | 0.803      | 269.761     |
| 5                                              | $x_1 \sin^2(a_0) \left[ \sin(\cos \omega) + \{ \alpha + a_0 + \alpha \omega^2 (2a_0 \cos(\omega t))^2 \} t \right] + a_0 x_1$                                                                                    | 0.774      | 0.854     | 0.837       | 0.811      | 291.594     |



Table 33: BSR results for the Feynman equation I\_50.26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                      | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $-1.947 + 0.754e^{\cos x_1} - 0.055e^{x_1} - 0.017 \sin\{(t + \omega)^3\} + 0.562x_1(1 + t)$                                                                                            | 1.842      | 1.883     | 0.163       | 0.112      | 191.627     |
| 2                                              | $35.478 - 37.326 \cos(e^{-t}) - 0.388 \cos \alpha - 2.534 \cos t + 0.457\alpha x_1$                                                                                                     | 1.734      | 1.727     | 0.299       | 0.239      | 190.812     |
| 3                                              | $-0.554 + 0.051e^{x_1} + 0.111e^\omega + 0.001 \frac{\cos x_1}{\sin(x_1^2)} + 0.309x_1$                                                                                                 | 1.837      | 1.853     | 0.127       | 0.101      | 193.801     |
| 4                                              | $-7.632 + 2.206\{t + e^{\cos t}\} + 0.089e^{x_1} + 1.114\omega - 0.213 \sin(x_1^2)$                                                                                                     | 1.759      | 1.863     | 0.220       | 0.131      | 190.078     |
| 5                                              | $-1.243 + 0.698\{\cos(\omega^{-1}) + \omega + \alpha\} - 0.315\omega^{-1} - 0.986\alpha^{-1} + 0.002(\alpha\omega)^3$                                                                   | 1.806      | 1.676     | 0.229       | 0.213      | 191.675     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $-2.506 + 1.531\alpha + 0.871 \frac{x_1}{\alpha} - 0.303 \sin(t^2) + 0.021 \sin^{-2}(x_1)$                                                                                              | 1.775      | 1.960     | 0.146       | 0.102      | 193.315     |
| 2                                              | $3.383 + 0.669 \cos(t^2) + 3.919e^{-\alpha} + 2.082 \cos(\sin \alpha) - 2.159e^{1/\alpha}$                                                                                              | 1.886      | 1.719     | 0.134       | 0.085      | 191.531     |
| 3                                              | $4.018 - 1.416e^{\sin t} - 1.264 \cos \alpha - 0.447 \sin(t^2) - 0.969 \cos x_1$                                                                                                        | 1.711      | 1.962     | 0.275       | 0.231      | 191.691     |
| 4                                              | $0.553 + 0.065\alpha e^\omega - 1.222 \sin(\sin \alpha) - 0.080 \sin^{-1}(x_1) + 0.093e^{x_1}$                                                                                          | 1.747      | 1.677     | 0.214       | 0.143      | 191.321     |
| 5                                              | $-2.954 - 0.051 \cos \omega - 0.922x_1^{-1} - 0.002(\omega + t) + 0.502(x_1 + \omega t + 2\alpha)$                                                                                      | 1.621      | 1.622     | 0.336       | 0.340      | 190.397     |
| <i>Noise level: <math>\sigma = 0.18</math></i> |                                                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $2.020 \times 10^{-8} + 7.707 \times 10^{-6} e^{\omega\alpha} - 2.830 \times 10^{-8} (-tx_1)^9 + 5.019 \times 10^{-7} e^{\omega t/\alpha} - 5.383 \times 10^{-8} \{\cos x_1 - \alpha\}$ | 2.357      | 2.195     | -0.379      | -0.384     | 190.049     |
| 2                                              | $1.025 \times 10^{-8} + 6.823 \times 10^{-9} \sin \omega - 5.033 \times 10^{-9} \cos x_1 + 9.683 \times 10^{-9} e^{e^\alpha} + 4.875 \times 10^{-8} t\alpha$                            | 2.449      | 2.642     | -0.527      | -0.526     | 191.367     |
| 3                                              | $3.473 - 1.635x_1^{-1} - 0.024 \sin(-t^3) - 1.294 \cos(\cos x_1) - 0.054 \sin\{\cos(x_1^2)\}$                                                                                           | 1.915      | 2.082     | 0.050       | 0.080      | 192.903     |
| 4                                              | $-0.640 + 1.373 \cos(\cos x_1) - 0.346 \cos^2\{\sin(\omega^2)\} + 0.085x_1^3 + 0.573 \sin^2\{\cos t + \omega\}$                                                                         | 1.871      | 2.064     | 0.063       | 0.083      | 204.162     |
| 5                                              | $0.116 - 1.696\alpha^{-1} + 0.965t^2 + 0.037e^{\cos(\alpha x_1) + \alpha} - 0.227t^3$                                                                                                   | 1.706      | 1.770     | 0.209       | 0.186      | 191.171     |

### C.5 Results for II\_36\_38 (FMMM)

Table 34: DSR results for the Feynman equation II\_36\_38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                    |            |           |             |            |             |
| 1                                              | $T^{-1} \exp\{k_b^{-1} + \exp[-\text{mom}^{-1}]\}$                                 | 0.731      | 0.842     | 0.561       | 0.537      | 41.900      |
| 2                                              | $\exp\{\sin[\text{mom}(Tk_b)^{-1}]\}$                                              | 0.794      | 0.731     | 0.456       | 0.450      | 40.600      |
| 3                                              | $\text{mom}\{T^{-1} + \sin[(k_b^2)^{-1}]\}$                                        | 0.668      | 0.650     | 0.625       | 0.635      | 41.300      |
| 4                                              | $(\alpha + \text{mom} \text{mom})k_b^{-1} T^{-1}$                                  | 0.710      | 0.678     | 0.570       | 0.580      | 41.100      |
| 5                                              | $H\{k_b \cos(T^{-1})\}^{-1}$                                                       | 0.949      | 0.808     | 0.417       | 0.450      | 42.200      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                    |            |           |             |            |             |
| 1                                              | $\sin(T^{-1})\{H + \text{mom} \sin(k_b^{-1})\}$                                    | 0.778      | 0.857     | 0.544       | 0.476      | 46.500      |
| 2                                              | $\{\text{mom}^{-1}\{k_b^{-1} \exp[-(-T)^{-1}]\}^{-1}\}^{-1}$                       | 0.710      | 0.699     | 0.566       | 0.582      | 58.100      |
| 3                                              | $\{\text{mom}^2 + \sin(\text{mom})\}(k_b T)^{-1}$                                  | 0.655      | 0.634     | 0.645       | 0.653      | 50.600      |
| 4                                              | $-[-\{k_b^{-1} + \exp(\sin[(-\exp\{(\exp k_b)^{-1})\}^{-1})\epsilon\} \text{mom}]$ | 0.837      | 0.877     | 0.415       | 0.394      | 84.700      |
| 5                                              | $\text{mom} + \cos[-\{(-k_b^3)^{-1} + T\}]$                                        | 0.851      | 0.739     | 0.477       | 0.538      | 56.700      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                    |            |           |             |            |             |
| 1                                              | $(\alpha + c + \text{mom}^2)(T + k_b T)^{-1}$                                      | 0.714      | 0.677     | 0.590       | 0.648      | 57.800      |
| 2                                              | $\text{mom} - \{k_b H^{-1} - T^{-1}\}$                                             | 0.750      | 0.759     | 0.516       | 0.524      | 82.300      |
| 3                                              | $\exp(k_b^{-1})T^{-1} \text{mom}$                                                  | 0.721      | 0.730     | 0.591       | 0.615      | 54.600      |
| 4                                              | $\text{mom} + \sin(T + k_b)$                                                       | 0.840      | 0.863     | 0.466       | 0.433      | 71.800      |
| 5                                              | $\exp\{\text{mom}(k_b T)^{-1}\}$                                                   | 0.718      | 0.732     | 0.579       | 0.580      | 63.400      |

Table 35: QLattice results for the Feynman equation II\_36\_38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                                                                                           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                                              |            |           |             |            |             |
| 1                        | $0.207\alpha + 0.257[0.648H + (0.366T - 2.270)(0.065\epsilon - 0.701) \\ \times \{-0.869T - 0.578c - 1.064k_b + 0.897\text{mom} + (0.868 - 0.162\epsilon) \\ \times (0.483M - 0.091) + 2.964\} + 0.238]^2 + 0.132$           | 0.270      | 0.262     | 0.940       | 0.955      | 89.144      |
| 2                        | $0.662 \exp[(0.527 - 0.058\text{mom})(0.065\epsilon - 1.061) \\ \times \{-0.395T + (1.643 - 0.424k_b)(-0.077M - 0.555) \\ \times (-0.141\alpha - 1.002) + 1.349\}][(-0.471H + 0.373c - 1.003\text{mom} \\ - 0.293) - 0.725]$ | 0.278      | 0.342     | 0.933       | 0.879      | 116.290     |

Continued on next page

Table 35: Q<sub>Lattice</sub> results for the Feynman equation II.36-38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | $13.534(1.128 - 0.227T)(2.260 - 0.422T)$<br>$\times \exp[0.151M + (0.134k_b + 0.226)(0.879H + 0.364\alpha - 0.709c$<br>$- 0.388\epsilon + 1.255\text{mom} - 8.372)] + 0.239$                                                       | 0.274      | 0.299     | 0.937       | 0.923      | 133.543     |
| 4                                              | $0.966 \exp[-0.171\epsilon - 0.585k_b + 0.494\text{mom} + (1.393 - 0.013T)$<br>$\times (0.253H - 0.448) + 0.223(-0.391M - 0.187)$<br>$\times (1.435c - 4.172) + 2.201(0.283T - 1)^2]$<br>$+ 0.038$                                 | 0.277      | 0.263     | 0.934       | 0.937      | 108.359     |
| 5                                              | $-3.622(0.056\alpha + 0.209)(0.480k_b - 1.923)(0.167\text{mom} - 0.040)$<br>$\times [-0.520k_b + (0.672T - 2.312)\{1.485T + (0.881M - 0.402)$<br>$\times (0.669c - 1.872) - 2.753\} + 2.860] \exp(0.383H - 0.191\epsilon) + 0.250$ | 0.312      | 0.264     | 0.937       | 0.941      | 121.090     |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                    |            |           |             |            |             |
| 1                                              | $-0.195k_b - 1.809[-0.715H + (0.124 - 0.546\alpha)(1.655 - 0.579c)$<br>$\times (0.553M - 0.110) + 0.207]$<br>$\times (-0.680\epsilon + 2.558\text{mom} + 0.449) \exp(-0.653T) \exp(-0.537k_b) + 0.649$                             | 0.284      | 0.347     | 0.939       | 0.914      | 112.151     |
| 2                                              | $6.936 \exp[0.170\alpha + (0.173M - 0.176\epsilon - 1.618)$<br>$\times \{-0.247H + 0.655T + (0.074T + 0.201)(-0.711T + 1.289k_b$<br>$- 0.911) + (-0.110c - 0.202)(0.897\text{mom} - 3.770) - 0.341\}] + 0.280$                     | 0.309      | 0.291     | 0.918       | 0.928      | 118.903     |
| 3                                              | $1.498(2.233 - 0.538k_b)(0.130T - 0.498)$<br>$\times (0.141\epsilon - 1.200)(1.421\text{mom} - 0.068)$<br>$\times [0.566H + (0.234 - 0.045k_b)(3.692 - 1.241c)$<br>$\times (1.074M + 0.961\alpha - 1.967) + 0.011] + 0.012$        | 0.307      | 0.310     | 0.922       | 0.917      | 110.390     |
| 4                                              | $2.162(0.055H + 0.055)(0.401k_b - 1.489)(-0.630T + 1.209\text{mom} + 1.071)$<br>$\times [0.901T - 0.171\epsilon + 2.238(1.291c - 3.960)(0.421M + 0.412\alpha - 0.628)$<br>$\times \exp(-0.699\epsilon) - 3.276] + 0.107$           | 0.342      | 0.341     | 0.902       | 0.909      | 104.237     |
| 5                                              | $3.706[-0.173H - 0.074M - 0.069\alpha + 0.078\epsilon$<br>$+ 0.635(0.104 - 0.340\text{mom})(1.711 - 0.346c)$<br>$+ 0.635(3.327 - 0.741k_b)(0.201T - 0.920) + 1]^2 + 0.577$                                                         | 0.354      | 0.317     | 0.910       | 0.915      | 105.278     |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                                                    |            |           |             |            |             |
| 1                                              | $5.648[-0.136H + 0.215T + 0.117c + 0.210k_b$<br>$+ 0.349(-0.088M - 0.260)\{1.171\text{mom} + (2.583 - 0.729\epsilon)$<br>$\times (0.307\alpha - 0.088) - 1.731\} - 1]^2 + 0.566$                                                   | 0.348      | 0.316     | 0.903       | 0.923      | 162.774     |
| 2                                              | $4.933(1 - 0.066\epsilon)^2(0.147H - 2.571)(1.322\text{mom} - 0.333)$<br>$\times (-0.800H - 0.260\alpha + 0.477) \exp(-0.337c) \exp(-0.666T - 0.646k_b)$<br>$+ 0.212$                                                              | 0.339      | 0.308     | 0.901       | 0.922      | 129.613     |
| 3                                              | $28.951 \exp[0.145M - 0.641T + (0.847 - 0.110H)$<br>$\times (-0.293\epsilon - 1.027k_b + 0.894\text{mom} - 3.160)$<br>$+ (-0.171\alpha - 0.065)(0.718c - 2.117)] + 0.156$                                                          | 0.308      | 0.344     | 0.925       | 0.915      | 121.915     |
| 4                                              | $-11.655(0.108 - 0.398\text{mom})(0.320 - 0.067c)(0.154M + 0.596)$<br>$\times (0.198\alpha + 0.693)(0.300\epsilon - 1.940)(0.718k_b - 3.043)$<br>$\times (-0.149H + 0.243T + 0.093k_b - 1)^2 + 0.381$                              | 0.385      | 0.429     | 0.888       | 0.860      | 93.531      |

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Table 35: **QLattice** results for the Feynman equation II.36-38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                         | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5    | $-26.832(-1.192H - 0.810)$ $\times \exp[-0.572T - 0.276c - 0.150\epsilon - 0.586k_b$ $+(1.373 - 0.347\text{mom})(0.209M - 1.849)] + 0.033$ | 0.340      | 0.352     | 0.905       | 0.903      | 95.179      |

Table 36: **SISSO++** results for the Feynman equation II.36-38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                            | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                               |            |           |             |            |             |
| 1                                              | $-0.487$ $+ (0.374) \{(M \text{ mom})/(c\epsilon)\}$ $+ (1.450) \{(\alpha/k_b)/(cT)\}$ $+ (0.995) \{(H \text{ mom})/(Tk_b)\}$ | 0.197      | 0.207     | 0.970       | 0.961      | 17.056      |
| 2                                              | $-0.459$ $+ (0.368) \{(\alpha \text{ mom})/(c\epsilon)\}$ $+ (1.400) \{(M/k_b)/(cT)\}$ $+ (0.993) \{(H \text{ mom})/(Tk_b)\}$ | 0.173      | 0.173     | 0.975       | 0.971      | 17.246      |
| 3                                              | $-0.467$ $+ (0.385) \{(M\alpha)/(c\epsilon)\}$ $+ (0.959) \{(H \text{ mom})/(Tk_b)\}$ $+ (1.492) \{(\text{mom}/T)/(ck_b)\}$   | 0.194      | 0.172     | 0.972       | 0.974      | 17.072      |
| 4                                              | $-0.451$ $+ (0.381) \{(M\alpha)/(c\epsilon)\}$ $+ (0.958) \{(H \text{ mom})/(Tk_b)\}$ $+ (1.445) \{(\text{mom}/c)/(Tk_b)\}$   | 0.195      | 0.226     | 0.966       | 0.965      | 17.246      |
| 5                                              | $-0.480$ $+ (0.394) \{(M\alpha)/(c\epsilon)\}$ $+ (0.961) \{(H \text{ mom})/(Tk_b)\}$ $+ (1.501) \{(\text{mom}/T)/(ck_b)\}$   | 0.192      | 0.197     | 0.970       | 0.971      | 17.013      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                               |            |           |             |            |             |
| 1                                              | $-0.493$ $+ (0.376) \{(M \text{ mom})/(c\epsilon)\}$ $+ (1.439) \{(\alpha/k_b)/(cT)\}$ $+ (1.001) \{(H \text{ mom})/(Tk_b)\}$ | 0.248      | 0.258     | 0.953       | 0.942      | 16.934      |
| 2                                              | $-0.466$ $+ (0.372) \{(\alpha \text{ mom})/(c\epsilon)\}$ $+ (1.418) \{(M/c)/(Tk_b)\}$ $+ (0.993) \{(H \text{ mom})/(Tk_b)\}$ | 0.227      | 0.229     | 0.958       | 0.951      | 17.099      |
| 3                                              | $-0.465$ $+ (0.388) \{(M\alpha)/(c\epsilon)\}$ $+ (0.959) \{(H \text{ mom})/(Tk_b)\}$ $+ (1.498) \{(\text{mom}/c)/(Tk_b)\}$   | 0.243      | 0.235     | 0.956       | 0.953      | 16.735      |

Continued on next page

Table 36: **SISSO++** results for the Feynman equation II.36.38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|--------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4                                              | − 0.464                                          |            |           |             |            |             |
|                                                | + (0.384) $\{(M\alpha)/(c\epsilon)\}$            |            |           |             |            |             |
|                                                | + (0.960) $\{(H \text{ mom})/(Tk_b)\}$           |            |           |             |            |             |
|                                                | + (1.465) $\{(\text{mom}/k_b)/(cT)\}$            | 0.242      | 0.265     | 0.949       | 0.952      | 16.932      |
| 5                                              | − 0.483                                          |            |           |             |            |             |
|                                                | + (0.396) $\{(M\alpha)/(c\epsilon)\}$            |            |           |             |            |             |
|                                                | + (0.959) $\{(H \text{ mom})/(Tk_b)\}$           |            |           |             |            |             |
|                                                | + (1.513) $\{(\text{mom}/c)/(Tk_b)\}$            | 0.244      | 0.245     | 0.952       | 0.955      | 16.868      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                  |            |           |             |            |             |
| 1                                              | − 0.495                                          |            |           |             |            |             |
|                                                | + (0.376) $\{(M \text{ mom})/(c\epsilon)\}$      |            |           |             |            |             |
|                                                | + (1.435) $\{(\alpha/c)/(Tk_b)\}$                |            |           |             |            |             |
|                                                | + (1.003) $\{(H \text{ mom})/(Tk_b)\}$           | 0.281      | 0.290     | 0.941       | 0.927      | 16.772      |
| 2                                              | − 0.469                                          |            |           |             |            |             |
|                                                | + (0.373) $\{(\alpha \text{ mom})/(c\epsilon)\}$ |            |           |             |            |             |
|                                                | + (1.424) $\{(M/c)/(Tk_b)\}$                     |            |           |             |            |             |
|                                                | + (0.994) $\{(H \text{ mom})/(Tk_b)\}$           | 0.262      | 0.263     | 0.945       | 0.936      | 16.868      |
| 3                                              | − 0.464                                          |            |           |             |            |             |
|                                                | + (0.389) $\{(M\alpha)/(c\epsilon)\}$            |            |           |             |            |             |
|                                                | + (0.959) $\{(H \text{ mom})/(Tk_b)\}$           |            |           |             |            |             |
|                                                | + (1.499) $\{(\text{mom}/k_b)/(cT)\}$            | 0.276      | 0.272     | 0.944       | 0.939      | 16.968      |
| 4                                              | − 0.469                                          |            |           |             |            |             |
|                                                | + (0.385) $\{(M\alpha)/(c\epsilon)\}$            |            |           |             |            |             |
|                                                | + (0.961) $\{(H \text{ mom})/(Tk_b)\}$           |            |           |             |            |             |
|                                                | + (1.471) $\{(\text{mom}/c)/(Tk_b)\}$            | 0.275      | 0.294     | 0.936       | 0.942      | 16.960      |
| 5                                              | − 0.484                                          |            |           |             |            |             |
|                                                | + (0.397) $\{(M\alpha)/(c\epsilon)\}$            |            |           |             |            |             |
|                                                | + (0.959) $\{(H \text{ mom})/(Tk_b)\}$           |            |           |             |            |             |
|                                                | + (1.517) $\{(\text{mom}/T)/(ck_b)\}$            | 0.278      | 0.278     | 0.939       | 0.943      | 16.558      |

Table 37: **gplearn** results for the Feynman equation II.36.38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                                                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                   |            |           |             |            |             |
| 1                        | $\{\text{mom} + \sin(e^{\sin(\cos^3 k_b)}) + \cos(k_b) - \cos[H + \sin(\cos c)]\}$<br>$\times \sin(e^{\cos T})$                                                                                   | 0.497      | 0.550     | 0.797       | 0.802      | 19.230      |
| 2                        | $\frac{H}{T} + \sin(k_b^{-4})$<br>$+ \frac{-\frac{H}{T} - \sin(k_b^{-2}) + k_b^{-2} e^{\sin(\cos T)} + k_b^{-1} \cos(\text{mom}) \cos(\text{mom})}{k_b}$<br>$\times \frac{\cos(\text{mom})}{k_b}$ | 0.485      | 0.558     | 0.797       | 0.679      | 20.785      |

*Continued on next page*

Table 37: **gplearn** results for the Feynman equation II\_36.38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                                                      | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | $T^{-1} \left[ \{ \text{mom} k_b^{-1} + (0.286 + \text{mom} + \cos k_b + \cos(H/k_b) \cos(0.286 + k_b + H/k_b)) \cos(\sin H) \} \cos(\sin H) + 0.286 + \text{mom} k_b^{-1} \right]$                                                                                                                                     | 0.484      | 0.464     | 0.803       | 0.814      | 19.168      |
| 4                                              | $\frac{H \text{mom}}{T k_b^2} + \frac{H}{T c k_b} + \frac{\text{mom}}{T c}$                                                                                                                                                                                                                                             | 0.360      | 0.324     | 0.889       | 0.904      | 19.868      |
| 5                                              | $\text{mom} + \cos(k_b) - \cos\{H + \cos(T)\}$                                                                                                                                                                                                                                                                          | 0.760      | 0.615     | 0.626       | 0.682      | 17.906      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $c^{-1} + T^{-1} \left[ \frac{H(\text{mom} - 0.859)}{k_b} + T^{-1} \left\{ \frac{H \text{mom} \cos(k_b^{-1})}{k_b^2} - 0.859 + \frac{2}{k_b} \right\} \right]$                                                                                                                                                          | 0.443      | 0.590     | 0.852       | 0.752      | 18.665      |
| 2                                              | $\cos\{\sin[\sin A_1]\} - \frac{c - \text{mom} + k_b^{-1}(T - \text{mom} - 1.799 \sin k_b) \sin k_b}{k_b} + T^{-1} - H^{-3}$<br>$A_3 = T - \text{mom} - 1.799 k_b^{-1} \sin k_b,$<br>$A_2 = \cos(1.799/k_b) - \text{mom}^{-3} - k_b^{-1} A_3 + T^{-1},$<br>$A_1 = \cos\{\sin(e^{\sin A_2})\} + 1.799 k_b^{-1} + T^{-1}$ | 0.532      | 0.544     | 0.757       | 0.747      | 22.552      |
| 3                                              | $\text{mom} \left[ \sin^2(e^{\cos T}) + \sin^2 \left( e^{\cos\{\sin^2(e^{\cos(k_b^{-1})})\}} \right) \right] \cos(H^{-1})$                                                                                                                                                                                              | 0.556      | 0.581     | 0.745       | 0.709      | 21.557      |
| 4                                              | $\frac{H}{T} + k_b^{-1} \left[ - \frac{0.588 k_b}{\text{mom} + \exp\{-2H \sin(k_b^{-1}) - \cos T\} - 0.635} + k_b^{-1} \left\{ \cos T + c^{-1} + T^{-1}(-0.480c + 3\text{mom} - 1.905) \right\} + c^{-1} \right]$                                                                                                       | 0.444      | 0.557     | 0.836       | 0.756      | 21.991      |
| 5                                              | $(H - T + \text{mom}) \sin\{\sin(k_b^{-1})\} + T^{-1} \left\{ \frac{H \text{mom}}{T c} + \frac{\text{mom}}{c} \right\} \sin(k_b^{-1})$                                                                                                                                                                                  | 0.443      | 0.381     | 0.858       | 0.877      | 17.420      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                                                                                                                                         |            |           |             |            |             |
| 1                                              | $\frac{\text{mom}}{T} \left[ \frac{0.605 \sin\{e^{1/k_b} \sin(M/k_b)\}}{k_b} + \frac{1}{c} \right] e^{1/k_b}$                                                                                                                                                                                                           | 0.552      | 0.469     | 0.755       | 0.831      | 27.758      |
| 2                                              | $c^{-1} \left[ k_b e^{-k_b} \left\{ e^{\text{mom}/T} + \frac{\text{mom}(M + 0.175) \sin(\epsilon^{-1})}{k_b^2} \right\} + \frac{\text{mom}\{-\cos H + c^{-1}\} \sin(\epsilon^{-1})}{k_b^2} \right] + \frac{-\cos H + \text{mom}/k_b}{T}$                                                                                | 0.385      | 0.346     | 0.873       | 0.901      | 22.410      |
| 3                                              | $\frac{\text{mom}/k_b + (0.446M - \cos \text{mom})/k_b}{T} + \frac{H/k_b + \text{mom}/k_b}{T c}$                                                                                                                                                                                                                        | 0.470      | 0.447     | 0.826       | 0.855      | 17.045      |
| 4                                              | $\frac{H}{T k_b} + T^{-1} \left[ \frac{1.193\{\text{mom} + c^{-2}(\text{mom} + \cos T)\}}{k_b} + c^{-2} \right]$                                                                                                                                                                                                        | 0.514      | 0.568     | 0.787       | 0.821      | 19.553      |
| 5                                              | $\frac{H}{T} \left[ \cos(k_b) + \cos\{\cos^3[\cos(\cos\{\text{mom} + \cos k_b + \cos(T^{-1}\{\cos(T^{-1}) + \cos k_b + 0.633\}) + 0.633 + T^{-3}\})]\} + 0.633 \right] - \cos(\text{mom})$                                                                                                                              | 0.516      | 0.540     | 0.854       | 0.849      | 19.143      |

Table 38: operon results for the Feynman equation II.36-38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $\frac{0.038 - 1.793 \left\{ \frac{(-1.402)\text{mom}}{2.718k_b} \right\} (2.303H)}{1.302T [\sin\{(0.673\epsilon + 0.568c - 0.232\alpha - 0.172M)(0.318H)\} + 0.248c]}$                                          | 0.192      | 0.207     | 0.970       | 0.972      | 25.131      |
| 2                                              | $\frac{-0.413 - 0.671 \left[ \left\{ \frac{-1.390\text{mom}}{-0.522c} + 1.389\text{mom} \right\} \frac{1.599H + 0.608M}{\cos(0.236\alpha)(5.930T)(-0.597k_b)} - \frac{-0.639}{-0.568\epsilon} \right]}{\times}$  | 0.184      | 0.285     | 0.971       | 0.917      | 24.610      |
| 3                                              | $\frac{-0.749 + 0.013 \left[ \frac{0.312M}{(-0.120\epsilon)^2} - \left\{ \frac{0.312M}{0.144c} + \frac{2.327H}{0.286T} \right\} \frac{0.298\text{mom}}{-0.029k_b} + \frac{3.453\alpha}{0.104c} \right]}{-}$      | 0.248      | 0.193     | 0.948       | 0.968      | 29.759      |
| 4                                              | $\frac{-0.074 - 0.127 \left[ \frac{\frac{1.443\alpha}{-2.307T}}{(0.026\epsilon)(2.551k_b)} - \frac{1.678\text{mom}}{-0.044T} \right]}{-0.318H \frac{1.678\text{mom}}{2.194k_b \sin\{\sin(-0.438c)\}}}$           | 0.270      | 0.237     | 0.938       | 0.949      | 29.578      |
| 5                                              | $\frac{0.028 \left[ \left\{ \frac{1.188\epsilon}{e^{0.186\epsilon}} - \cos(0.619M) + 1.286H \right\} \frac{0.707\text{mom}}{e^{-1.611\epsilon} - 0.828T} \right]}{-0.087 (2.718c)(0.022k_b)}$                    | 0.278      | 0.222     | 0.950       | 0.958      | 28.538      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $\frac{0.134 - 4.891(0.495H)}{\times \frac{0.771\text{mom} + 0.656c + 0.366\alpha + 0.729\text{mom} - \cos(0.448M) - 0.628\epsilon}{(-1.588T)(2.398c)(0.424k_b)}}$                                               | 0.292      | 0.338     | 0.936       | 0.919      | 25.042      |
| 2                                              | $\frac{0.084 - 3.079 \left[ \frac{-0.060H}{(0.130c)(1.994k_b)} \frac{1.583\text{mom}}{0.709T} - \frac{-1.278\alpha}{2.450k_b} \frac{0.521\text{mom}}{(-0.686T)(1.559\epsilon)} \right]}{-}$                      | 0.325      | 0.303     | 0.909       | 0.921      | 31.595      |
| 3                                              | $\frac{AB(-0.900\alpha)(-0.821\text{mom}) + 0.697\text{mom}}{A = \sin\{\sin(-0.367M)\}, B = -0.368H + 0.292c - 0.188, C = (0.330T)(0.435k_b) - 0.004 + 0.394}$                                                   | 0.324      | 0.357     | 0.913       | 0.890      | 25.952      |
| 4                                              | $\frac{-0.001 + 63.119 (0.930H)(0.862\text{mom})}{\times \frac{2.718k_b \left[ 2.008c + \frac{1.078H - 1.772\alpha}{1.367\epsilon} + 1.333\epsilon + 0.910H \right] (1.772T)}$                                   | 0.262      | 0.299     | 0.943       | 0.930      | 24.575      |
| 5                                              | $\frac{-0.027 + 0.083 \left( \frac{3.628\alpha}{-0.661c} - 1.488M + (1.178\text{mom})(-1.809H) + \frac{1.746\alpha}{0.738\text{mom}} + 1.737\epsilon \right)}{\times \frac{(1.443T) \sin(-0.120k_b)}$            | 0.278      | 0.260     | 0.944       | 0.943      | 25.120      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                                  |            |           |             |            |             |
| 1                                              | $\frac{-0.160 + 0.069 \left( \frac{2.349\text{mom}}{0.130T} \left[ 0.465M + \frac{0.533\text{mom}}{0.408c} - 0.481\text{mom} + (0.558H)^2 \right] + \frac{1.548\alpha}{0.130\epsilon} \right)}{\times 1.432k_b}$ | 0.316      | 0.288     | 0.920       | 0.936      | 37.939      |
| 2                                              | $\frac{0.043 - 0.057 \left[ \frac{0.176\text{mom}}{0.112c} - \frac{-0.524T}{2.170M} \right]}{- \frac{-0.541\text{mom}}{\left\{ \frac{-0.536T}{1.417H} \right\} (-0.594k_b + 0.106\alpha)} - 0.102c}$             | 0.316      | 0.265     | 0.914       | 0.942      | 31.098      |

Continued on next page

Table 38: **operon** results for the Feynman equation II.36\_38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                                                           | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | $-0.015 + 2.075 \frac{\frac{\sin(-0.413H)}{0.184T}}{\frac{1.862k_b}{0.434\text{mom}} \left[ \sin\{\sin(-0.509\epsilon)\} - \frac{2.186M+2.135\alpha}{(1.507\epsilon)(2.350c)} \right]^{-1}}$ | 0.235      | 0.260     | 0.957       | 0.951      | 27.554      |
| 4    | $\frac{-0.010 - 22.478}{\frac{(-1.210\text{mom})(1.414H)}{-0.690c e^{-0.237\alpha}} \frac{1}{1.790/(0.318M)+2.003\epsilon} + 0.460\text{mom}} \times \frac{1}{(6.921T)(-1.302k_b)}$          | 0.277      | 0.366     | 0.942       | 0.898      | 23.024      |
| 5    | $1.650 + 2.162 \left[ \sin\{\sin(-0.586c)\} + \frac{2.238\epsilon \sin(-0.514c) + 1.737M + 1.951\alpha + 5.167H}{(2.410T)\{3.182k_b/(0.679\text{mom})\}} \right]$                            | 0.288      | 0.294     | 0.932       | 0.932      | 27.426      |

Table 39: **PySR** results for the Feynman equation II.36\_38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                          | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                             |            |           |             |            |             |
| 1                                              | $\frac{\text{mom}}{1.174 T k_b} \left\{ H + \frac{\alpha M}{\epsilon c} \right\}$           | 0.173      | 0.174     | 0.975       | 0.980      | 66.425      |
| 2                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{M}{c^2} \right\}$                               | 0.249      | 0.306     | 0.946       | 0.904      | 97.007      |
| 3                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{\alpha}{\epsilon(c - 0.431)} \right\}$          | 0.216      | 0.185     | 0.961       | 0.970      | 76.063      |
| 4                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{M\alpha}{\epsilon c^2} \right\}$                | 0.000      | 0.000     | 1.000       | 1.000      | 73.380      |
| 5                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{1}{\epsilon(c + \alpha^{-1} - 1.262)} \right\}$ | 0.231      | 0.194     | 0.965       | 0.968      | 76.818      |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                             |            |           |             |            |             |
| 1                                              | $\frac{\text{mom}}{T k_b} \left\{ H + M\alpha e^{-c} \right\}$                              | 0.300      | 0.408     | 0.932       | 0.882      | 73.086      |
| 2                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{1}{c - 0.493} \right\}$                         | 0.360      | 0.330     | 0.889       | 0.907      | 133.422     |
| 3                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{M\{\cos(\epsilon) + 1.423\}}{c^2} \right\}$     | 0.246      | 0.246     | 0.950       | 0.948      | 71.924      |
| 4                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{\alpha}{\epsilon(c - 0.411)} \right\}$          | 0.261      | 0.298     | 0.943       | 0.930      | 89.639      |
| 5                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{M}{c^2} \right\}$                               | 0.340      | 0.286     | 0.917       | 0.931      | 86.361      |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                             |            |           |             |            |             |
| 1                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{M}{c^2} \right\}$                               | 0.331      | 0.298     | 0.912       | 0.932      | 208.305     |
| 2                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{\alpha + M - \epsilon}{c^2} \right\}$           | 0.226      | 0.228     | 0.956       | 0.957      | 110.782     |
| 3                                              | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{M}{c^2} \right\}$                               | 0.331      | 0.356     | 0.914       | 0.909      | 62.105      |

Continued on next page



Table 39: PySR results for the Feynman equation II.36.38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                     | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4    | $\frac{\text{mom}}{T k_b} \left\{ H + \frac{\alpha}{c^2} \right\}$     | 0.367      | 0.441     | 0.898       | 0.852      | 70.073      |
| 5    | $0.356 \frac{\text{mom}}{T} \{ \sin(c) + H \} \{ \cos(k_b) + 1.859 \}$ | 0.430      | 0.464     | 0.849       | 0.831      | 141.019     |

Table 40: BMS results for the Feynman equation II.36.38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} H$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set. Here,  $a_0$  denotes a fitted constant learned by BMS.

| Rep.                                           | Learned expression                                                                                      | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                         |            |           |             |            |             |
| 1                                              | $-\text{mom } a_0 \{ a_0 (H + k_b) + \cos T \} \times [-\{a_0 + a_0\}^2]^3 \exp\{\sin(\cos k_b)\}$      | 0.519      | 0.509     | 0.779       | 0.831      | 2414.000    |
| 2                                              | $\text{mom } H \exp \left[ -\{c a_0\}^3 + T a_0 c (a_0 M + k_b) \right]$                                | 0.445      | 0.484     | 0.829       | 0.759      | 2980.680    |
| 3                                              | $\text{mom } H \exp [T k_b (H + T) a_0 (c + \epsilon)]$                                                 | 0.457      | 0.446     | 0.824       | 0.828      | 2263.260    |
| 4                                              | $\exp^3 \left( a_0 \{ H + \cos a_0 + \sin(c + \epsilon) + \text{mom} - T k_b + \alpha^2 a_0 \} \right)$ | 0.397      | 0.394     | 0.866       | 0.858      | 2485.010    |
| 5                                              | $-2 a_0 \text{mom } H \exp \{ \cos(a_0) (c + T + k_b) \}$                                               | 0.561      | 0.416     | 0.797       | 0.855      | 2289.430    |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                         |            |           |             |            |             |
| 1                                              | $\text{mom } H \exp \left[ (a_0 - T) k_b a_0^2 c \right] + a_0$                                         | 0.504      | 0.561     | 0.809       | 0.775      | 2309.130    |
| 2                                              | $\text{mom} [1 - \sin\{a_0(k_b + c + \cos(a_0 \cos H + \epsilon) + \epsilon + T)\}]$                    | 0.560      | 0.546     | 0.731       | 0.746      | 2713.060    |
| 3                                              | $\{ \text{mom}(\alpha + H) + \cos c + M \} \times \exp\{a_0(T + a_0 + k_b)\} + a_0$                     | 0.422      | 0.399     | 0.852       | 0.862      | 2417.780    |
| 4                                              | $\text{mom } e^{-T} [a_0^3 + \epsilon + k_b + c + k_b - \{H + T + \alpha\} + k_b]$                      | 0.501      | 0.569     | 0.790       | 0.745      | 2479.190    |
| 5                                              | $a_0^2 \{ H + \cos c \} \{ \text{mom} + \cos T \} \exp(\cos k_b) + a_0^2$                               | 0.513      | 0.454     | 0.810       | 0.826      | 2426.710    |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                         |            |           |             |            |             |
| 1                                              | $\{ \text{mom}(H - a_0 c M) \} \exp^2 \left[ -a_0 (k_b + T)^3 \right] + M^3 \sin(a_0 \alpha) + 3 a_0$   | 0.541      | 0.477     | 0.765       | 0.825      | 3558.890    |
| 2                                              | $\text{mom } H \exp \left[ -a_0^2 \{ e^T + k_b^2 c \} + a_0 \right] + a_0$                              | 0.488      | 0.452     | 0.795       | 0.831      | 2786.200    |
| 3                                              | $e^{a_0} 2 a_0 \{ H + \cos T \} \{ \cos k_b + \text{mom} \} + a_0$                                      | 0.565      | 0.654     | 0.749       | 0.691      | 1977.010    |
| 4                                              | $a_0 \{ -a_0 \epsilon^2 + k_b c + k_b H \} + \text{mom} + \cos T + H$                                   | 0.621      | 0.639     | 0.708       | 0.689      | 2035.710    |
| 5                                              | $(H - c + \text{mom}) \exp \left[ \cos \{ a_0^3 (k_b + a_0 + T) \} \right]$                             | 0.474      | 0.488     | 0.817       | 0.814      | 2647.510    |

Table 41: BSR results for the Feynman equation II\_36\_38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$  across noise levels and repetitions. Train RMSE and  $R^2$  are computed on the 90% training set, while test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                                    | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                       |            |           |             |            |             |
| 1                                              | $1.319 - 0.130 \exp\{\cos(\alpha)\} + 0.002 \exp(\alpha) + 0.011 \sin\{(c + \epsilon)^3\} + 0.068(\alpha + c \text{ mom})$                                                                            | 1.093      | 1.237     | 0.019       | 0.001      | 2028.980    |
| 2                                              | $10.235 - 9.025 \cos\{\exp(-c)\} + 0.013 \cos(M) + 0.262 \cos(c) + 0.092 M \alpha$                                                                                                                    | 1.031      | 0.968     | 0.082       | 0.036      | 2046.020    |
| 3                                              | $-0.048 + 0.026 \exp(\alpha) - 0.016 \exp(\epsilon) - 0.003 \frac{\cos(\alpha)}{\sin(\alpha \text{ mom})} + 0.802 \text{ mom}$                                                                        | 0.977      | 1.010     | 0.197       | 0.118      | 1910.030    |
| 4                                              | $4.692 - 1.532\{k_b + \exp[\cos(k_b)]\} + 0.017 \exp(\alpha) + 0.552 H - 0.157 \sin(\text{mom}^2)$                                                                                                    | 0.956      | 0.922     | 0.218       | 0.224      | 1912.210    |
| 5                                              | $-2.822 + 0.388\{\cos(H^{-1}) + H + M\} + 1.020 \epsilon^{-1} + 3.579 T^{-1} - 6.204 \times 10^{-4} (T\epsilon)^3$                                                                                    | 1.007      | 0.907     | 0.345       | 0.307      | 1932.020    |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                       |            |           |             |            |             |
| 1                                              | $3.574 - 0.927 T - 0.053 \frac{\alpha}{M} + 0.516 \sin(c k_b) + 0.010 \sin^{-2}(\alpha)$                                                                                                              | 0.957      | 1.022     | 0.310       | 0.256      | 1923.980    |
| 2                                              | $-1.097 - 0.117 \cos(k_b^2) - 1.122 \exp(-M) + 0.357 \cos\{\sin(M)\} + 1.523 \exp(T^{-1})$                                                                                                            | 0.947      | 0.917     | 0.229       | 0.280      | 1935.810    |
| 3                                              | $0.387 + 0.340 \exp\{\sin(c)\} - 0.249 \cos(M) + 0.272 \sin(k_b^2) - 0.979 \cos(\text{mom})$                                                                                                          | 0.954      | 0.905     | 0.247       | 0.293      | 1921.380    |
| 4                                              | $-0.293 + 0.028 M \exp(H) + 1.805 \sin\{\sin(T)\} - 0.054 \sin^{-1}(\alpha) + 0.034 \exp(\alpha)$                                                                                                     | 0.979      | 0.974     | 0.199       | 0.254      | 1960.230    |
| 5                                              | $6.922 - 1.301 \cos(H) - 2.606 \text{mom}^{-1} - 0.467(H + c) - 0.243(T + \alpha + \epsilon k_b + M)$                                                                                                 | 0.855      | 0.707     | 0.473       | 0.577      | 1924.410    |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                       |            |           |             |            |             |
| 1                                              | $2.398 \times 10^{-8} + 8.053 \times 10^{-6} \exp(HM) - 1.081 \times 10^{-8} (-k_b \alpha)^9 + 5.583 \times 10^{-7} \exp\left(\frac{k_b H}{M}\right) - 6.071 \times 10^{-8} \{\cos(\text{mom}) - M\}$ | 1.938      | 2.022     | -2.020      | -2.138     | 1931.390    |
| 2                                              | $1.100 \times 10^{-8} + 8.718 \times 10^{-9} \sin(\epsilon) - 4.275 \times 10^{-9} \cos(\alpha) + 6.393 \times 10^{-9} \exp\{\exp(M)\} + 3.798 \times 10^{-8} c T$                                    | 1.881      | 1.826     | -2.043      | -1.753     | 1914.320    |
| 3                                              | $4.287 - 0.692 \alpha^{-1} - 0.185 \sin(-c^3) - 2.741 \cos\{\cos(\text{mom})\} + 0.130 \sin\{\cos(\text{mom}\alpha)\}$                                                                                | 1.011      | 1.093     | 0.195       | 0.138      | 1921.150    |
| 4                                              | $3.873 - 2.517 \cos\{\cos(\text{mom})\} + 0.210 \cos^2\{\sin(H^2)\} + 0.016 \alpha^3 - 0.633 \sin^2\{\cos(k_b) + H\}$                                                                                 | 1.059      | 1.063     | 0.151       | 0.139      | 1959.312    |
| 5                                              | $1.530 + 3.028 T^{-1} - 0.907 k_b^2 + 0.023 \exp\{\cos(M \text{mom}) + M\} + 0.218 k_b^3$                                                                                                             | 0.766      | 0.773     | 0.520       | 0.532      | 1933.380    |

## D Results of HierBOSSS for Learning Feynman Equations

### D.1 Experimental Settings of HierBOSSS

We evaluate HierBOSSS on the Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). Each experimental run uses a random subsample of  $n = 2000$  observations out of  $10^5$  observations, followed by a 90/10 train-test split. We consider the noiseless setting (with the original target) and noisy regimes, where independent Gaussian noise with different standard deviations  $\sigma$ 's is added to the target, as given below. Test RMSE and  $R^2$  are computed on the held-out 10% test split. For each Feynman equation and noise level, we run 5 independent repetitions. RMSE and  $R^2$  denotes the root mean square error and coefficient of determination, i.e.,  $\text{RMSE} = \sqrt{n^{-1} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$  and  $R^2 = 1 - \text{RMSE}^2 / [n^{-1} \sum_{i=1}^n (y_i - \bar{y})^2]$  ( $\bar{y} = n^{-1} \sum_{i=1}^n y_i$ ), respectively.

#### HierBOSSS noise configurations and data splitting for Feynman equations

```

1  # Noiseless and noisy regimes
2  # 0.00 -> noiseless setting
3
4  feynman_equations = {
5      "I_12_2": {      # CL
6          "noise_levels": [0.00, 0.15, 0.20, 0.25],
7      },
8      "I_12_11": {    # FCE
9          "noise_levels": [0.00, 0.25, 1.00, 2.00],
10     },
11     "I_24_6": {      # EHFO
12         "noise_levels": [0.00, 0.15, 0.25, 1.00],
13     },
14     "I_50_26": {     # HOQN
15         "noise_levels": [0.00, 0.15, 0.18, 0.20],
16     },
17     "II_36_38": {    # FMMM
18         "noise_levels": [0.00, 0.15, 0.20, 0.25],
19     },
20 }
21
22 # Data split settings
23 # y_clean -> original target
24 data_split_settings = {
25     "base_seed": ...,
26     "nreps": 5,
27     "sample_n": 2000,
28     "test_size": 0.10,
29     "train_size": 0.90,
30     "noise_model": "y = y_clean + Normal(0, sigma^2)",
31     "split_seed_rule": "...",
32 }
```

The operator set  $\mathbb{O}$  is kept common across every Feynman equation. The set consists of two binary operators and seven unary operators, as given below. In the numerical implementation,

the inverse operator is protected using a threshold  $10^{-8}$ , while the exponential operator is evaluated after clipping its input to  $[-20, 20]$  for numerical stability.

#### HierBOSSS operator set

```

1 opset = [
2     "add", # (x, y) -> x + y
3     "mul", # (x, y) -> x*y
4     "neg", # x -> -x
5     "inv", # x -> 1/x, protected near zero
6     "sin", # x -> sin(x)
7     "cos", # x -> cos(x)
8     "exp", # x -> exp(x), with input clipped to [-20, 20]
9     "sq",  # x -> x^2
10    "cu",  # x -> x^3
11 ]

```

The prior hyperparameters are set uniformly across all Feynman equations unless otherwise stated. The Dirichlet concentration parameters for both operator and feature assignments are set to one, i.e.,  $\alpha_{\text{op}} = (1, \dots, 1)$  and  $\alpha_{\text{ft}} = (1, \dots, 1)$ . The depth-dependent split probability is controlled by  $(\alpha_0, \delta_0) = (0.95, 1.20)$ . For the outer regression coefficient vector  $\beta$  with an intercept, we set the Gaussian prior mean vector to  $\mu_\beta = \mathbf{0}$ , use Gaussian prior covariance matrix  $\Sigma_\beta = 10\mathbf{I}_{K+1}$ , and assign the hyperparameters of the Inverse-Gamma prior on the model noise variance  $\sigma^2$  to  $\nu = 0.05 = \lambda$ .

#### HierBOSSS prior configuration

```

1 HierBOSSS_prior = {
2     # operator and feature Dirichlet weight prior concentration
3     "alpha_op": "ones(len(opset))",
4     "alpha_ft": "ones(number_of_features)",
5
6     # split probability
7     "alpha_0": 0.95,
8     "delta_0": 1.20,
9
10    # Gaussian prior parameters of outer regression coefficient vector
11    "add_intercept": True,
12    "beta0": "zeros(K + 1)",
13    "V0": "10 * I_(K + 1)",
14
15    # Inverse-Gamma prior parameters of model noise variance
16    "a0": 0.05,
17    "b0": 0.05,
18    "sigma2_prior": "Inverse-Gamma(a0 / 2, b0 / 2)",
19 }

```

For each Feynman equation, the symbolic forest size  $K$  and number of MCMC iterations are chosen according to the symbolic complexity of the underlying target law, as given below. However, one can choose a suitable  $K$  balancing the expressiveness and computational efficiency for learning the underlying Feynman equations; see [Section G](#).

**HierBOSSS** equation-specific  $K$  and MCMC iterations

```

1 # maxiter -> number of MCMC iterations within each parallel chain of HierBOSSS
2 # K -> symbolic forest size
3
4 HierBOSSS_feynman_settings = {
5   "I_12_2": {           # CL
6     "K": 4,
7     "maxiter": 2000,
8   },
9   "I_12_11": {         # FCE
10    "K": 3,
11    "maxiter": 2000,
12  },
13  "I_24_6": {           # EHFO
14    "K": 4,
15    "maxiter": 2000,
16  },
17  "I_50_26": {         # HOQN
18    "K": 4,
19    "maxiter": 2000,
20  },
21  "II_36_38": {        # FMMM
22    "K": 4,
23    "maxiter": 20000,
24  },
25 }

```

Each repetition is fit using 5 independent parallel MCMC chains of **HierBOSSS**. The final symbolic model selection is performed by pooling sampled forests across chains and ranking them by their joint marginal posterior over  $\mathcal{T}$ , i.e.,  $\text{JMP}(\mathcal{T})$ .

**HierBOSSS** parallel chain configuration

```

1 parallel_mcmc_settings = {
2   "nchains": 5,          # number of parallel MCMC chains
3   "burnin": 0,
4   "thin": 1,
5   "n_jobs": "nchains",
6   "chain_seed_rule": "...",
7   "target": "JMP(T)",
8 }

```

The MH kernel uses seven local symbolic tree proposal moves: grow (g), prune (p), change feature (cf), change operator (co), subtree replacement (st), delete node (del), and insert node (ins). Unless a move is unavailable for the current tree state, all moves are assigned equal proposal weight through the move weight vector  $\pi_{\text{move}}$ . At each tree update, the sampler first identifies the set of valid moves and valid node addresses, normalizes the proposal weights over the available moves, samples a move type, and then samples a valid node address uniformly for that move.

## HierBOSSS local symbolic tree proposal moves

```

1 # local symbolic tree proposal move weights
2
3 move_weights = {
4     "grow": 1.0,          # g
5     "prune": 1.0,        # p
6     "change_feature": 1.0, # cf
7     "change_operator": 1.0, # co
8     "subtree_replace": 1.0, # st
9     "delete_node": 1.0,   # del
10    "insert_node": 1.0,   # ins
11 }

```

The initial symbolic forests and proposal-generated subtrees use uniform operator and feature weights, i.e., uniform choices for  $(\mathbf{w}_{\text{op,init}}, \mathbf{w}_{\text{op,prop}}, \mathbf{w}_{\text{ft,init}}, \mathbf{w}_{\text{ft,prop}})$ . These weights are used only for initialization and proposal generation; the posterior target itself uses the Dirichlet-collapsed symbolic forest prior  $\Pi_{\text{forest},K}(\mathcal{T} \mid \alpha_0, \delta_0, \boldsymbol{\alpha}_{\text{op}}, \boldsymbol{\alpha}_{\text{ft}})$ . Thus, operator and feature preferences are learned through the marginal prior contribution in the joint marginal posterior rather than fixed at the proposal level.

## HierBOSSS initialization and proposal weights

```

1 # initialization of operator and feature weight vectors with uniform choice
2 # proposal operator and feature weight vectors with uniform choice
3
4 initialization_and_proposal = {
5     "wts_init_op": "ones(len(opset))",
6     "wts_init_ft": "ones(number_of_features)",
7     "wts_prop_op": "wts_init_op",
8     "wts_prop_ft": "wts_init_ft",
9     "proposal_normalization": "normalize over available moves and valid labels",
10 }

```

After sampling, all retained symbolic forests from the parallel MCMC chains are pooled and ranked by their log joint marginal posterior over  $\mathcal{T}$ , i.e.,  $\log \text{JMP}(\mathcal{T})$ . For each repetition, the top  $r = 10$  symbolic forests are retained as the representative posterior set. For each retained forest, the posterior mean of the outer regression coefficient vector is computed under the conjugate Gaussian layer, yielding the corresponding raw symbolic expression (*Raw*).

HierBOSSS posterior ranking configuration and obtaining *Raw* expressions

```

1 posterior_ranking = {
2     "ranking_score": "log_JMP",
3     "pooling": "all retained forests from all parallel chains",
4     "top_r": 10,
5     "raw_expression_coefficients": "posterior mean under NIG layer",
6     "raw_expression_digits": 6, # rounding off decimals in raw expression
7 }

```

Finally, each top-ranked symbolic forest is passed through the post-MCMC symbolic model refine-

ment step in **HierBOSSS**. The symbolic trees in a fixed forest are treated as candidate additive components, and a smaller subset is selected by minimizing BIC. After selecting the subset, the outer regression coefficient vector is re-estimated, and the retained expression is simplified using SymPy [Meurer et al., 2017]. The final reported expression corresponds to the *Final* expression after this refinement step with effective symbolic forest size  $K_{\text{eff}} \leq K$ .

**HierBOSSS** post-MCMC symbolic model refinement to obtain *Final* expressions

```

1 post_mcmc_refinement = {
2     "candidate_models": "top_r forests ranked by log_JMP",
3     "criterion": "BIC",
4     "coefficient_refit": "posterior mean on selected symbolic design",
5     "simplification": "SymPy",
6     "final_expression_digits": 2, # rounding off decimals in final expression
7     "coefficient_tolerance": 1e-4,
8     "effective_forest_size": "K_eff = number of retained non-intercept trees",
9 }
```

## D.2 Results for I\_12\_2 (CL)

### D.2.1 Expressions Learned by **HierBOSSS**

Table 42: **HierBOSSS** results for the Feynman equation I\_12\_2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HierBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                                                                      | Train RMSE             | Test RMSE              | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                                                                         |                        |                        |             |            |             |
| 1                        | <b>Raw:</b> $-5.280 \times 10^{-6} + 0.080 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 8.378 \times 10^{-6} \frac{1}{r} + 8.378 \times 10^{-6} \frac{1}{r}$<br>$+ 8.378 \times 10^{-6} \frac{1}{r}$             | $5.395 \times 10^{-6}$ | $5.504 \times 10^{-6}$ | 1.000       | 1.000      | 49.369      |
|                          | <b>Final:</b> $0.080 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                                                                      | $5.394 \times 10^{-6}$ | $5.503 \times 10^{-6}$ | 1.000       | 1.000      | —           |
| 2                        | <b>Raw:</b> $-1.422 \times 10^{-6} + 2.295 \times 10^{-6} \frac{1}{r}$<br>$+ 2.295 \times 10^{-6} \frac{1}{r} + 2.295 \times 10^{-6} \frac{1}{r}$<br>$+ 0.040 \frac{q_2(q_1 + q_1)}{\epsilon r^2}$      | $1.441 \times 10^{-6}$ | $1.592 \times 10^{-6}$ | 1.000       | 1.000      | 47.098      |
|                          | <b>Final:</b> $0.040 \frac{2q_1 q_2}{\epsilon r^2}$                                                                                                                                                     | $8.700 \times 10^{-7}$ | $9.738 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 3                        | <b>Raw:</b> $5.501 \times 10^{-7} + 0.040 \frac{q_2(q_1 + q_1)}{\epsilon r^2}$<br>$- 5.504 \times 10^{-8} \cos(q_2) - 5.504 \times 10^{-8} \cos(q_2)$<br>$- 5.504 \times 10^{-8} \cos(q_2)$             | $9.758 \times 10^{-7}$ | $8.354 \times 10^{-7}$ | 1.000       | 1.000      | 47.762      |
|                          | <b>Final:</b> $0.040 \frac{2q_1 q_2}{\epsilon r^2}$                                                                                                                                                     | $9.758 \times 10^{-7}$ | $8.354 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 4                        | <b>Raw:</b> $-4.565 \times 10^{-6} + 7.093 \times 10^{-6} \frac{1}{r}$<br>$+ 0.080 \frac{q_1 q_2}{\epsilon r^2} + 7.093 \times 10^{-6} \frac{1}{r}$<br>$+ 7.093 \times 10^{-6} \frac{1}{r}$             | $4.921 \times 10^{-6}$ | $3.756 \times 10^{-6}$ | 1.000       | 1.000      | 62.565      |
|                          | <b>Final:</b> $0.080 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                                                                      | $4.918 \times 10^{-6}$ | $3.753 \times 10^{-6}$ | 1.000       | 1.000      | —           |
| 5                        | <b>Raw:</b> $1.220 \times 10^{-6} - 4.183 \times 10^{-7} \frac{1}{q_1}$<br>$- 4.183 \times 10^{-7} \frac{1}{q_1} + 0.040 \frac{q_1(q_2 + q_2)}{\epsilon r^2}$<br>$- 4.183 \times 10^{-7} \frac{1}{q_1}$ | $1.038 \times 10^{-6}$ | $9.866 \times 10^{-7}$ | 1.000       | 1.000      | 50.158      |
|                          | <b>Final:</b> $0.040 \frac{2q_1 q_2}{\epsilon r^2}$                                                                                                                                                     | $1.038 \times 10^{-6}$ | $9.866 \times 10^{-7}$ | 1.000       | 1.000      | —           |

*Noise level:  $\sigma = 0.15$*

*Continued on next page*



Table 42: **HIERBOSSS** results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                          | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 1                                              | <b>Raw:</b> $0.014 - 0.011 \frac{1}{q_1}$<br>$- 0.011 \frac{1}{q_1} - 0.011 \frac{1}{q_1}$<br>$+ 0.083 \frac{q_1 q_2}{\epsilon r^2}$        | 0.149      | 0.143     | 0.281       | 0.238      | 48.198      |
|                                                | <b>Final:</b> $0.084 \frac{q_1 q_2}{\epsilon r^2}$                                                                                          | 0.149      | 0.144     | 0.279       | 0.235      | —           |
| 2                                              | <b>Raw:</b> $-0.007 + 0.019 \frac{1}{\epsilon}$<br>$+ 0.019 \frac{1}{\epsilon} + 0.074 \frac{q_1 q_2}{\epsilon r^2}$<br>$- 0.020 \cos(r^3)$ | 0.152      | 0.147     | 0.229       | 0.135      | 46.409      |
|                                                | <b>Final:</b> $0.079 \frac{q_1 q_2}{\epsilon r^2} - 0.020 \cos(r^3)$                                                                        | 0.152      | 0.147     | 0.226       | 0.142      | —           |
| 3                                              | <b>Raw:</b> $0.006 - 0.003 \frac{1}{q_1}$<br>$- 0.003 \frac{1}{q_1} - 0.003 \frac{1}{q_1}$<br>$+ 0.081 \frac{q_1 q_2}{\epsilon r^2}$        | 0.152      | 0.156     | 0.276       | 0.273      | 62.039      |
|                                                | <b>Final:</b> $0.082 \frac{q_1 q_2}{\epsilon r^2}$                                                                                          | 0.152      | 0.156     | 0.276       | 0.274      | —           |
| 4                                              | <b>Raw:</b> $-0.022 + 0.085 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.014 \frac{1}{q_2} + 0.014 \frac{1}{q_2}$<br>$+ 0.014 \frac{1}{q_2}$       | 0.152      | 0.148     | 0.231       | 0.274      | 63.390      |
|                                                | <b>Final:</b> $0.081 \frac{q_1 q_2}{\epsilon r^2}$                                                                                          | 0.152      | 0.148     | 0.228       | 0.275      | —           |
| 5                                              | <b>Raw:</b> $-0.133 + 0.075 \frac{1}{r}$<br>$+ 0.075 \frac{1}{r} + 0.083 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.024 r$                       | 0.149      | 0.140     | 0.234       | 0.292      | 61.748      |
|                                                | <b>Final:</b> $0.084 \frac{q_1 q_2}{\epsilon r^2}$                                                                                          | 0.149      | 0.140     | 0.229       | 0.291      | —           |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                             |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $-0.045 + 0.035 \frac{1}{r}$<br>$+ 0.035 \frac{1}{r} + 0.035 \frac{1}{r}$<br>$+ 0.085 \frac{q_1 q_2}{\epsilon r^2}$             | 0.198      | 0.198     | 0.127       | 0.205      | 56.024      |
|                                                | <b>Final:</b> $-0.045 + 0.110 \frac{1}{r}$<br>$+ 0.085 \frac{q_1 q_2}{\epsilon r^2}$                                                        | 0.198      | 0.198     | 0.127       | 0.205      | —           |

Continued on next page

Table 42: **HIERBOSSS** results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                  | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 2                                              | <b>Raw:</b> $0.003 - 0.017 \frac{1}{\epsilon}$<br>$+ 0.083 \frac{q_1 q_2}{\epsilon r^2} - 0.017 \frac{1}{\epsilon}$<br>$- 0.017 \frac{1}{\epsilon}$ | 0.200      | 0.193     | 0.167       | 0.244      | 60.635      |
|                                                | <b>Final:</b> $-0.044 \frac{1}{\epsilon} + 0.083 \frac{q_1 q_2}{\epsilon r^2}$                                                                      | 0.200      | 0.193     | 0.167       | 0.244      | —           |
| 3                                              | <b>Raw:</b> $0.010 - 0.009 \frac{1}{r}$<br>$- 0.009 \frac{1}{r} + 0.078 \frac{q_1 q_2}{\epsilon r^2}$<br>$- 0.009 \frac{1}{r}$                      | 0.201      | 0.199     | 0.148       | 0.289      | 58.254      |
|                                                | <b>Final:</b> $0.075 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                  | 0.201      | 0.199     | 0.148       | 0.285      | —           |
| 4                                              | <b>Raw:</b> $0.019 + 0.078 \frac{q_1 q_2}{\epsilon r^2}$<br>$- 0.006 \frac{1}{r} - 0.006 \frac{1}{r}$<br>$- 0.006 \frac{1}{r}$                      | 0.197      | 0.211     | 0.107       | 0.134      | 55.818      |
|                                                | <b>Final:</b> $0.082 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                  | 0.197      | 0.211     | 0.104       | 0.129      | —           |
| 5                                              | <b>Raw:</b> $-0.063 + 0.081 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.039 \frac{1}{q_1} + 0.072 \frac{1}{\epsilon}$<br>$+ 0.039 \frac{1}{q_1}$          | 0.203      | 0.197     | 0.171       | 0.098      | 58.478      |
|                                                | <b>Final:</b> $0.080 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                  | 0.204      | 0.195     | 0.162       | 0.117      | —           |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                     |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $-0.067 + 0.082 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.060 \frac{1}{r} + 0.060 \frac{1}{r}$<br>$+ 0.060 \frac{1}{r}$                     | 0.252      | 0.236     | 0.097       | 0.108      | 55.903      |
|                                                | <b>Final:</b> $-0.067 + 0.082 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.180 \frac{1}{r}$                                                                | 0.252      | 0.236     | 0.097       | 0.108      | —           |
| 2                                              | <b>Raw:</b> $0.026 + 0.088 \frac{q_1 q_2}{\epsilon r^2}$<br>$- 0.024 \frac{1}{r} - 0.024 \frac{1}{r}$<br>$- 0.024 \frac{1}{r}$                      | 0.249      | 0.264     | 0.109       | 0.176      | 58.066      |
|                                                | <b>Final:</b> $0.081 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                  | 0.250      | 0.263     | 0.107       | 0.177      | —           |

*Continued on next page*

Table 42: **HIERBOSSS** results for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | <b>Raw:</b> $0.018 - 0.019 \frac{1}{q_2}$<br>$- 0.019 \frac{1}{q_2} - 0.019 \frac{1}{q_2}$<br>$+ 0.078 \frac{q_1 q_2}{\epsilon r^2}$                 | 0.254      | 0.242     | 0.123       | 0.049      | 58.756      |
|      | <b>Final:</b> $0.078 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                   | 0.254      | 0.241     | 0.121       | 0.063      | —           |
| 4    | <b>Raw:</b> $-0.014 + 0.082 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.004 \frac{1}{\epsilon} + 0.004 \frac{1}{\epsilon}$<br>$+ 0.004 \frac{1}{\epsilon}$ | 0.247      | 0.211     | 0.118       | 0.159      | 58.926      |
|      | <b>Final:</b> $0.079 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                   | 0.247      | 0.210     | 0.117       | 0.166      | —           |
| 5    | <b>Raw:</b> $-0.108 + 0.084 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.075 \frac{1}{r} + 0.075 \frac{1}{r}$<br>$+ 0.048 \frac{1}{q_1}$                    | 0.246      | 0.245     | 0.134       | -0.009     | 58.167      |
|      | <b>Final:</b> $-0.089 + 0.081 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.160 \frac{1}{r}$                                                                 | 0.246      | 0.245     | 0.133       | -0.007     | —           |

### D.2.2 Top Expressions Ranked by JMP

Table 43: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rank                                   | Learned expression                                                                                                                                                                          | Test RMSE              | Test $R^2$ | log JMP  |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------|----------|
| <i>Noiseless setting, Repetition 1</i> |                                                                                                                                                                                             |                        |            |          |
| 1–10                                   | <b>Raw:</b> $-5.280 \times 10^{-6} + 0.080 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 8.378 \times 10^{-6} \frac{1}{r} + 8.378 \times 10^{-6} \frac{1}{r}$<br>$+ 8.378 \times 10^{-6} \frac{1}{r}$ | $5.504 \times 10^{-6}$ | 1.000      | 8436.284 |
|                                        | <b>Final:</b> $0.080 \frac{q_1 q_2}{\epsilon r^2}$                                                                                                                                          | $5.503 \times 10^{-6}$ | 1.000      | —        |

*Noise level:  $\sigma = 0.15$ , Repetition 1*

*Continued on next page*

Table 43: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I.12.2 (CL):  $F = \frac{q_1 q_2}{4\pi\epsilon r^2}$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rank                                                         | Learned expression                                                                                                                   | Test RMSE | Test $R^2$ | log JMP  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|----------|
| 1–5                                                          | <b>Raw:</b> $0.014 - 0.011 \frac{1}{q_1}$<br>$- 0.011 \frac{1}{q_1} - 0.011 \frac{1}{q_1}$<br>$+ 0.083 \frac{q_1 q_2}{\epsilon r^2}$ | 0.143     | 0.238      | 2432.541 |
|                                                              | <b>Final:</b> $0.084 \frac{q_1 q_2}{\epsilon r^2}$                                                                                   | 0.144     | 0.235      | —        |
| 6–10                                                         | <b>Raw:</b> $0.014 - 0.011 \frac{1}{q_1}$<br>$- 0.011 \frac{1}{q_1} + 0.083 \frac{q_1 q_2}{\epsilon r^2}$<br>$- 0.011 \frac{1}{q_1}$ | 0.143     | 0.238      | 2432.541 |
|                                                              | <b>Final:</b> $0.084 \frac{q_1 q_2}{\epsilon r^2}$                                                                                   | 0.144     | 0.235      | —        |
| <i>Noise level: <math>\sigma = 0.20</math>, Repetition 1</i> |                                                                                                                                      |           |            |          |
| 1–9                                                          | <b>Raw:</b> $- 0.045 + 0.035 \frac{1}{r}$<br>$+ 0.035 \frac{1}{r} + 0.035 \frac{1}{r}$<br>$+ 0.085 \frac{q_1 q_2}{\epsilon r^2}$     | 0.198     | 0.205      | 1922.118 |
|                                                              | <b>Final:</b> $- 0.045 + 0.110 \frac{1}{r}$<br>$+ 0.085 \frac{q_1 q_2}{\epsilon r^2}$                                                | 0.198     | 0.205      | —        |
| 10                                                           | <b>Raw:</b> $- 0.172 + 0.121 \frac{1}{r}$<br>$+ 0.024 r + 0.121 \frac{1}{r}$<br>$+ 0.085 \frac{q_1 q_2}{\epsilon r^2}$               | 0.198     | 0.202      | 1921.041 |
|                                                              | <b>Final:</b> $- 0.045 + 0.110 \frac{1}{r}$<br>$+ 0.085 \frac{q_1 q_2}{\epsilon r^2}$                                                | 0.198     | 0.205      | —        |
| <i>Noise level: <math>\sigma = 0.25</math>, Repetition 1</i> |                                                                                                                                      |           |            |          |
| 1                                                            | <b>Raw:</b> $- 0.067 + 0.082 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.060 \frac{1}{r} + 0.060 \frac{1}{r}$<br>$+ 0.060 \frac{1}{r}$     | 0.236     | 0.108      | 1490.524 |
|                                                              | <b>Final:</b> $- 0.067 + 0.082 \frac{q_1 q_2}{\epsilon r^2}$<br>$+ 0.180 \frac{1}{r}$                                                | 0.236     | 0.108      | —        |

*Continued on next page*

Table 43: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I\_12\_2 (CL):  $F = \frac{q_1 q_2}{4\pi \epsilon r^2}$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rank | Learned expression                                   | Test RMSE | Test $R^2$ | log JMP  |
|------|------------------------------------------------------|-----------|------------|----------|
| 2–10 | <b>Raw:</b> $-0.064 - 0.033 \sin(q_1)$               | 0.240     | 0.074      | 1490.074 |
|      | $+ 0.127 \frac{q_2}{\epsilon r} + 0.088 \frac{1}{r}$ |           |            |          |
|      | $+ 0.088 \frac{1}{r}$                                |           |            |          |
|      | <b>Final:</b> $-0.064 - 0.033 \sin(q_1)$             | 0.240     | 0.074      | —        |
|      | $+ 0.130 \frac{q_2}{\epsilon r} + 0.180 \frac{1}{r}$ |           |            |          |

### D.3 Results for I\_12\_11 (FCE)

#### D.3.1 Expressions Learned by **HierBOSSS**

Table 44: **HierBOSSS** results for the Feynman equation I\_12\_11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HierBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                         | Train RMSE             | Test RMSE              | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------|------------------------|------------------------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                            |                        |                        |             |            |             |
| 1                                              | <b>Raw:</b> $E_f q + Bqv \sin(\theta) + 3.088 \times 10^{-7}$              | $1.705 \times 10^{-7}$ | $1.782 \times 10^{-7}$ | 1.000       | 1.000      | 15.484      |
|                                                | <b>Final:</b> $E_f q + Bqv \sin(\theta)$                                   | $8.627 \times 10^{-8}$ | $8.641 \times 10^{-8}$ | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $E_f q + Bqv \sin(\theta) + 7.311 \times 10^{-7}$              | $4.202 \times 10^{-7}$ | $4.186 \times 10^{-7}$ | 1.000       | 1.000      | 14.780      |
|                                                | <b>Final:</b> $E_f q + Bqv \sin(\theta) + 1.196 \times 10^{-6}$            | $6.817 \times 10^{-7}$ | $6.778 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 3                                              | <b>Raw:</b> $E_f q + Bqv \sin(\theta) + 2.381 \times 10^{-7}$              | $1.340 \times 10^{-7}$ | $1.308 \times 10^{-7}$ | 1.000       | 1.000      | 15.398      |
|                                                | <b>Final:</b> $E_f q + Bqv \sin(\theta) + 2.453 \times 10^{-7}$            | $1.381 \times 10^{-7}$ | $1.347 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 4                                              | <b>Raw:</b> $E_f q + Bqv \sin(\theta) + 5.151 \times 10^{-7}$              | $2.838 \times 10^{-7}$ | $2.975 \times 10^{-7}$ | 1.000       | 1.000      | 15.166      |
|                                                | <b>Final:</b> $E_f q + Bqv \sin(\theta)$                                   | $1.460 \times 10^{-7}$ | $1.446 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 5                                              | <b>Raw:</b> $E_f q + Bqv \sin(\theta) + 1.896 \times 10^{-7}$              | $1.063 \times 10^{-7}$ | $1.085 \times 10^{-7}$ | 1.000       | 1.000      | 14.862      |
|                                                | <b>Final:</b> $E_f q + Bqv \sin(\theta) + 1.907 \times 10^{-7}$            | $1.076 \times 10^{-7}$ | $1.097 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                            |                        |                        |             |            |             |
| 1                                              | <b>Raw:</b> $E_f q + Bqv \sin(\theta) - 0.091 \frac{1}{B} + 0.038$         | 0.248                  | 0.240                  | 1.000       | 1.000      | 15.472      |
|                                                | <b>Final:</b> $1.001 E_f q + Bqv \sin(\theta)$                             | 0.248                  | 0.238                  | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $1.002 E_f q + Bqv \sin(\theta) + 0.057 \frac{1}{E_f} - 0.033$ | 0.255                  | 0.244                  | 1.000       | 1.000      | 13.165      |
|                                                | <b>Final:</b> $1.001 E_f q + Bqv \sin(\theta)$                             | 0.255                  | 0.244                  | 1.000       | 1.000      | —           |
| 3                                              | <b>Raw:</b> $1.001 E_f q + Bqv \sin(\theta) + 0.062 \frac{1}{E_f} - 0.032$ | 0.254                  | 0.261                  | 1.000       | 1.000      | 13.696      |
|                                                | <b>Final:</b> $E_f q + Bqv \sin(\theta)$                                   | 0.254                  | 0.260                  | 1.000       | 1.000      | —           |

Continued on next page

Table 44: **HIERBOSSS** results for the Feynman equation I\_12\_11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                               | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4                                              | <b>Raw:</b> $E_f q + 1.001 B q v \sin(\theta) - 0.041 \frac{1}{B} + 0.009$       | 0.252      | 0.246     | 1.000       | 1.000      | 14.391      |
|                                                | <b>Final:</b> $E_f q + 1.001 B q v \sin(\theta)$                                 | 0.253      | 0.246     | 1.000       | 1.000      | —           |
| 5                                              | <b>Raw:</b> $E_f q + B q v \sin(\theta) + 0.041 \frac{1}{B} - 0.018$             | 0.249      | 0.231     | 1.000       | 1.000      | 13.686      |
|                                                | <b>Final:</b> $0.999 E_f q + B q v \sin(\theta)$                                 | 0.249      | 0.231     | 1.000       | 1.000      | —           |
| <i>Noise level: <math>\sigma = 1.00</math></i> |                                                                                  |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $1.002 E_f q + 0.999 B q v \sin(\theta) + 0.342 \frac{1}{B} - 0.125$ | 0.985      | 0.983     | 0.999       | 0.999      | 13.525      |
|                                                | <b>Final:</b> $1.003 E_f q + 0.999 B q v \sin(\theta)$                           | 0.987      | 0.978     | 0.999       | 0.999      | —           |
| 2                                              | <b>Raw:</b> $E_f q + 1.002 B q v \sin(\theta) - 0.174 \frac{1}{q} - 0.001$       | 1.001      | 0.969     | 0.998       | 0.998      | 13.876      |
|                                                | <b>Final:</b> $0.995 E_f q + 1.002 B q v \sin(\theta)$                           | 1.003      | 0.968     | 0.998       | 0.998      | —           |
| 3                                              | <b>Raw:</b> $0.994 E_f q + 1.001 B q v \sin(\theta) - 0.415 \frac{1}{q} + 0.208$ | 1.004      | 0.990     | 0.999       | 0.999      | 14.962      |
|                                                | <b>Final:</b> $E_f q + 1.001 B q v \sin(\theta)$                                 | 1.007      | 0.996     | 0.999       | 0.999      | —           |
| 4                                              | <b>Raw:</b> $1.005 E_f q + 0.999 B q v \sin(\theta) - 0.109 \sin(e^q) + 0.008$   | 0.982      | 1.060     | 0.999       | 0.998      | 17.241      |
|                                                | <b>Final:</b> $1.005 E_f q + 0.999 B q v \sin(\theta) - 0.110 \sin(e^q)$         | 0.982      | 1.061     | 0.999       | 0.998      | —           |
| 5                                              | <b>Raw:</b> $0.996 E_f q + B q v \sin(\theta) + 0.022$                           | 1.021      | 0.973     | 0.998       | 0.999      | 15.910      |
|                                                | <b>Final:</b> $0.998 E_f q + B q v \sin(\theta)$                                 | 1.021      | 0.974     | 0.998       | 0.999      | —           |
| <i>Noise level: <math>\sigma = 2.00</math></i> |                                                                                  |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $1.002 E_f q + B q v \sin(\theta) + 0.270 \frac{1}{E_f} - 0.054$     | 2.014      | 1.849     | 0.994       | 0.995      | 15.336      |
|                                                | <b>Final:</b> $1.005 E_f q + B q v \sin(\theta)$                                 | 2.016      | 1.846     | 0.994       | 0.995      | —           |
| 2                                              | <b>Raw:</b> $1.012 E_f q + B q v \sin(\theta) + 0.247 \frac{1}{q} - 0.174$       | 1.995      | 2.103     | 0.994       | 0.992      | 15.370      |
|                                                | <b>Final:</b> $1.004 E_f q + B q v \sin(\theta)$                                 | 1.996      | 2.107     | 0.994       | 0.992      | —           |

Continued on next page

Table 44: **HIERBOSSS** results for the Feynman equation I.12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                             | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3    | <b>Raw:</b> $1.008 E_f q + 1.001 B q v \sin(\theta) + 0.235 \cos(B^2) - 0.105$ | 2.027      | 1.927     | 0.994       | 0.995      | 14.927      |
|      | <b>Final:</b> $0.999 E_f q + 1.001 B q v \sin(\theta) + 0.237 \cos(B^2)$       | 2.027      | 1.926     | 0.994       | 0.995      | —           |
| 4    | <b>Raw:</b> $1.001 E_f q + B q v \sin(\theta) + 0.345 \frac{1}{q} - 0.199$     | 1.977      | 1.691     | 0.994       | 0.996      | 15.981      |
|      | <b>Final:</b> $0.994 E_f q + B q v \sin(\theta)$                               | 1.978      | 1.689     | 0.994       | 0.996      | —           |
| 5    | <b>Raw:</b> $E_f q + 1.004 B q v \sin(\theta) + 0.236 \cos(B) + 0.036$         | 1.962      | 1.958     | 0.995       | 0.994      | 15.997      |
|      | <b>Final:</b> $0.994 E_f q + 1.004 B q v \sin(\theta)$                         | 1.965      | 1.958     | 0.995       | 0.994      | —           |

### D.3.2 Top Expressions Ranked by JMP

Table 45: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I.12.11 (FCE):  $F = q(E_f + Bv \sin \theta)$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rank                                                         | Learned expression                                                               | Test RMSE              | Test $R^2$ | log JMP  |
|--------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------|------------|----------|
| <i>Noiseless setting, Repetition 1</i>                       |                                                                                  |                        |            |          |
| 1–10                                                         | <b>Raw:</b> $E_f q + B q v \sin(\theta) + 3.088 \times 10^{-7}$                  | $1.782 \times 10^{-7}$ | 1.000      | 8367.026 |
|                                                              | <b>Final:</b> $E_f q + B q v \sin(\theta)$                                       | $8.641 \times 10^{-8}$ | 1.000      | —        |
| <i>Noise level: <math>\sigma = 0.25</math>, Repetition 1</i> |                                                                                  |                        |            |          |
| 1–10                                                         | <b>Raw:</b> $E_f q + B q v \sin(\theta) - 0.091 \frac{1}{B} + 0.038$             | 0.240                  | 1.000      | 1517.584 |
|                                                              | <b>Final:</b> $1.001 E_f q + B q v \sin(\theta)$                                 | 0.238                  | 1.000      | —        |
| <i>Noise level: <math>\sigma = 1.00</math>, Repetition 1</i> |                                                                                  |                        |            |          |
| 1–10                                                         | <b>Raw:</b> $1.002 E_f q + 0.999 B q v \sin(\theta) + 0.342 \frac{1}{B} - 0.125$ | 0.983                  | 0.999      | -964.878 |
|                                                              | <b>Final:</b> $1.003 E_f q + 0.999 B q v \sin(\theta)$                           | 0.978                  | 0.999      | —        |
| <i>Noise level: <math>\sigma = 2.00</math>, Repetition 1</i> |                                                                                  |                        |            |          |

*Continued on next page*



Table 45: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I\_12\_11 (**FCE**):  $F = q(E_f + Bv \sin \theta)$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rank | Learned expression                               | Test RMSE | Test $R^2$ | log JMP   |
|------|--------------------------------------------------|-----------|------------|-----------|
| 1-10 | <b>Raw:</b> $1.002 E_f q + B q v \sin(\theta)$   | 1.849     | 0.995      | -2252.841 |
|      | $+ 0.270 \frac{1}{E_f} - 0.054$                  |           |            |           |
|      | <b>Final:</b> $1.005 E_f q + B q v \sin(\theta)$ | 1.846     | 0.995      | —         |

## D.4 Results for I\_24\_6 (EHFO)

### D.4.1 Expressions Learned by **HierBOSSS**

Table 46: **HierBOSSS** results for the Feynman equation I\_24\_6 (EHFO):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HierBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                            | Train RMSE             | Test RMSE              | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                               |                        |                        |             |            |             |
| 1                                              | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2 - 4.500 \times 10^{-9} + 4.780 \times 10^{-8} \frac{1}{\omega}$                                                                      | $2.817 \times 10^{-8}$ | $2.595 \times 10^{-8}$ | 1.000       | 1.000      | 66.103      |
|                                                | <b>Final:</b> $0.248 m\omega^2 x^2 + 0.248 m\omega_0^2 x^2$                                                                                                                                   | $2.816 \times 10^{-8}$ | $2.595 \times 10^{-8}$ | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $0.250 mx^2(\omega^2 + \omega_0^2) - 3.780 \times 10^{-7} \sin(\omega) + 6.420 \times 10^{-7}$                                                                                    | $2.551 \times 10^{-7}$ | $2.464 \times 10^{-7}$ | 1.000       | 1.000      | 70.628      |
|                                                | <b>Final:</b> $0.250 mx^2(\omega^2 + \omega_0^2)$                                                                                                                                             | $2.004 \times 10^{-7}$ | $2.021 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 3                                              | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2 + 8.120 \times 10^{-8} - 6.460 \times 10^{-8} \frac{1}{m}$                                                                           | $3.260 \times 10^{-8}$ | $3.451 \times 10^{-8}$ | 1.000       | 1.000      | 62.323      |
|                                                | <b>Final:</b> $0.248 m\omega^2 x^2 + 0.248 m\omega_0^2 x^2$                                                                                                                                   | $2.589 \times 10^{-8}$ | $2.687 \times 10^{-8}$ | 1.000       | 1.000      | —           |
| 4                                              | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2 + 6.090 \times 10^{-9} \sin(\omega_0) + 2.530 \times 10^{-8}$                                                                        | $2.645 \times 10^{-8}$ | $2.542 \times 10^{-8}$ | 1.000       | 1.000      | 63.981      |
|                                                | <b>Final:</b> $0.248 m\omega^2 x^2 + 0.252 m\omega_0^2 x^2$                                                                                                                                   | $2.594 \times 10^{-8}$ | $2.524 \times 10^{-8}$ | 1.000       | 1.000      | —           |
| 5                                              | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2 - 1.150 \times 10^{-7} \sin(\omega) + 1.250 \times 10^{-7}$                                                                          | $3.481 \times 10^{-8}$ | $3.788 \times 10^{-8}$ | 1.000       | 1.000      | 60.097      |
|                                                | <b>Final:</b> $0.248 m\omega^2 x^2 + 0.248 m\omega_0^2 x^2$                                                                                                                                   | $2.323 \times 10^{-8}$ | $2.547 \times 10^{-8}$ | 1.000       | 1.000      | —           |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                               |                        |                        |             |            |             |
| 1                                              | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2 - 0.034 \sin(x) + 0.038$                                                                                                             | 0.149                  | 0.143                  | 1.000       | 1.000      | 61.153      |
|                                                | <b>Final:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$                                                                                                                                   | 0.149                  | 0.143                  | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2 - 0.012 + 0.016 \frac{1}{\omega_0}$                                                                                                  | 0.153                  | 0.146                  | 1.000       | 1.000      | 63.261      |
|                                                | <b>Final:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$                                                                                                                                   | 0.153                  | 0.146                  | 1.000       | 1.000      | —           |
| 3                                              | <b>Raw:</b> $0.129 m^2 x^2 e^{-\omega_0^2} - 0.514 m\omega x^2 \cos(\omega) + 0.925 m\omega_0 \left(x^2 + \frac{1}{\omega_0}\right) - 1.310 mx \sin(\sin(\omega_0)) - 0.514 \omega_0 + 0.929$ | 0.483                  | 0.496                  | 0.999       | 0.999      | 71.026      |
|                                                |                                                                                                                                                                                               |                        |                        |             |            |             |

Continued on next page

Table 46: **HIERBOSSS** results for the Feynman equation I.24\_6 (**EHFO**):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep. | Learned expression                                                                                                                                                                   | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4    | <b>Final:</b> $0.130 m^2 x^2 e^{-\omega_0^2} - 0.510 m \omega x^2 \cos(\omega)$<br>$- 1.300 m x \sin(\sin(\omega_0))$<br>$+ 0.920 m(\omega_0 x^2 + 1)$<br>$- 0.510 \omega_0 + 0.930$ | 0.483      | 0.496     | 0.999       | 0.999      | —           |
|      | <b>Raw:</b> $- 0.360 m^3 + 0.541 m^2 \omega_0 x$<br>$+ 0.251 m \omega^2 x^2 + 0.541 m$<br>$+ 0.035 \omega_0^3 x^3 - 0.360 \omega_0$<br>$- 0.360 x + 0.494$                           | 0.571      | 0.622     | 0.998       | 0.998      | 63.494      |
|      | <b>Final:</b> $- 0.360 m^3 + 0.250 m \omega^2 x^2$<br>$+ 0.540 m(m \omega_0 x + 1) + 0.035 \omega_0^3 x^3$<br>$- 0.360 \omega_0 - 0.360 x$<br>$+ 0.490$                              | 0.571      | 0.622     | 0.998       | 0.998      | —           |
|      | <b>Raw:</b> $0.665 m \omega^2 x + 0.885 m \omega_0 x^2$<br>$- 0.018 x^4 (m + 2\omega) \cos(\omega)$<br>$- 0.032 (m + \omega + x + \sin(\sin(\omega_0)))^3$<br>$+ 0.821$              | 0.633      | 0.586     | 0.998       | 0.998      | 61.686      |
| 5    | <b>Final:</b> $0.660 m \omega^2 x + 0.880 m \omega_0 x^2$<br>$- 0.018 x^4 (m + 2\omega) \cos(\omega)$<br>$- 0.032 (m + \omega + x + \sin(\sin(\omega_0)))^3$<br>$+ 0.820$            | 0.633      | 0.586     | 0.998       | 0.998      | —           |
|      | <b>Raw:</b> $0.250 m \omega^2 x^2 + 0.250 m \omega_0^2 x^2$<br>$+ 0.050 - 0.048 \frac{1}{\omega_0}$                                                                                  | 0.247      | 0.245     | 1.000       | 1.000      | 62.123      |
|      | <b>Final:</b> $0.250 m \omega^2 x^2 + 0.250 m \omega_0^2 x^2$                                                                                                                        | 0.247      | 0.245     | 1.000       | 1.000      | —           |
|      | <b>Raw:</b> $- 0.497 m \omega \omega_0 x^2 + 0.249 m x^2 (\omega + \omega_0)^2$<br>$- 0.031 + 0.029 \frac{1}{\omega}$                                                                | 0.251      | 0.242     | 1.000       | 1.000      | 65.230      |
| 2    | <b>Final:</b> $- 0.500 m \omega \omega_0 x^2 + 0.250 m x^2 (\omega + \omega_0)^2$                                                                                                    | 0.251      | 0.242     | 1.000       | 1.000      | —           |
|      | <b>Raw:</b> $3.980 m \omega x + 1.030 m \omega_0 x^2$<br>$- 7.600 m x + 3.720 m \left( \cos(\omega) + \frac{1}{\omega \omega_0} \right)$<br>$+ 1.030 \omega_0 - 2.290$               | 0.875      | 0.879     | 0.996       | 0.996      | 63.661      |
|      | <b>Final:</b> $4.000 m \omega x - 7.600 m x$<br>$+ 3.700 m \frac{\omega \omega_0 \cos(\omega) + 1}{\omega \omega_0} + \omega_0 (m x^2 + 1)$<br>$- 2.300$                             | 0.875      | 0.879     | 0.996       | 0.996      | —           |
|      |                                                                                                                                                                                      |            |           |             |            |             |

Noise level:  $\sigma = 0.25$

Continued on next page

Table 46: **HIERBOSSS** results for the Feynman equation I.24\_6 (**EHFO**):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 4                                              | <b>Raw:</b> $0.250 m \omega^2 x^2 + 0.250 m x \left( \omega_0^2 x + \frac{1}{x} \right) - 0.426 m - 0.209 \cos(m) + 0.289$                         | 0.246      | 0.263     | 1.000       | 1.000      | 64.655      |
|                                                | <b>Final:</b> $0.250 m \omega^2 x^2 + 0.250 m (\omega_0^2 x^2 + 1) - 0.250 m$                                                                      | 0.246      | 0.264     | 1.000       | 1.000      | —           |
| 5                                              | <b>Raw:</b> $0.250 m \omega^2 x^2 + 0.250 m \omega_0^2 x^2 - 0.033 + 0.041 \frac{1}{\omega_0}$                                                     | 0.255      | 0.244     | 1.000       | 1.000      | 62.556      |
|                                                | <b>Final:</b> $0.250 m \omega^2 x^2 + 0.250 m \omega_0^2 x^2$                                                                                      | 0.255      | 0.243     | 1.000       | 1.000      | —           |
| <b>Noise level: <math>\sigma = 1.00</math></b> |                                                                                                                                                    |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $0.250 m x^2 (\omega^2 + \omega_0^2) - 0.098 + 0.220 \frac{1}{\omega_0}$                                                               | 1.007      | 0.925     | 0.995       | 0.995      | 66.461      |
|                                                | <b>Final:</b> $0.250 m x^2 (\omega^2 + \omega_0^2)$                                                                                                | 1.008      | 0.922     | 0.995       | 0.995      | —           |
| 2                                              | <b>Raw:</b> $0.250 m \omega^2 x^2 + 0.247 m \omega_0^2 x^2 - 0.386 \sin(x) + 0.667 - 0.412 \frac{1}{\omega_0}$                                     | 0.995      | 1.061     | 0.995       | 0.992      | 63.424      |
|                                                | <b>Final:</b> $0.250 m \omega^2 x^2 + 0.250 m \omega_0^2 x^2$                                                                                      | 0.998      | 1.054     | 0.995       | 0.992      | —           |
| 3                                              | <b>Raw:</b> $0.251 m \omega^2 x^2 + 0.130 m \omega_0^2 e^x - 0.259 m \omega_0^2 \cos^6(x) - 0.177 \omega_0^2 + 0.251 \omega_0 x^2 \sin(x) - 0.687$ | 1.033      | 0.971     | 0.994       | 0.995      | 67.736      |
|                                                | <b>Final:</b> $0.130 m \omega_0^2 e^x - 0.260 m \omega_0^2 \cos^6(x) - 0.180 \omega_0^2 + 0.250 x^2 (m \omega^2 + \omega_0 \sin(x)) - 0.690$       | 1.033      | 0.971     | 0.994       | 0.995      | —           |
| 4                                              | <b>Raw:</b> $0.250 m \omega^2 x^2 + 0.049 \omega_0^3 x^2 + 0.066 x^2 (m \omega_0 + \sin(m))^2 + 0.570 - 0.991 \frac{1}{\omega_0}$                  | 1.001      | 0.878     | 0.995       | 0.996      | 67.552      |
|                                                | <b>Final:</b> $0.250 m \omega^2 x^2 + 0.049 \omega_0^3 x^2 + 0.066 x^2 (m \omega_0 + \sin(m))^2 + 0.570 - 0.990 \frac{1}{\omega_0}$                | 1.001      | 0.878     | 0.995       | 0.996      | —           |
| 5                                              | <b>Raw:</b> $0.249 m \omega^2 x^2 + 0.249 m \omega_0^2 x^2 + 0.197 - 0.315 \frac{1}{x}$                                                            | 0.983      | 0.975     | 0.996       | 0.995      | 67.711      |
|                                                | <b>Final:</b> $0.250 m \omega^2 x^2 + 0.250 m \omega_0^2 x^2$                                                                                      | 0.984      | 0.974     | 0.996       | 0.995      | —           |

### D.4.2 Top Expressions Ranked by JMP

Table 47: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I.24.6 (**EHFO**):  $E_n = \frac{1}{4}m(\omega^2 + \omega_0^2)x^2$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rank                                                         | Learned expression                                                                                                            | Test RMSE              | Test $R^2$ | log JMP   |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------|------------|-----------|
| <i>Noiseless setting, Repetition 1</i>                       |                                                                                                                               |                        |            |           |
| 1–7                                                          | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$<br>$- 4.500 \times 10^{-9} + 4.780 \times 10^{-8} \frac{1}{\omega}$ | $2.595 \times 10^{-8}$ | 1.000      | 8391.666  |
|                                                              | <b>Final:</b> $0.248 m\omega^2 x^2 + 0.248 m\omega_0^2 x^2$                                                                   | $2.595 \times 10^{-8}$ | 1.000      | —         |
| 8–10                                                         | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$<br>$- 1.160 \times 10^{-7} \sin(\omega_0) + 1.270 \times 10^{-7}$   | $3.259 \times 10^{-8}$ | 1.000      | 8391.327  |
|                                                              | <b>Final:</b> $0.248 m\omega^2 x^2 + 0.248 m\omega_0^2 x^2$                                                                   | $3.258 \times 10^{-8}$ | 1.000      | —         |
| <i>Noise level: <math>\sigma = 0.15</math>, Repetition 1</i> |                                                                                                                               |                        |            |           |
| 1–10                                                         | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$<br>$- 0.034 \sin(x) + 0.038$                                        | 0.143                  | 1.000      | 2414.742  |
|                                                              | <b>Final:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$                                                                   | 0.143                  | 1.000      | —         |
| <i>Noise level: <math>\sigma = 0.25</math>, Repetition 1</i> |                                                                                                                               |                        |            |           |
| 1–10                                                         | <b>Raw:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$<br>$+ 0.050 - 0.048 \frac{1}{\omega_0}$                             | 0.245                  | 1.000      | 1508.432  |
|                                                              | <b>Final:</b> $0.250 m\omega^2 x^2 + 0.250 m\omega_0^2 x^2$                                                                   | 0.245                  | 1.000      | —         |
| <i>Noise level: <math>\sigma = 1.00</math>, Repetition 1</i> |                                                                                                                               |                        |            |           |
| 1                                                            | <b>Raw:</b> $0.250 mx^2(\omega^2 + \omega_0^2) - 0.098$<br>$+ 0.220 \frac{1}{\omega_0}$                                       | 0.925                  | 0.995      | -1017.126 |
|                                                              | <b>Final:</b> $0.250 mx^2(\omega^2 + \omega_0^2)$                                                                             | 0.922                  | 0.995      | —         |
| 2–6                                                          | <b>Raw:</b> $0.250 mx^2(\omega^2 + \omega_0^2) + 0.916 \omega$<br>$+ 1.110 \cos(\omega) - 1.400$                              | 0.920                  | 0.995      | -1017.169 |
|                                                              | <b>Final:</b> $0.250 mx^2(\omega^2 + \omega_0^2)$                                                                             | 0.922                  | 0.995      | —         |
| 7–10                                                         | <b>Raw:</b> $0.250 mx^2(\omega^2 + \omega_0^2) - 0.165 \sin(m)$<br>$+ 0.182$                                                  | 0.921                  | 0.995      | -1017.247 |
|                                                              | <b>Final:</b> $0.250 mx^2(\omega^2 + \omega_0^2)$                                                                             | 0.922                  | 0.995      | —         |

## D.5 Results for I\_50\_26 (HOQN)

### D.5.1 Expressions Learned by **HierBOSSS**

Table 48: **HierBOSSS** results for the Feynman equation I\_50\_26 (HOQN):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HierBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                     | Learned expression                                                                                                                                  | Train RMSE             | Test RMSE              | Train $R^2$ | Test $R^2$ | Runtime (s) |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-------------|------------|-------------|
| <i>Noiseless setting</i> |                                                                                                                                                     |                        |                        |             |            |             |
| 1                        | <b>Raw:</b> $0.999 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t) + 2.458 \times 10^{-5} - 3.023 \times 10^{-5} \frac{1}{t}$                | $1.511 \times 10^{-5}$ | $1.556 \times 10^{-5}$ | 1.000       | 1.000      | 61.611      |
|                          | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                    | $1.336 \times 10^{-5}$ | $1.354 \times 10^{-5}$ | 1.000       | 1.000      | —           |
| 2                        | <b>Raw:</b> $0.999 x_1 \{\alpha + \cos(\omega t)\} - 1.000 \alpha x_1 \sin^2(\omega t) - 9.780 \times 10^{-6} \sin(\omega) + 1.710 \times 10^{-5}$  | $7.647 \times 10^{-6}$ | $7.587 \times 10^{-6}$ | 1.000       | 1.000      | 59.610      |
|                          | <b>Final:</b> $x_1 \{-0.996 \alpha \sin^2(\omega t) + \alpha + \cos(\omega t)\}$                                                                    | $7.647 \times 10^{-6}$ | $7.586 \times 10^{-6}$ | 1.000       | 1.000      | —           |
| 3                        | <b>Raw:</b> $0.999 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t) - 6.102 \times 10^{-6} \sin(\omega) + 1.358 \times 10^{-5}$               | $8.878 \times 10^{-6}$ | $9.173 \times 10^{-6}$ | 1.000       | 1.000      | 58.390      |
|                          | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                    | $7.959 \times 10^{-6}$ | $8.139 \times 10^{-6}$ | 1.000       | 1.000      | —           |
| 4                        | <b>Raw:</b> $-1.000 \alpha x_1 \sin^2(\omega t) + 1.000 \alpha x_1 + 1.000 x_1 \cos(\omega t) + 5.479 \times 10^{-5} \omega - 5.040 \times 10^{-5}$ | $8.840 \times 10^{-5}$ | $9.163 \times 10^{-5}$ | 1.000       | 1.000      | 62.377      |
|                          | <b>Final:</b> $-\alpha x_1 \sin^2(\omega t) + \alpha x_1 + x_1 \cos(\omega t)$                                                                      | $8.840 \times 10^{-5}$ | $9.163 \times 10^{-5}$ | 1.000       | 1.000      | —           |
| 5                        | <b>Raw:</b> $0.999 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t) + 2.642 \times 10^{-5} - 3.337 \times 10^{-5} \frac{1}{t}$                | $1.494 \times 10^{-5}$ | $1.378 \times 10^{-5}$ | 1.000       | 1.000      | 60.502      |
|                          | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                    | $1.319 \times 10^{-5}$ | $1.195 \times 10^{-5}$ | 1.000       | 1.000      | —           |

*Noise level:  $\sigma = 0.15$*

|   |                                                                                                                                                                             |       |       |       |       |        |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|--------|
| 1 | <b>Raw:</b> $0.184 \alpha \omega t x_1 \cos^2(\omega t) + 0.184 \alpha \{x_1 + \sin(t + x_1)\} + 0.167 t \sin(\omega + x_1) - 1.374 \cos\{\exp(\cos(\omega t))\} + 0.326$   | 0.413 | 0.405 | 0.954 | 0.962 | 58.388 |
|   | <b>Final:</b> $0.180 \alpha \omega t x_1 \cos^2(\omega t) + 0.180 \alpha \{x_1 + \sin(t + x_1)\} + 0.170 t \sin(\omega + x_1) - 1.400 \cos\{\exp(\cos(\omega t))\} + 0.330$ | 0.413 | 0.405 | 0.954 | 0.962 | —      |
| 2 | <b>Raw:</b> $1.002 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t) + 0.008 t - 0.015$                                                                                | 0.153 | 0.146 | 0.994 | 0.993 | 63.595 |

*Continued on next page*

Table 48: **HIERBOSSS** results for the Feynman equation I\_50\_26 (**HOQN**):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                                              | <b>Final:</b> $x_1\{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                   | 0.154      | 0.146     | 0.994       | 0.993      | —           |
|                                                | <b>Raw:</b> $-0.095 x_1^2 + 0.243 (\alpha + x_1)^2 \cos^2(\omega t)$<br>$-1.720 \sin\{\cos(t)\} \cos(\omega) \cos(x_1)$<br>$+1.801 \cos(\omega t) + 0.275$        | 0.326      | 0.333     | 0.974       | 0.978      | 66.467      |
|                                                | <b>Final:</b> $-0.095 x_1^2 + 0.240 (\alpha + x_1)^2 \cos^2(\omega t)$<br>$-1.700 \sin\{\cos(t)\} \cos(\omega) \cos(x_1)$<br>$+1.800 \cos(\omega t) + 0.280$      | 0.326      | 0.333     | 0.974       | 0.978      | —           |
| 4                                              | <b>Raw:</b> $0.097 \alpha x_1^2 + 0.494 \alpha x_1 \cos(2\omega t)$<br>$+0.588 \alpha + 0.588 \sin\{x_1 + \cos(\omega)\}$<br>$+1.979 \cos(\omega t) - 0.473$      | 0.394      | 0.379     | 0.960       | 0.956      | 63.475      |
|                                                | <b>Final:</b> $0.097 \alpha x_1^2 + 0.490 \alpha x_1 \cos(2\omega t)$<br>$+0.590 \alpha + 0.590 \sin\{x_1 + \cos(\omega)\}$<br>$+2.000 \cos(\omega t) - 0.470$    | 0.394      | 0.379     | 0.960       | 0.956      | —           |
|                                                | <b>Raw:</b> $0.529 \alpha x_1 + 0.819 \sin^6(t) \cos(x_1)$<br>$+1.984 \cos(\omega t) + 2.025 \cos(2\omega t)$<br>$+0.819 \frac{\sin^6(t)}{\alpha \omega} - 0.066$ | 0.637      | 0.613     | 0.897       | 0.906      | 65.792      |
| 5                                              | <b>Final:</b> $0.530 \alpha x_1 + 0.820 \sin^6(t) \cos(x_1)$<br>$+2.000 \cos(\omega t) + 2.000 \cos(2\omega t)$<br>$+0.820 \frac{\sin^6(t)}{\alpha \omega}$       | 0.637      | 0.613     | 0.897       | 0.906      | —           |
| <b>Noise level: <math>\sigma = 0.18</math></b> |                                                                                                                                                                   |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $0.118 \alpha^2 x_1 + 0.294 \exp\{x_1 \cos(\omega t)\}$<br>$+1.095 \sin\{\cos(\omega t)\} + 1.558 \cos(2\omega t)$<br>$+0.242$                        | 0.546      | 0.548     | 0.926       | 0.914      | 60.667      |
|                                                | <b>Final:</b> $0.120 \alpha^2 x_1 + 0.290 \exp\{x_1 \cos(\omega t)\}$<br>$+1.100 \sin\{\cos(\omega t)\} + 1.550 \cos(2\omega t)$<br>$+0.250$                      | 0.546      | 0.548     | 0.926       | 0.914      | —           |
|                                                | <b>Raw:</b> $1.000 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t)$<br>$-0.001 - 0.022 \frac{1}{\omega}$                                                   | 0.181      | 0.174     | 0.992       | 0.993      | 65.954      |
| 2                                              | <b>Final:</b> $x_1\{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                   | 0.181      | 0.172     | 0.992       | 0.993      | —           |
|                                                | <b>Raw:</b> $0.999 \alpha x_1 \cos^2(\omega t) + 1.002 x_1 \cos(\omega t)$<br>$+0.016 \frac{1}{x_1} - 0.007$                                                      | 0.181      | 0.179     | 0.992       | 0.993      | 64.002      |
|                                                | <b>Final:</b> $x_1\{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                   | 0.181      | 0.179     | 0.992       | 0.993      | —           |
| 3                                              | <b>Raw:</b> $0.996 \alpha x_1 \cos^2(\omega t) + 0.998 x_1 \cos(\omega t)$<br>$+0.025 \sin(\omega) - 0.002$                                                       | 0.177      | 0.189     | 0.992       | 0.992      | 59.533      |
|                                                |                                                                                                                                                                   |            |           |             |            |             |
|                                                |                                                                                                                                                                   |            |           |             |            |             |

Continued on next page

Table 48: **HIERBOSSS** results for the Feynman equation I.50.26 (**HOQN**):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5                                              | <b>Final:</b> $\alpha x_1 \cos^2(\omega t) + x_1 \cos(\omega t) + 0.023 \sin(\omega)$                                                                                                                                                                                              | 0.177      | 0.189     | 0.992       | 0.992      | —           |
|                                                | <b>Raw:</b> $-1.592 \alpha \sin^2(\omega t) + 1.829 \alpha$<br>$+ 0.884 x_1 \exp\{\cos(\omega t)\} - 0.098 x_1$<br>$- 2.114$                                                                                                                                                       | 0.422      | 0.447     | 0.952       | 0.948      | 61.797      |
|                                                | <b>Final:</b> $-1.600 \alpha \sin^2(\omega t) + 1.800 \alpha$<br>$+ 0.880 x_1 \exp\{\cos(\omega t)\} - 0.098 x_1$<br>$- 2.100$                                                                                                                                                     | 0.422      | 0.447     | 0.952       | 0.948      | —           |
| <i>Noise level: <math>\sigma = 0.20</math></i> |                                                                                                                                                                                                                                                                                    |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $-0.013 \alpha^2 \omega \left(t + \frac{1}{t}\right) + 0.204 \alpha \sin(x_1)$<br>$+ 0.204 \alpha + 0.204 x_1$<br>$+ 0.054 (\alpha x_1 + \omega + t)^2 \cos^2(\omega t)$<br>$+ 0.203 \left(t + x_1^2 + x_1 + \frac{1}{t}\right) \sin\{\cos(\omega t)\}$<br>$- 0.816$   | 0.255      | 0.264     | 0.984       | 0.984      | 61.204      |
|                                                | <b>Final:</b> $-0.013 \alpha^2 \omega \left(t + \frac{1}{t}\right) + 0.200 \alpha \sin(x_1)$<br>$+ 0.200 \alpha + 0.200 x_1$<br>$+ 0.054 (\alpha x_1 + \omega + t)^2 \cos^2(\omega t)$<br>$+ 0.200 \left(t + x_1^2 + x_1 + \frac{1}{t}\right) \sin\{\cos(\omega t)\}$<br>$- 0.820$ | 0.255      | 0.264     | 0.984       | 0.984      | —           |
| 2                                              | <b>Raw:</b> $0.501 \alpha x_1 \cos(2\omega t) + 0.503 \alpha x_1$<br>$+ 1.002 x_1 \cos(\omega t) - 0.007$                                                                                                                                                                          | 0.200      | 0.211     | 0.990       | 0.984      | 62.652      |
|                                                | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                                                                                                                                   | 0.201      | 0.211     | 0.990       | 0.984      | —           |
| 3                                              | <b>Raw:</b> $1.001 \alpha x_1 \cos^2(\omega t) + 1.005 x_1 \cos(\omega t)$<br>$+ 0.039 \frac{1}{x_1} - 0.025$                                                                                                                                                                      | 0.203      | 0.192     | 0.990       | 0.991      | 62.807      |
|                                                | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                                                                                                                                   | 0.203      | 0.192     | 0.990       | 0.991      | —           |
| 4                                              | <b>Raw:</b> $0.999 \alpha x_1 \cos^2(\omega t) + 0.996 x_1 \cos(\omega t)$<br>$+ 0.009 t - 0.023$                                                                                                                                                                                  | 0.198      | 0.169     | 0.989       | 0.993      | 64.214      |
|                                                | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                                                                                                                                   | 0.198      | 0.169     | 0.989       | 0.993      | —           |
| 5                                              | <b>Raw:</b> $1.000 \alpha x_1 \cos^2(\omega t) + 0.999 x_1 \cos(\omega t)$<br>$- 0.010 t + 0.013$                                                                                                                                                                                  | 0.197      | 0.195     | 0.990       | 0.990      | 68.162      |
|                                                | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                                                                                                                                                   | 0.197      | 0.195     | 0.990       | 0.990      | —           |



### D.5.2 Top Expressions Ranked by JMP

Table 49: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation I.50.26 (**HOQN**):  $x = x_1[\cos(\omega t) + \alpha \cos^2(\omega t)]$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rank                                                         | Learned expression                                                                                                                                            | Test RMSE              | Test $R^2$ | log JMP  |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------|----------|
| <i>Noiseless setting, Repetition 1</i>                       |                                                                                                                                                               |                        |            |          |
| 1–10                                                         | <b>Raw:</b> $0.999 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t)$<br>$+ 2.458 \times 10^{-5} - 3.023 \times 10^{-5} \frac{1}{t}$                     | $1.556 \times 10^{-5}$ | 1.000      | 7792.650 |
|                                                              | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                              | $1.354 \times 10^{-5}$ | 1.000      | —        |
| <i>Noise level: <math>\sigma = 0.15</math>, Repetition 2</i> |                                                                                                                                                               |                        |            |          |
| 1–8                                                          | <b>Raw:</b> $1.002 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t)$<br>$+ 0.004 t - 0.004 (-t)$<br>$- 0.015$                                           | 0.146                  | 0.993      | 2364.333 |
|                                                              | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                              | 0.146                  | 0.993      | —        |
| 9                                                            | <b>Raw:</b> $1.002 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t)$<br>$+ 0.004 t + 0.004 t$<br>$- 0.015$                                              | 0.146                  | 0.993      | 2364.119 |
|                                                              | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                              | 0.146                  | 0.993      | —        |
| 10                                                           | <b>Raw:</b> $1.002 \alpha x_1 \cos^2(\omega t) + 1.001 x_1 \cos(\omega t)$<br>$- 0.002 \cos(x_1) - 0.002 \cos(x_1)$<br>$+ 8.634 \times 10^{-4}$               | 0.146                  | 0.993      | 2364.089 |
|                                                              | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                              | 0.146                  | 0.993      | —        |
| <i>Noise level: <math>\sigma = 0.18</math>, Repetition 2</i> |                                                                                                                                                               |                        |            |          |
| 1–10                                                         | <b>Raw:</b> $1.000 \alpha x_1 \cos^2(\omega t) + 1.000 x_1 \cos(\omega t)$<br>$- 0.011 \frac{1}{\omega} - 0.011 \frac{1}{\omega}$<br>$- 8.373 \times 10^{-4}$ | 0.174                  | 0.993      | 2067.444 |
|                                                              | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                              | 0.172                  | 0.994      | —        |
| <i>Noise level: <math>\sigma = 0.20</math>, Repetition 2</i> |                                                                                                                                                               |                        |            |          |
| 1–10                                                         | <b>Raw:</b> $0.501 \alpha x_1 \cos(2\omega t) + 1.002 x_1 \cos(\omega t)$<br>$+ 0.251 \alpha x_1 + 0.251 \alpha x_1$<br>$- 0.007$                             | 0.211                  | 0.984      | 1880.029 |
|                                                              | <b>Final:</b> $x_1 \{\alpha \cos^2(\omega t) + \cos(\omega t)\}$                                                                                              | 0.211                  | 0.983      | —        |

## D.6 Results for II\_36\_38 (FMMM)

### D.6.1 Expressions Learned by **HierBOSSS**

Table 50: **HIERBOSSS** results for the Feynman equation II\_36\_38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rep.                                           | Learned expression                                                                                                                                                                                                                                                            | Train RMSE             | Test RMSE              | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-------------|------------|-------------|
| <i>Noiseless setting</i>                       |                                                                                                                                                                                                                                                                               |                        |                        |             |            |             |
| 1                                              | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 8.3 \times 10^{-7} - \frac{3.92 \times 10^{-7}}{H}$                                                                                                                            | $4.840 \times 10^{-7}$ | $4.782 \times 10^{-7}$ | 1.000       | 1.000      | 1116.153    |
|                                                | <b>Final:</b> $\frac{1.0H\text{mom}}{Tk_b} + \frac{1.0M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                                                                                   | $4.840 \times 10^{-7}$ | $4.782 \times 10^{-7}$ | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} - 1.45 \times 10^{-6} + \frac{4.18 \times 10^{-6}}{c}$                                                                                                                           | $9.941 \times 10^{-7}$ | $1.161 \times 10^{-6}$ | 1.000       | 1.000      | 1166.543    |
|                                                | <b>Final:</b> $\frac{1.0H\text{mom}}{Tk_b} + \frac{0.996M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                                                                                 | $9.936 \times 10^{-7}$ | $1.160 \times 10^{-6}$ | 1.000       | 1.000      | —           |
| 3                                              | <b>Raw:</b> $-1.9 \times 10^{-7} + 0.990 \frac{H\text{mom}}{Tk_b} + 0.190 \frac{M^2\text{mom}}{T\epsilon k_b c^2} + 0.0036 \frac{M^2\alpha\text{mom}}{T\epsilon k_b} + 0.00025 \epsilon \left\{ T\epsilon - \alpha^2 - \alpha k_b + ck_b + k_b - \epsilon\text{mom} \right\}$ | 0.035                  | 0.030                  | 0.999       | 0.999      | 1217.604    |
|                                                | <b>Final:</b> $\frac{0.990H\text{mom}}{Tk_b} + \frac{0.187M^2\text{mom}}{T\epsilon k_b c^2} + \frac{0.0033M^2\alpha\text{mom}}{T\epsilon k_b}$                                                                                                                                | 0.035                  | 0.030                  | 0.999       | 0.999      | —           |
| 4                                              | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 3.15 \times 10^{-7} + \frac{2.31 \times 10^{-6}}{c}$                                                                                                                           | $1.043 \times 10^{-6}$ | $1.013 \times 10^{-6}$ | 1.000       | 1.000      | 1097.652    |
|                                                | <b>Final:</b> $\frac{0.990H\text{mom}}{Tk_b} + \frac{0.996M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                                                                               | $1.043 \times 10^{-6}$ | $1.013 \times 10^{-6}$ | 1.000       | 1.000      | —           |
| 5                                              | <b>Raw:</b> $-1.7 \times 10^{-5} + 0.995 \frac{H\text{mom}}{Tk_b} + 0.156 \frac{M^2\text{mom}}{T\epsilon c^2 k_b} + 0.0025 \frac{(M^2 + \alpha)\alpha\text{mom}}{T\epsilon k_b} + 0.00006 \frac{\sin(\cos(\epsilon)) + \alpha}{T + c + \epsilon}$                             | 0.049                  | 0.039                  | 0.998       | 0.999      | 1113.009    |
|                                                | <b>Final:</b> $\frac{0.995H\text{mom}}{Tk_b} + \frac{0.154M^2\text{mom}}{T\epsilon c^2 k_b} + \frac{0.0025(M^2 + \alpha)\alpha\text{mom}}{T\epsilon k_b}$                                                                                                                     | 0.049                  | 0.039                  | 0.998       | 0.999      | —           |
| <i>Noise level: <math>\sigma = 0.15</math></i> |                                                                                                                                                                                                                                                                               |                        |                        |             |            |             |
| 1                                              | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.995M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.0488 \sin(M) - 0.0361$                                                                                                                                                       | 0.148                  | 0.143                  | 1.000       | 1.000      | 1039.296    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                                                                               | 0.149                  | 0.144                  | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $-0.0173 + \frac{1.000\text{mom} \left( H + \frac{M\alpha}{c^2\epsilon} \right)}{Tk_b} + \frac{0.0271}{M}$                                                                                                                                                        | 0.153                  | 0.146                  | 1.000       | 1.000      | 1049.078    |
|                                                | <b>Final:</b> $\frac{1.000\text{mom} \left( Hc^2\epsilon + M\alpha \right)}{Tc^2\epsilon k_b}$                                                                                                                                                                                | 0.153                  | 0.146                  | 1.000       | 1.000      | —           |

Continued on next page

Table 50: **HIERBOSSS** results for the Feynman equation II.36-38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                         | Learned expression                                                                                                                                                                                                                 | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 3                            | <b>Raw:</b> $\frac{0.995M\alpha\text{mom}}{Tc^2\epsilon k_b} + \frac{0.153}{k_b^3}$<br>$+ 0.159H\text{mom} + 0.159T + 0.159k_b \cos(T)$<br>$- 0.256 + 5.84H\text{mom}e^{-T}e^{-k_b}$                                               | 0.160      | 0.162     | 0.999       | 0.999      | 1201.265    |
|                              | <b>Final:</b> $\frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b} + \frac{0.15}{k_b^3}$<br>$+ 0.16H\text{mom} + 0.16T + 0.16k_b \cos(T)$<br>$- 0.26 + 5.7H\text{mom}e^{-T}e^{-k_b}$                                                   | 0.160      | 0.162     | 0.999       | 0.999      | —           |
| 4                            | <b>Raw:</b> $\frac{0.970H\text{mom}}{Tk_b} - 2.16 + \frac{0.538M\alpha}{Tc^2\epsilon k_b} + \frac{0.538M\text{mom}}{Tc^2\epsilon k_b}$                                                                                             | 0.156      | 0.157     | 0.999       | 0.999      | 1222.962    |
|                              | <b>Final:</b> $\frac{0.97H\text{mom}}{Tk_b} - 2.2 + \frac{0.54M(\alpha + \text{mom})}{Tc^2\epsilon k_b}$                                                                                                                           | 0.156      | 0.157     | 0.999       | 0.999      | —           |
| 5                            | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$+ 0.0349 \sin(M) - 0.0342$                                                                                                       | 0.149      | 0.140     | 1.000       | 1.000      | 1041.973    |
|                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                                    | 0.149      | 0.139     | 1.000       | 1.000      | —           |
| Noise level: $\sigma = 0.20$ |                                                                                                                                                                                                                                    |            |           |             |            |             |
| 1                            | <b>Raw:</b> $\frac{1.010H\text{mom}}{Tk_b} + \frac{1.27M\text{mom}(2\alpha + \epsilon)}{c^2 k_b (T\epsilon + 1)^2}$<br>$+ \frac{0.00429\alpha^2 (M\text{mom} + M \cos(k_b) + 2)^2}{c^2 \epsilon}$<br>$- 0.0444 + \frac{0.0773}{c}$ | 0.199      | 0.204     | 1.000       | 1.000      | 1177.871    |
|                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.3M\text{mom}(2\alpha + \epsilon)}{c^2 k_b (T\epsilon + 1)^2}$<br>$+ \frac{0.0043\alpha^2 (M\text{mom} + M \cos(k_b) + 2)^2}{c^2 \epsilon}$<br>$+ \frac{0.077}{c}$           | 0.199      | 0.202     | 1.000       | 1.000      | —           |
| 2                            | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.01M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$+ 0.00215 - \frac{0.0429}{k_b}$                                                                                                   | 0.200      | 0.193     | 1.000       | 1.000      | 1082.862    |
|                              | <b>Final:</b> $\frac{0.990H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                                    | 0.201      | 0.193     | 1.000       | 1.000      | —           |
| 3                            | <b>Raw:</b> $\frac{1.020H\text{mom}}{Tk_b} + \frac{0.999M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$+ 2.48 \times 10^{-5} - \frac{0.0365}{\alpha}$                                                                                   | 0.201      | 0.200     | 1.000       | 1.000      | 1003.359    |
|                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$- \frac{0.036}{\alpha}$                                                                                                        | 0.201      | 0.200     | 1.000       | 1.000      | —           |
| 4                            | <b>Raw:</b> $\frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.972 + \frac{0.415H(k_b \cos(T) + \text{mom})}{k_b}$<br>$+ 0.296H + 0.296T - 1.74 \cos\left(\frac{\text{mom} \sin(T)}{T^2 k_b}\right)$                             | 0.214      | 0.222     | 1.000       | 1.000      | 1149.019    |
|                              | <b>Final:</b> $\frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.97 + \frac{0.41H(k_b \cos(T) + \text{mom})}{k_b}$<br>$+ 0.3H + 0.3T - 1.7 \cos\left(\frac{\text{mom} \sin(T)}{T^2 k_b}\right)$                                  | 0.214      | 0.222     | 1.000       | 1.000      | —           |

Continued on next page

Table 50: **HIERBOSSS** results for the Feynman equation II.36-38 (FMMM):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$  across noise levels and repetitions. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. The expressions correspond to the top-ranked by JMP. Train RMSE and  $R^2$  are computed on the 90% training set, while the test RMSE and  $R^2$  are computed on the held-out 10% test set (*continued*).

| Rep.                                           | Learned expression                                                                                                                                                                                                         | Train RMSE | Test RMSE | Train $R^2$ | Test $R^2$ | Runtime (s) |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|-------------|------------|-------------|
| 5                                              | <b>Raw:</b> $\frac{1.010H\text{mom}}{Tk_b} + \frac{1.22(M+\alpha)\cos(\alpha+\text{mom})}{Tc^2\epsilon k_b}$<br>$- \frac{1.68(M^2+c)}{T\alpha^2c^2\epsilon k_b\text{mom}}$<br>$- 0.00638 + \frac{5.73M}{Tc^2\epsilon k_b}$ | 0.205      | 0.198     | 1.000       | 1.000      | 1173.493    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.2(M+\alpha)\cos(\alpha+\text{mom})}{Tc^2\epsilon k_b}$<br>$- \frac{1.7(M^2+c)}{T\alpha^2c^2\epsilon k_b\text{mom}}$<br>$+ \frac{5.8M}{Tc^2\epsilon k_b}$            | 0.205      | 0.197     | 1.000       | 1.000      | —           |
| <i>Noise level: <math>\sigma = 0.25</math></i> |                                                                                                                                                                                                                            |            |           |             |            |             |
| 1                                              | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$- 0.0288 + \frac{0.0723}{M}$                                                                                             | 0.251      | 0.230     | 1.000       | 1.000      | 1018.236    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                            | 0.252      | 0.231     | 1.000       | 1.000      | —           |
| 2                                              | <b>Raw:</b> $\frac{1.010H\text{mom}}{Tk_b} + \frac{0.979M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$+ 0.0707\sin(\epsilon) - 0.0536$                                                                                         | 0.249      | 0.266     | 1.000       | 1.000      | 1086.514    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                            | 0.249      | 0.263     | 1.000       | 1.000      | —           |
| 3                                              | <b>Raw:</b> $\frac{0.988H\text{mom}}{Tk_b} + \frac{0.977M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$- 0.0183 + \frac{0.0663}{k_b}$                                                                                           | 0.254      | 0.243     | 1.000       | 1.000      | 1073.973    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.980M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                            | 0.254      | 0.242     | 1.000       | 1.000      | —           |
| 4                                              | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$<br>$+ 0.0587\sin(\alpha) - 0.0482$                                                                                           | 0.247      | 0.212     | 1.000       | 1.000      | 1078.759    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                                                                                            | 0.247      | 0.211     | 1.000       | 1.000      | —           |
| 5                                              | <b>Raw:</b> $\frac{1.020H\text{mom}}{Tk_b} + \frac{4.6M\alpha\text{mom}}{c\epsilon(Tc^2 + 2Tck_b + Tk_b^2 + c)}$<br>$+ 0.00862 - \frac{0.0695}{M}$                                                                         | 0.247      | 0.249     | 1.000       | 1.000      | 1042.153    |
|                                                | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} - \frac{0.058}{M} + \frac{4.6M\alpha\text{mom}}{c\epsilon(Tc^2 + 2Tck_b + Tk_b^2 + c)}$                                                                                       | 0.247      | 0.249     | 1.000       | 1.000      | —           |

### D.6.2 Top Expressions Ranked by JMP

Table 51: Top  $r = 10$  expressions ranked by JMP learned by **HIERBOSSS** for the Feynman equation II.36-38 (**FMMM**):  $f = \frac{\text{mom}H}{k_b T} + \frac{\text{mom}\alpha}{\epsilon c^2 k_b T} M$  across noise levels. **Raw** expression correspond to the symbolic forest learned by **HIERBOSSS** followed by the **Final** expression after post-MCMC symbolic model refinement. Test RMSE and  $R^2$  are computed on the held-out 10% test set.

| Rank                                                         | Learned expression                                                                                                                                           | Test RMSE              | Test $R^2$ | log JMP  |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------|----------|
| <i>Noiseless setting, Repetition 1</i>                       |                                                                                                                                                              |                        |            |          |
| 1-2                                                          | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 8.300 \times 10^{-7} - 3.920 \times 10^{-7} \frac{1}{H}$      | $4.782 \times 10^{-7}$ | 1.000      | 8296.095 |
|                                                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                              | $4.782 \times 10^{-7}$ | 1.000      | —        |
| 3-10                                                         | <b>Raw:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b} + 1.051 \times 10^{-6} - 5.877 \times 10^{-7} \sin(\text{mom})$ | $4.729 \times 10^{-7}$ | 1.000      | 8295.793 |
|                                                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{1.005M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                              | $3.377 \times 10^{-7}$ | 1.000      | —        |
| <i>Noise level: <math>\sigma = 0.15</math>, Repetition 1</i> |                                                                                                                                                              |                        |            |          |
| 1-10                                                         | <b>Raw:</b> $\frac{1.004H\text{mom}}{Tk_b} + \frac{0.995M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.049 \sin(M) - 0.036$                                        | 0.143                  | 0.985      | 2358.727 |
|                                                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                              | 0.144                  | 0.985      | —        |
| <i>Noise level: <math>\sigma = 0.20</math>, Repetition 2</i> |                                                                                                                                                              |                        |            |          |
| 1-10                                                         | <b>Raw:</b> $\frac{1.003H\text{mom}}{Tk_b} + \frac{1.006M\alpha\text{mom}}{Tc^2\epsilon k_b} + 0.002 - 0.043 \frac{1}{k_b}$                                  | 0.193                  | 0.969      | 1816.438 |
|                                                              | <b>Final:</b> $\frac{0.990H\text{mom}}{Tk_b} + \frac{1.000M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                              | 0.193                  | 0.969      | —        |
| <i>Noise level: <math>\sigma = 0.25</math>, Repetition 1</i> |                                                                                                                                                              |                        |            |          |
| 1-10                                                         | <b>Raw:</b> $\frac{1.002H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b} - 0.029 + 0.072 \frac{1}{M}$                                    | 0.230                  | 0.951      | 1419.573 |
|                                                              | <b>Final:</b> $\frac{1.000H\text{mom}}{Tk_b} + \frac{0.990M\alpha\text{mom}}{Tc^2\epsilon k_b}$                                                              | 0.231                  | 0.951      | —        |

## E Effective Symbolic Forest Size

Figure 1 displays the effective symbolic forest size  $K_{\text{eff}}$  selected by the post-MCMC symbolic model refinement in **HIERBOSSS**. Since  $K_{\text{eff}}$  is determined by the BIC-based subset selection step, it measures how many symbolic tree components are retained in the *Final* expression after removing collinear and weakly contributing tree-structured symbolic expressions. The results show that **HIERBOSSS** often selects effective forests smaller than the nominal value of  $K$ , particularly for the simpler Feynman equations such as I\_12\_2 (CL) and I\_12\_11 (FCE). In contrast, structurally richer equations such as I\_24\_6 (EHFO), II\_36\_38 (FMMM), and I\_50\_26 (HOQN) sometimes retain 3 or 4 components (closer to the nominal  $K$  used), especially under increased noise levels. This illustrates that the post-MCMC symbolic model refinement provides a data-adaptive notion of symbolic complexity, balancing parsimony with the need to preserve multiple meaningful additive components capturing the symbolic structure underlying the data.

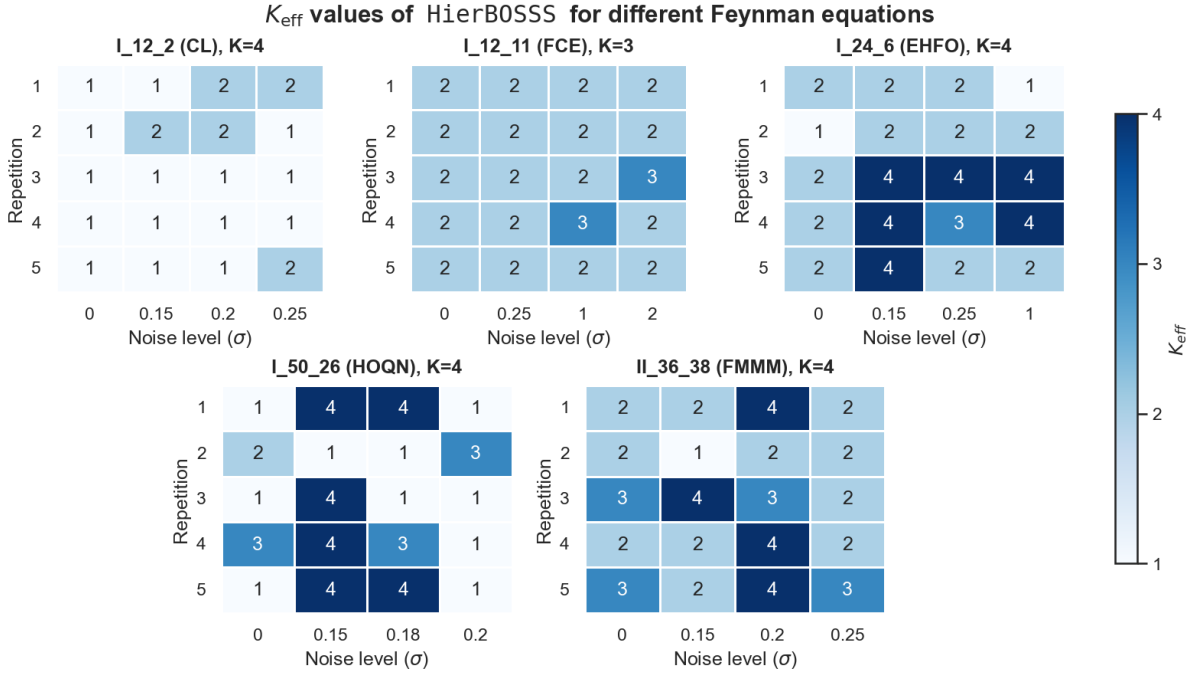


Figure 1: Effective symbolic forest size ( $K_{\text{eff}}$ ) of **HIERBOSSS** for the Feynman equations in I\_12\_2 (CL)-II\_36\_38 (FMMM), across 5 independent repetitions and different noise levels.

## F Expression Catching, Complexity, Accuracy, and Runtime Summaries for Learning Feynman Equations

### F.1 Structural Recovery

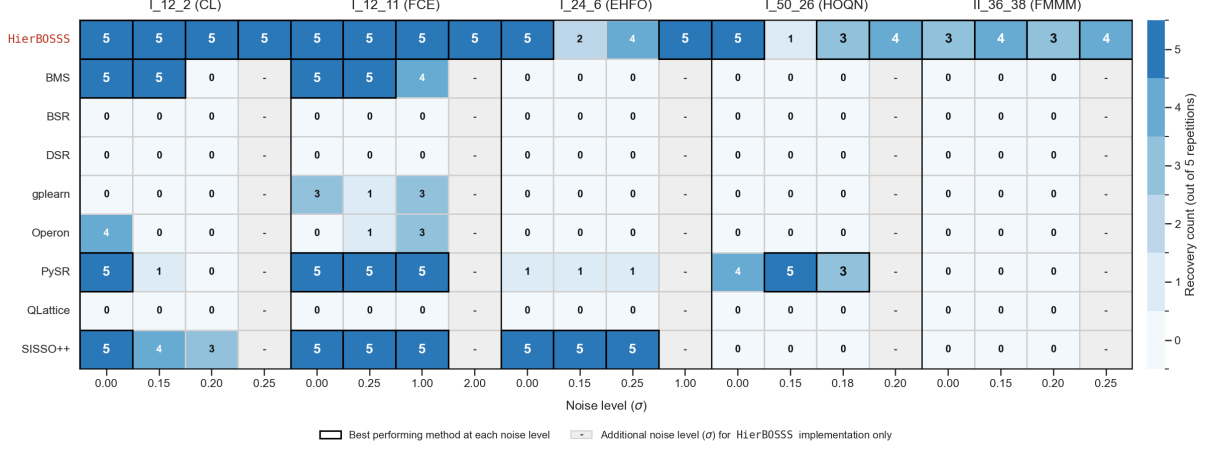
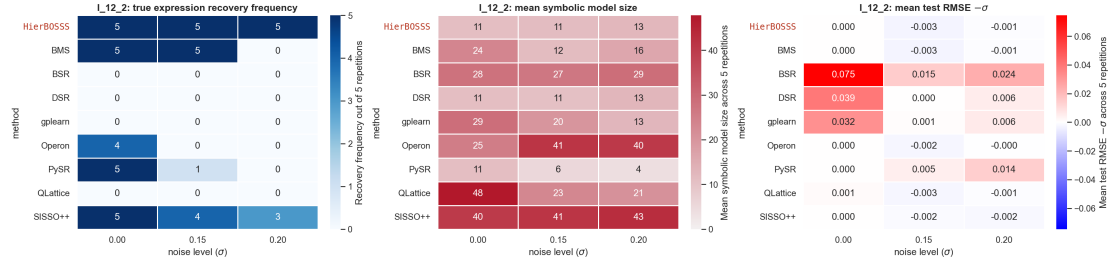


Figure 2: Structural recovery performance (out of 5 independent repetitions) of **HierBOSSS** and competitors across Feynman equations in L\_12\_2 (CL)-II\_36\_38 (FMMM) and different noise levels.

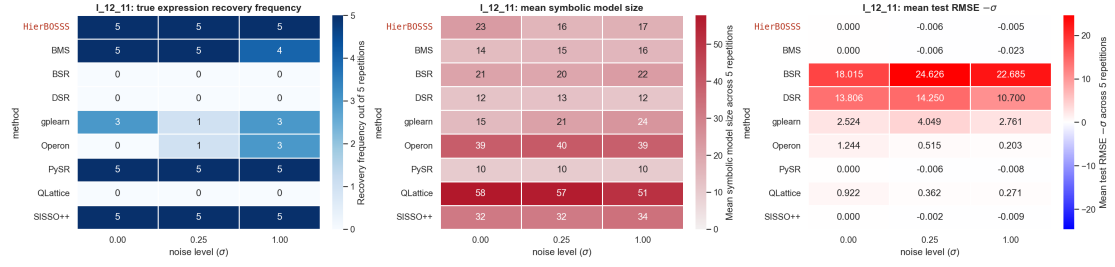
### F.2 Expression Complexity

We evaluate the complexity of the symbolic expressions learned by **HierBOSSS** and the competing SR methods, for the Feynman equations in L\_12\_2 (CL)-II\_36\_38 (FMMM). Predictive accuracy alone does not fully characterize the quality of a learned scientific expression, since an accurate expression may still be difficult to interpret if it contains redundant operators, unnecessary constants, deeply nested transformation, or unwieldy algebraic structure. This behavior is observed for several competitors (see Section C), where good numerical fit is sometimes achieved through complex nested output expressions rather than compact, interpretable scientific laws. We therefore examine expression complexity jointly with structural recovery and held-out predictive error. Experimental configurations for **HierBOSSS** and the competing methods are kept the same as those described in Sections D.1 and B.

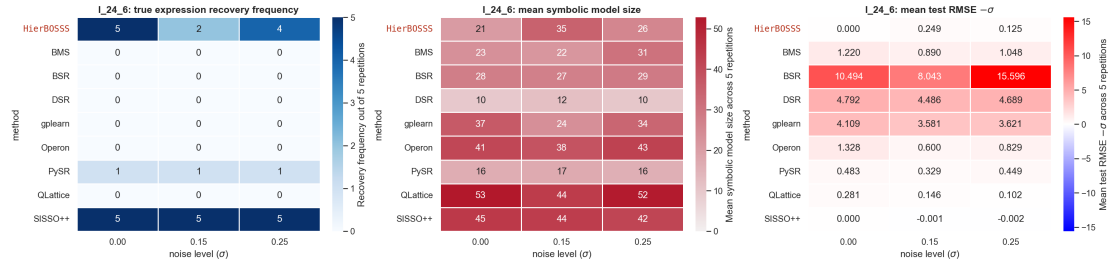
For a learned expression, we calculate its symbolic model size as the total number of nodes appearing in its expression tree [Imai Aldeia et al., 2025, La Cava et al., 2021]. Nodes include numerical constants, symbolic constants, variables or features, and mathematical operators. This gives a common symbolic model size measure across heterogeneous SR methods. For the competing methods, symbolic model size is computed from the expression returned (learned) by the corresponding method. For **HierBOSSS**, symbolic model size is computed from the *Final* expression. For each Feynman equation in L\_12\_2 (CL)-II\_36\_38 (FMMM), noise level, and method, we average the symbolic model size across the 5 independent repetitions. This average is reported together with the structural recovery frequency (out of the same 5 repetitions) and the mean (across the same 5 repetitions) test error gap, i.e., mean test RMSE  $-\sigma$ .



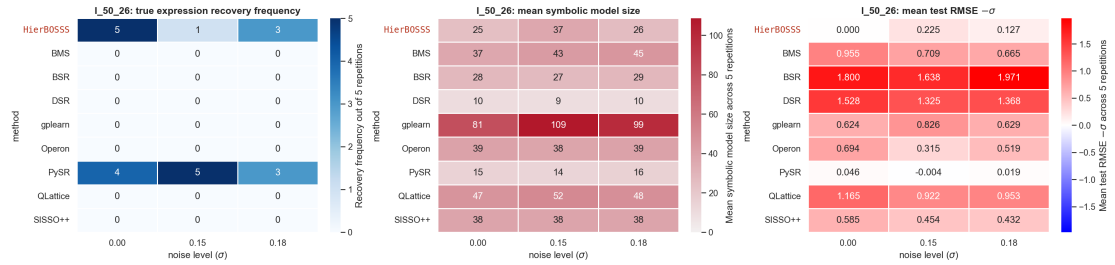
(a) I12.2 (CL)



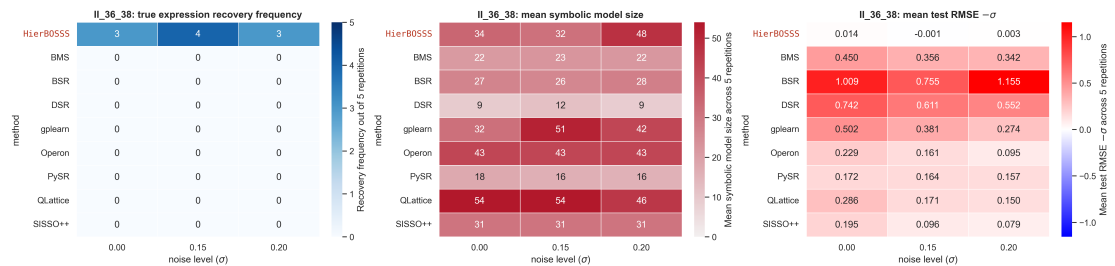
(b) I12.11 (FCE)



(c) I24.6 (EHFO)



(d) I50.26 (HOQN)



(e) II36.38 (FMMM)

Figure 3: Expression recovery, symbolic model size, and predictive error gap. For each Feynman equation in I12.2 (CL)-II36.38 (FMMM), the three panels report structurally correct recovery frequency, mean symbolic model size, and mean test RMSE $-\sigma$ , respectively, across 5 independent repetitions.



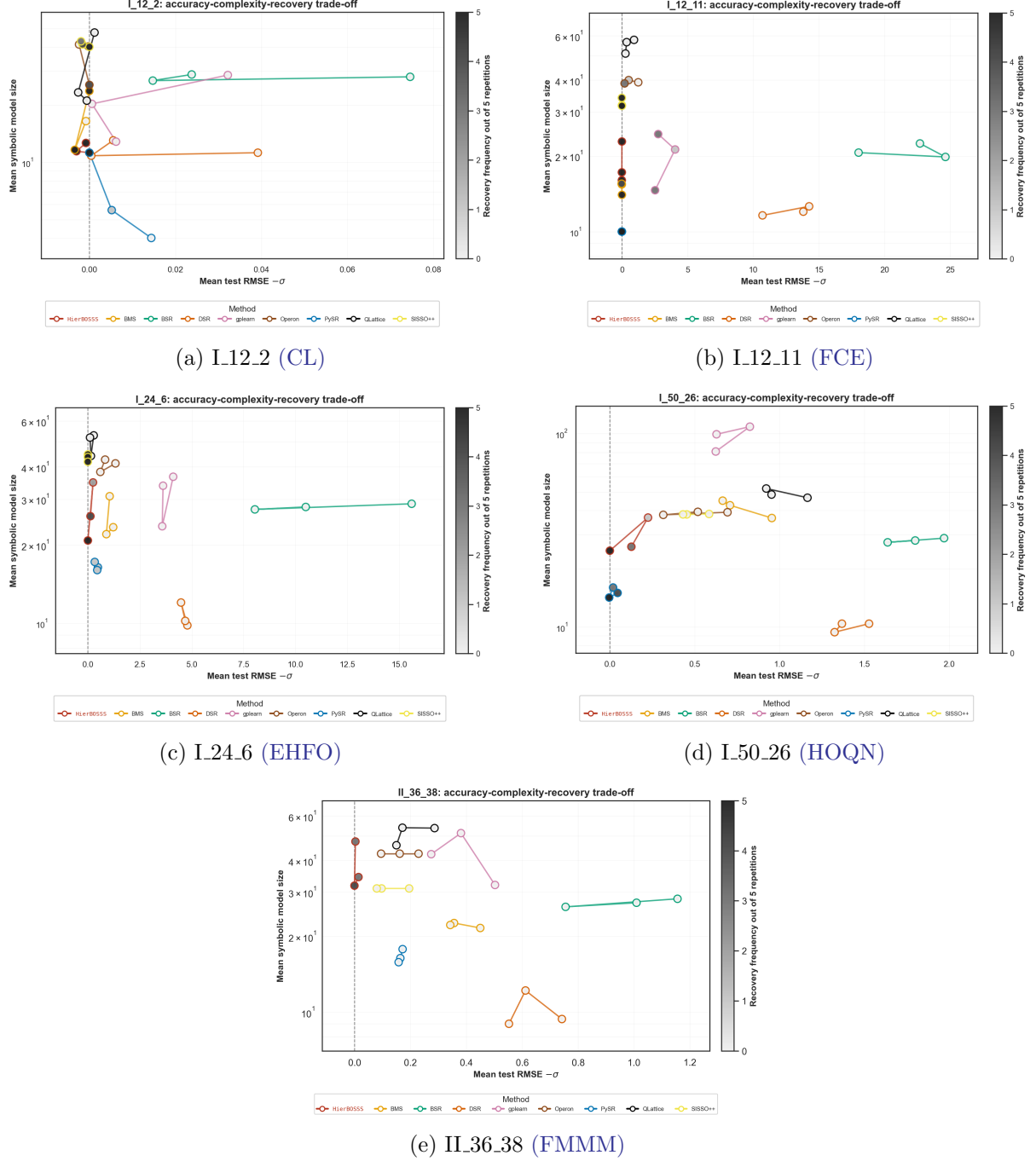


Figure 4: Accuracy-complexity-recovery trade-off plots for the Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). Each point corresponds to a method at a given noise level. The horizontal axis reports mean test RMSE  $-\sigma$ , the vertical axis reports mean symbolic model size, and point shading indicates structurally correct recovery frequency, across 5 independent repetitions.

Figure 3 summarize these three diagnostics for each of the Feynman equations in I.12.2 (CL)-II.36.38 (FMMM). These heatmaps distinguish methods that recover the true symbolic structure of the underlying Feynman equation from methods that achieve reasonable prediction through larger or only partially correct expressions.

To complement the preceding diagnostic heatmaps, we also provide accuracy-complexity-recovery

trade-off plots in Figure 4. Each point represents one method at one noise level, with the horizontal axis measuring the mean test-error gap viz., mean test RMSE  $-\sigma$ , the vertical axis measuring the mean symbolic model size, and the point shading indicating the structurally correct recovery frequency, across 5 independent repetitions. Hence, the most desirable region is near the vertical zero-error reference line, with small model size and darker point shading. Such points correspond to expressions that are compact, predictive near the noise floor, and structurally correct.

Across the Feynman equations in I.12.2 (CL)-II.36.38 (FMMM), **HIERBOSSS** generally exhibits a favorable balance of the accuracy-complexity-recovery trade-off. For the simpler equations I.12.2 (CL) and I.12.11 (FCE), **HIERBOSSS** recovers the true symbolic structure across all displayed noise levels while maintaining compact model sizes and test errors close to the noise level. Some competitors, such as BMS, PySR, and SISSO++, can also perform well for selected equations and noise levels, but their behavior is less uniform across the Feynman benchmark. Other methods often either fail to recover the true structure, produce substantially larger expressions, or exhibit larger test error gaps.

For more challenging equations, such as I.24.6 (EHFO), I.50.26 (HOQN), and II.36.38 (FMMM), the distinction between prediction and structural discovery becomes more pronounced. Several competing SR methods (e.g., DSR, BMS, PySR, operon, and SISSO++) obtain reasonable prediction errors or compact expressions in some settings, but this does not necessarily imply recovery of the underlying Feynman equation. In particular, a method may lie close to the zero-error line while still having low recovery frequency, indicating that it has found a predictive approximation rather than the correct symbolic structure. Conversely, some methods return large expressions with many redundant or nested components, making the resulting formulas less interpretable even when predictive performance is acceptable.

The behavior of **HIERBOSSS** is consistent. The depth-dependent splitting rule favors parsimonious symbolic tree structures, encoding the Occam’s razor principle [Jefferys and Berger, 1992] directly into the symbolic forest prior over  $\mathcal{T}$ . In addition, post-MCMC symbolic model refinement applies an Occam’s window-based [Madigan and Raftery, 1994] symbolic model selection using the joint marginal posterior of the symbolic forest,  $\text{JMP}(\mathcal{T})$ , retaining high-posterior expressions while discarding unnecessary symbolic components. This complete probabilistic SR framework of **HIERBOSSS** allows it balance expressiveness, predictive accuracy, structural recovery, and symbolic parsimony.

### F.3 Test RMSE and $R^2$

Table 52: Summary of test RMSE (computed on a held-out 10% test set) for **HIERBOSSS** and competitors across Feynman equations in I.12.2 (CL)-II.36.38 (FMMM) and different noise levels. Each entry reports the mean  $\pm$  standard deviation over 5 repetitions. — denotes additional noise level ( $\sigma$ ) for **HIERBOSSS** implementation only.

| Noise ( $\sigma$ ) | <b>HIERBOSSS</b>  | BMS               | BSR               | DSR               | gplearn           | operon            | PySR              | QLattice          | SISSO++           |
|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| I.12.2 (CL)        |                   |                   |                   |                   |                   |                   |                   |                   |                   |
| 0.00               | 0.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 | 0.075 $\pm$ 0.005 | 0.039 $\pm$ 0.011 | 0.032 $\pm$ 0.013 | 0.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 | 0.001 $\pm$ 0.001 | 0.000 $\pm$ 0.000 |

*Continued on next page*

Table 52: Summary of test RMSE (computed on a held-out 10% test set) for **HierBOSSS** and competitors across Feynman equations in I.12.2 (CL)-II.36.38 (FMMM) and different noise levels. Each entry reports the mean  $\pm$  standard deviation over 5 repetitions. — denotes additional noise level ( $\sigma$ ) for **HierBOSSS** implementation only (*continued*).

| Noise ( $\sigma$ )     | <b>HierBOSSS</b>  | BMS               | BSR                | DSR                | gplearn           | operon            | PySR              | QLattice          | SISSO + +         |
|------------------------|-------------------|-------------------|--------------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 0.15                   | 0.147 $\pm$ 0.006 | 0.147 $\pm$ 0.006 | 0.165 $\pm$ 0.008  | 0.150 $\pm$ 0.005  | 0.151 $\pm$ 0.005 | 0.148 $\pm$ 0.007 | 0.155 $\pm$ 0.003 | 0.147 $\pm$ 0.006 | 0.148 $\pm$ 0.002 |
| 0.20                   | 0.199 $\pm$ 0.007 | 0.199 $\pm$ 0.007 | 0.224 $\pm$ 0.015  | 0.205 $\pm$ 0.006  | 0.206 $\pm$ 0.008 | 0.200 $\pm$ 0.007 | 0.214 $\pm$ 0.013 | 0.199 $\pm$ 0.005 | 0.198 $\pm$ 0.003 |
| 0.25                   | 0.239 $\pm$ 0.019 | —                 | —                  | —                  | —                 | —                 | —                 | —                 | —                 |
| <b>I.12.11 (FCE)</b>   |                   |                   |                    |                    |                   |                   |                   |                   |                   |
| 0.00                   | 0.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 | 18.015 $\pm$ 1.724 | 13.805 $\pm$ 4.072 | 2.524 $\pm$ 3.543 | 1.244 $\pm$ 0.868 | 0.000 $\pm$ 0.000 | 0.922 $\pm$ 0.451 | 0.000 $\pm$ 0.000 |
| 0.25                   | 0.244 $\pm$ 0.011 | 0.244 $\pm$ 0.011 | 24.876 $\pm$ 3.416 | 14.500 $\pm$ 1.855 | 4.298 $\pm$ 3.260 | 0.765 $\pm$ 0.435 | 0.244 $\pm$ 0.011 | 0.612 $\pm$ 0.285 | 0.248 $\pm$ 0.003 |
| 1.00                   | 0.995 $\pm$ 0.038 | 0.977 $\pm$ 0.014 | 23.685 $\pm$ 4.119 | 11.700 $\pm$ 4.319 | 3.760 $\pm$ 3.785 | 1.203 $\pm$ 0.304 | 0.992 $\pm$ 0.038 | 1.271 $\pm$ 0.236 | 0.991 $\pm$ 0.011 |
| 2.00                   | 1.905 $\pm$ 0.153 | —                 | —                  | —                  | —                 | —                 | —                 | —                 | —                 |
| <b>I.24.6 (EHFO)</b>   |                   |                   |                    |                    |                   |                   |                   |                   |                   |
| 0.00                   | 0.000 $\pm$ 0.000 | 1.219 $\pm$ 0.090 | 10.494 $\pm$ 2.528 | 4.792 $\pm$ 0.087  | 4.109 $\pm$ 1.872 | 1.328 $\pm$ 0.536 | 0.482 $\pm$ 0.377 | 0.281 $\pm$ 0.012 | 0.000 $\pm$ 0.000 |
| 0.15                   | 0.399 $\pm$ 0.236 | 1.040 $\pm$ 0.325 | 8.193 $\pm$ 1.748  | 4.637 $\pm$ 0.227  | 3.731 $\pm$ 1.025 | 0.750 $\pm$ 0.613 | 0.479 $\pm$ 0.261 | 0.296 $\pm$ 0.036 | 0.149 $\pm$ 0.002 |
| 0.25                   | 0.375 $\pm$ 0.282 | 1.298 $\pm$ 0.370 | 15.846 $\pm$ 5.789 | 4.939 $\pm$ 0.258  | 3.871 $\pm$ 0.612 | 1.080 $\pm$ 0.387 | 0.699 $\pm$ 0.274 | 0.352 $\pm$ 0.066 | 0.248 $\pm$ 0.003 |
| 1.00                   | 0.960 $\pm$ 0.066 | —                 | —                  | —                  | —                 | —                 | —                 | —                 | —                 |
| <b>I.50.26 (HOQN)</b>  |                   |                   |                    |                    |                   |                   |                   |                   |                   |
| 0.00                   | 0.000 $\pm$ 0.000 | 0.955 $\pm$ 0.139 | 1.800 $\pm$ 0.093  | 1.528 $\pm$ 0.151  | 0.624 $\pm$ 0.391 | 0.694 $\pm$ 0.217 | 0.046 $\pm$ 0.102 | 1.165 $\pm$ 0.134 | 0.585 $\pm$ 0.015 |
| 0.15                   | 0.375 $\pm$ 0.167 | 0.859 $\pm$ 0.239 | 1.788 $\pm$ 0.162  | 1.475 $\pm$ 0.122  | 0.975 $\pm$ 0.585 | 0.465 $\pm$ 0.297 | 0.146 $\pm$ 0.006 | 1.072 $\pm$ 0.097 | 0.603 $\pm$ 0.012 |
| 0.18                   | 0.307 $\pm$ 0.178 | 0.845 $\pm$ 0.277 | 2.151 $\pm$ 0.316  | 1.548 $\pm$ 0.170  | 0.809 $\pm$ 0.198 | 0.699 $\pm$ 0.357 | 0.199 $\pm$ 0.029 | 1.133 $\pm$ 0.139 | 0.612 $\pm$ 0.012 |
| 0.20                   | 0.206 $\pm$ 0.036 | —                 | —                  | —                  | —                 | —                 | —                 | —                 | —                 |
| <b>II.36.38 (FMMM)</b> |                   |                   |                    |                    |                   |                   |                   |                   |                   |
| 0.00                   | 0.014 $\pm$ 0.019 | 0.450 $\pm$ 0.047 | 1.009 $\pm$ 0.134  | 0.742 $\pm$ 0.082  | 0.502 $\pm$ 0.113 | 0.229 $\pm$ 0.035 | 0.172 $\pm$ 0.110 | 0.286 $\pm$ 0.035 | 0.195 $\pm$ 0.023 |
| 0.15                   | 0.150 $\pm$ 0.010 | 0.506 $\pm$ 0.075 | 0.905 $\pm$ 0.120  | 0.761 $\pm$ 0.104  | 0.531 $\pm$ 0.086 | 0.311 $\pm$ 0.038 | 0.314 $\pm$ 0.061 | 0.321 $\pm$ 0.023 | 0.246 $\pm$ 0.015 |
| 0.20                   | 0.203 $\pm$ 0.011 | 0.542 $\pm$ 0.096 | 1.355 $\pm$ 0.538  | 0.752 $\pm$ 0.069  | 0.474 $\pm$ 0.087 | 0.295 $\pm$ 0.042 | 0.357 $\pm$ 0.098 | 0.350 $\pm$ 0.048 | 0.279 $\pm$ 0.013 |
| 0.25                   | 0.239 $\pm$ 0.020 | —                 | —                  | —                  | —                 | —                 | —                 | —                 | —                 |

Table 53: Summary of test  $R^2$  (computed on a held-out 10% test set) for **HIERBOSSS** and competitors across Feynman equations in I.12.2 (CL)-II.36.38 (FMMM) and different noise levels. Each entry reports the mean  $\pm$  standard deviation over 5 repetitions. — denotes additional noise level ( $\sigma$ ) for **HIERBOSSS** implementation only.

| Noise ( $\sigma$ )     | <b>HIERBOSSS</b>  | BMS               | BSR                | DSR               | gplearn           | operon            | PySR              | QLattice          | SISSO + +         |
|------------------------|-------------------|-------------------|--------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| <b>I.12.2 (CL)</b>     |                   |                   |                    |                   |                   |                   |                   |                   |                   |
| 0.00                   | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.210 $\pm$ 0.096  | 0.776 $\pm$ 0.092 | 0.827 $\pm$ 0.113 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 |
| 0.15                   | 0.243 $\pm$ 0.060 | 0.246 $\pm$ 0.063 | 0.053 $\pm$ 0.081  | 0.206 $\pm$ 0.060 | 0.202 $\pm$ 0.080 | 0.235 $\pm$ 0.059 | 0.152 $\pm$ 0.085 | 0.234 $\pm$ 0.068 | 0.249 $\pm$ 0.030 |
| 0.20                   | 0.196 $\pm$ 0.073 | 0.196 $\pm$ 0.073 | -0.011 $\pm$ 0.078 | 0.145 $\pm$ 0.058 | 0.140 $\pm$ 0.047 | 0.190 $\pm$ 0.074 | 0.073 $\pm$ 0.033 | 0.194 $\pm$ 0.077 | 0.155 $\pm$ 0.024 |
| 0.25                   | 0.101 $\pm$ 0.076 | —                 | —                  | —                 | —                 | —                 | —                 | —                 | —                 |
| <b>I.12.11 (FCE)</b>   |                   |                   |                    |                   |                   |                   |                   |                   |                   |
| 0.00                   | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.499 $\pm$ 0.047  | 0.691 $\pm$ 0.166 | 0.976 $\pm$ 0.037 | 0.997 $\pm$ 0.004 | 1.000 $\pm$ 0.000 | 0.999 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| 0.25                   | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.090 $\pm$ 0.223  | 0.687 $\pm$ 0.092 | 0.961 $\pm$ 0.041 | 0.999 $\pm$ 0.001 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| 1.00                   | 0.999 $\pm$ 0.001 | 0.999 $\pm$ 0.001 | 0.111 $\pm$ 0.329  | 0.774 $\pm$ 0.139 | 0.962 $\pm$ 0.050 | 0.998 $\pm$ 0.001 | 0.999 $\pm$ 0.001 | 0.998 $\pm$ 0.001 | 0.998 $\pm$ 0.000 |
| 2.00                   | 0.994 $\pm$ 0.002 | —                 | —                  | —                 | —                 | —                 | —                 | —                 | —                 |
| <b>I.24.6 (EHFO)</b>   |                   |                   |                    |                   |                   |                   |                   |                   |                   |
| 0.00                   | 1.000 $\pm$ 0.000 | 0.993 $\pm$ 0.001 | 0.459 $\pm$ 0.240  | 0.892 $\pm$ 0.007 | 0.909 $\pm$ 0.080 | 0.991 $\pm$ 0.007 | 0.998 $\pm$ 0.002 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 |
| 0.15                   | 0.999 $\pm$ 0.001 | 0.994 $\pm$ 0.003 | 0.657 $\pm$ 0.147  | 0.894 $\pm$ 0.013 | 0.927 $\pm$ 0.040 | 0.996 $\pm$ 0.005 | 0.999 $\pm$ 0.001 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 |
| 0.25                   | 0.999 $\pm$ 0.002 | 0.991 $\pm$ 0.005 | -0.397 $\pm$ 0.980 | 0.880 $\pm$ 0.013 | 0.925 $\pm$ 0.024 | 0.994 $\pm$ 0.005 | 0.997 $\pm$ 0.002 | 0.999 $\pm$ 0.001 | 1.000 $\pm$ 0.000 |
| 1.00                   | 0.995 $\pm$ 0.002 | —                 | —                  | —                 | —                 | —                 | —                 | —                 | —                 |
| <b>I.50.26 (HOQN)</b>  |                   |                   |                    |                   |                   |                   |                   |                   |                   |
| 0.00                   | 1.000 $\pm$ 0.000 | 0.758 $\pm$ 0.078 | 0.159 $\pm$ 0.063  | 0.390 $\pm$ 0.120 | 0.869 $\pm$ 0.162 | 0.867 $\pm$ 0.088 | 0.997 $\pm$ 0.007 | 0.643 $\pm$ 0.087 | 0.908 $\pm$ 0.003 |
| 0.15                   | 0.959 $\pm$ 0.033 | 0.792 $\pm$ 0.151 | 0.180 $\pm$ 0.106  | 0.434 $\pm$ 0.122 | 0.716 $\pm$ 0.258 | 0.923 $\pm$ 0.074 | 0.994 $\pm$ 0.001 | 0.704 $\pm$ 0.050 | 0.903 $\pm$ 0.004 |
| 0.18                   | 0.968 $\pm$ 0.036 | 0.804 $\pm$ 0.150 | -0.112 $\pm$ 0.320 | 0.433 $\pm$ 0.085 | 0.838 $\pm$ 0.083 | 0.865 $\pm$ 0.140 | 0.990 $\pm$ 0.004 | 0.691 $\pm$ 0.083 | 0.900 $\pm$ 0.005 |
| 0.20                   | 0.988 $\pm$ 0.004 | —                 | —                  | —                 | —                 | —                 | —                 | —                 | —                 |
| <b>II.36.38 (FMMM)</b> |                   |                   |                    |                   |                   |                   |                   |                   |                   |
| 0.00                   | 1.000 $\pm$ 0.001 | 0.826 $\pm$ 0.040 | 0.137 $\pm$ 0.128  | 0.530 $\pm$ 0.081 | 0.776 $\pm$ 0.096 | 0.953 $\pm$ 0.022 | 0.964 $\pm$ 0.036 | 0.927 $\pm$ 0.029 | 0.968 $\pm$ 0.005 |
| 0.15                   | 1.000 $\pm$ 0.001 | 0.791 $\pm$ 0.052 | 0.332 $\pm$ 0.138  | 0.529 $\pm$ 0.099 | 0.768 $\pm$ 0.064 | 0.921 $\pm$ 0.020 | 0.920 $\pm$ 0.026 | 0.917 $\pm$ 0.007 | 0.951 $\pm$ 0.005 |
| 0.20                   | 1.000 $\pm$ 0.000 | 0.770 $\pm$ 0.073 | -0.616 $\pm$ 1.231 | 0.560 $\pm$ 0.085 | 0.851 $\pm$ 0.031 | 0.932 $\pm$ 0.020 | 0.896 $\pm$ 0.053 | 0.905 $\pm$ 0.026 | 0.937 $\pm$ 0.006 |
| 0.25                   | 1.000 $\pm$ 0.000 | —                 | —                  | —                 | —                 | —                 | —                 | —                 | —                 |

## F.4 Compute Runtime

Table 54: Summary of runtime in seconds for **HIERBOSS** and competitors across Feynman equations in I.12.2 (CL)-II.36-38 (FMMM) and different noise levels. Each entry reports the mean  $\pm$  standard deviation over 5 repetitions. — denotes additional noise level ( $\sigma$ ) for **HIERBOSS** implementation only.

| Noise ( $\sigma$ )     | <b>HIERBOSS</b>       | BMS                    | BSR                   | DSR                  | gplearn            | operon             | PySR                 | QLattice             | SISSO + +          |
|------------------------|-----------------------|------------------------|-----------------------|----------------------|--------------------|--------------------|----------------------|----------------------|--------------------|
| <b>I.12.2 (CL)</b>     |                       |                        |                       |                      |                    |                    |                      |                      |                    |
| 0.00                   | 51.390 $\pm$ 6.365    | 210.905 $\pm$ 14.077   | 188.939 $\pm$ 0.420   | 76.420 $\pm$ 15.339  | 17.349 $\pm$ 1.267 | 19.513 $\pm$ 0.976 | 69.953 $\pm$ 4.908   | 93.850 $\pm$ 1.890   | 16.639 $\pm$ 1.111 |
| 0.15                   | 56.357 $\pm$ 8.312    | 361.300 $\pm$ 6.939    | 189.356 $\pm$ 1.034   | 107.400 $\pm$ 7.225  | 15.700 $\pm$ 0.688 | 24.173 $\pm$ 1.647 | 72.777 $\pm$ 6.282   | 83.086 $\pm$ 2.549   | 17.552 $\pm$ 0.298 |
| 0.20                   | 57.842 $\pm$ 1.986    | 399.125 $\pm$ 15.148   | 189.955 $\pm$ 2.828   | 123.560 $\pm$ 24.557 | 15.695 $\pm$ 0.783 | 25.032 $\pm$ 2.637 | 81.711 $\pm$ 15.020  | 73.609 $\pm$ 4.531   | 17.585 $\pm$ 0.505 |
| 0.25                   | 57.964 $\pm$ 1.210    | —                      | —                     | —                    | —                  | —                  | —                    | —                    | —                  |
| <b>I.12.11 (FCE)</b>   |                       |                        |                       |                      |                    |                    |                      |                      |                    |
| 0.00                   | 15.138 $\pm$ 0.313    | 272.269 $\pm$ 30.810   | 165.651 $\pm$ 1.424   | 48.720 $\pm$ 8.014   | 16.342 $\pm$ 1.109 | 26.616 $\pm$ 2.265 | 127.246 $\pm$ 8.985  | 93.075 $\pm$ 2.206   | 43.668 $\pm$ 1.667 |
| 0.25                   | 14.082 $\pm$ 0.891    | 293.312 $\pm$ 17.077   | 165.657 $\pm$ 1.107   | 42.600 $\pm$ 0.911   | 16.877 $\pm$ 0.777 | 24.694 $\pm$ 1.669 | 121.428 $\pm$ 9.605  | 92.105 $\pm$ 1.032   | 42.753 $\pm$ 0.650 |
| 1.00                   | 15.103 $\pm$ 1.520    | 295.094 $\pm$ 40.632   | 165.410 $\pm$ 1.454   | 42.500 $\pm$ 0.283   | 16.751 $\pm$ 1.640 | 24.779 $\pm$ 1.212 | 123.313 $\pm$ 13.838 | 93.531 $\pm$ 4.214   | 43.789 $\pm$ 1.669 |
| 2.00                   | 15.522 $\pm$ 0.460    | —                      | —                     | —                    | —                  | —                  | —                    | —                    | —                  |
| <b>I.24.6 (EHFO)</b>   |                       |                        |                       |                      |                    |                    |                      |                      |                    |
| 0.00                   | 64.626 $\pm$ 4.014    | 235.951 $\pm$ 19.032   | 196.386 $\pm$ 4.105   | 31.800 $\pm$ 1.131   | 16.797 $\pm$ 2.877 | 26.912 $\pm$ 2.841 | 69.940 $\pm$ 10.127  | 95.689 $\pm$ 2.282   | 19.535 $\pm$ 0.596 |
| 0.15                   | 64.124 $\pm$ 3.986    | 232.742 $\pm$ 17.890   | 192.569 $\pm$ 3.252   | 32.720 $\pm$ 1.410   | 14.813 $\pm$ 0.947 | 23.552 $\pm$ 3.040 | 76.673 $\pm$ 8.929   | 94.665 $\pm$ 1.604   | 19.015 $\pm$ 0.130 |
| 0.25                   | 63.645 $\pm$ 1.326    | 242.652 $\pm$ 6.945    | 194.237 $\pm$ 6.542   | 31.640 $\pm$ 2.182   | 15.715 $\pm$ 1.215 | 23.836 $\pm$ 1.224 | 67.912 $\pm$ 9.139   | 94.928 $\pm$ 2.089   | 19.163 $\pm$ 0.330 |
| 1.00                   | 66.577 $\pm$ 1.839    | —                      | —                     | —                    | —                  | —                  | —                    | —                    | —                  |
| <b>I.50.26 (HOQN)</b>  |                       |                        |                       |                      |                    |                    |                      |                      |                    |
| 0.00                   | 60.498 $\pm$ 1.581    | 274.056 $\pm$ 27.935   | 191.599 $\pm$ 1.395   | 44.880 $\pm$ 1.018   | 19.290 $\pm$ 1.877 | 24.758 $\pm$ 3.164 | 108.801 $\pm$ 15.951 | 85.273 $\pm$ 1.281   | 26.372 $\pm$ 0.552 |
| 0.15                   | 63.543 $\pm$ 3.170    | 294.648 $\pm$ 29.925   | 191.651 $\pm$ 1.057   | 45.760 $\pm$ 1.031   | 20.649 $\pm$ 5.276 | 26.250 $\pm$ 1.756 | 103.283 $\pm$ 3.208  | 86.189 $\pm$ 7.616   | 26.119 $\pm$ 0.638 |
| 0.18                   | 62.391 $\pm$ 2.588    | 300.061 $\pm$ 24.232   | 193.930 $\pm$ 5.809   | 46.000 $\pm$ 1.304   | 20.935 $\pm$ 2.589 | 28.398 $\pm$ 1.562 | 100.307 $\pm$ 11.418 | 85.358 $\pm$ 5.604   | 25.999 $\pm$ 0.704 |
| 0.20                   | 63.808 $\pm$ 2.657    | —                      | —                     | —                    | —                  | —                  | —                    | —                    | —                  |
| <b>II.36-38 (FMMM)</b> |                       |                        |                       |                      |                    |                    |                      |                      |                    |
| 0.00                   | 1142.192 $\pm$ 49.481 | 2486.476 $\pm$ 290.751 | 1965.850 $\pm$ 66.240 | 41.420 $\pm$ 0.638   | 19.391 $\pm$ 1.055 | 27.523 $\pm$ 2.473 | 77.939 $\pm$ 11.421  | 113.685 $\pm$ 16.482 | 17.127 $\pm$ 0.111 |
| 0.15                   | 1110.915 $\pm$ 92.768 | 2469.174 $\pm$ 149.699 | 1933.160 $\pm$ 16.120 | 59.320 $\pm$ 14.941  | 20.437 $\pm$ 2.257 | 26.457 $\pm$ 2.915 | 90.886 $\pm$ 25.039  | 110.192 $\pm$ 5.903  | 16.914 $\pm$ 0.131 |
| 0.20                   | 1117.321 $\pm$ 74.164 | 2601.064 $\pm$ 644.774 | 1931.850 $\pm$ 17.040 | 65.980 $\pm$ 11.217  | 21.182 $\pm$ 4.144 | 29.408 $\pm$ 5.562 | 118.457 $\pm$ 59.492 | 120.602 $\pm$ 28.463 | 16.825 $\pm$ 0.169 |

Continued on next page

Table 54: Summary of runtime in seconds for **HIERBOSSS** and competitors across Feynman equations in I\_12.2 (CL)-II\_36\_38 (FMMM) and different noise levels. Each entry reports the mean  $\pm$  standard deviation over 5 repetitions. — denotes additional noise level ( $\sigma$ ) for **HIERBOSSS** implementation only (*continued*).

| Noise ( $\sigma$ ) | <b>HIERBOSSS</b>      | BMS | BSR | DSR | gplearn | operon | PySR | QLattice | SISSO++ |
|--------------------|-----------------------|-----|-----|-----|---------|--------|------|----------|---------|
| 0.25               | 1059.927 $\pm$ 28.779 | —   | —   | —   | —       | —      | —    | —        | —       |

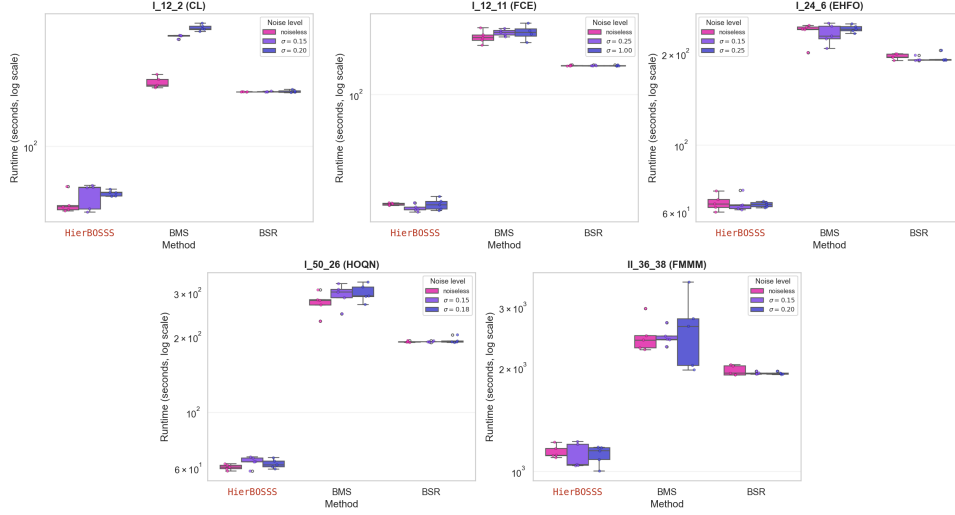


Figure 5: Compute runtimes (in log-scale) of **HIERBOSSS**, BMS (5 parallel MCMC chains), and BSR (5 restarts) for the Feynman equations in I\_12.2 (CL)-II\_36\_39 (FMMM), across different noise levels and 5 independent repetitions. Implementation details of **HIERBOSSS** and the Bayesian competitors for each Feynman equation follow from Sections D.1, B.7, and B.8.

## G Ablation Study for HierBOSSS and Practical Diagnostic for Choosing the Symbolic Forest Size

We conduct an ablation study to assess the sensitivity of HierBOSSS to the nominal number of symbolic trees  $K$  in the symbolic forest component  $\mathcal{T}$ . At the same time, this study provides a practical diagnostic for choosing  $K$ . The study is performed on two representative Feynman equations, I\_12.2 (CL) and I\_12.11 (FCE), under a fixed noise level  $\sigma = 0.25$ . For each equation, we vary  $K$  over  $\{2, 3, 4, 5, 6, 8, 10, 12\}$  and run HierBOSSS with 5 independent repetitions under the experimental settings described in Section D.1.

Figures 6 and 7 summarize the effect of  $K$  on the predictive accuracy of HierBOSSS. Across the considered values of  $K$ , both the *Raw* and *Final* expressions exhibit stable average test RMSE and test  $R^2$  over 5 repetitions. For I\_12.2 (CL), increasing  $K$  beyond small to moderate values yields little change in predictive performance. A similar pattern is observed for I\_12.11 (FCE), where the test RMSE and test  $R^2$  remain highly stable across the full range of  $K$ . These results indicate that, once  $K$  is large enough to represent the main additive symbolic components, additional trees do not materially improve predictive performance.

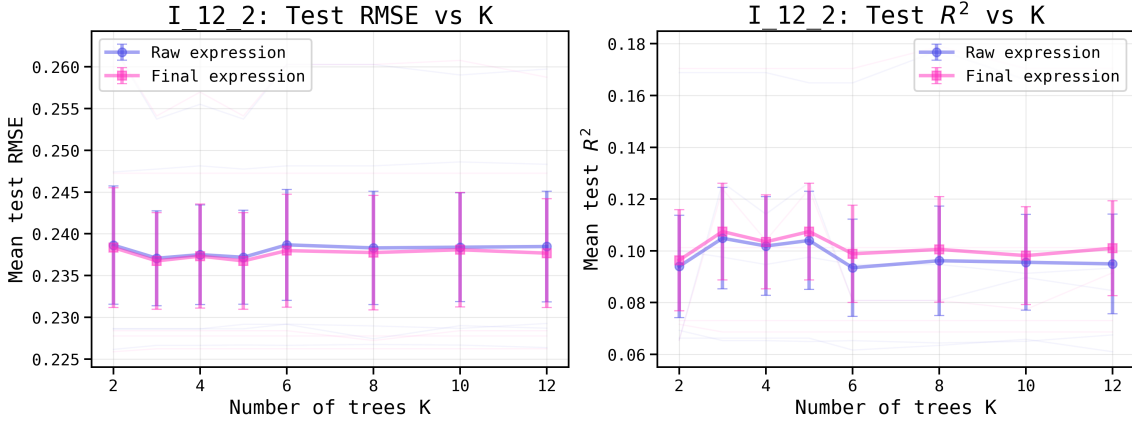


Figure 6: Average (over 5 independent repetitions) test RMSE and  $R^2$  of HierBOSSS across different values of  $K$  for learning I\_12.2 (CL) under noise level  $\sigma = 0.25$ .

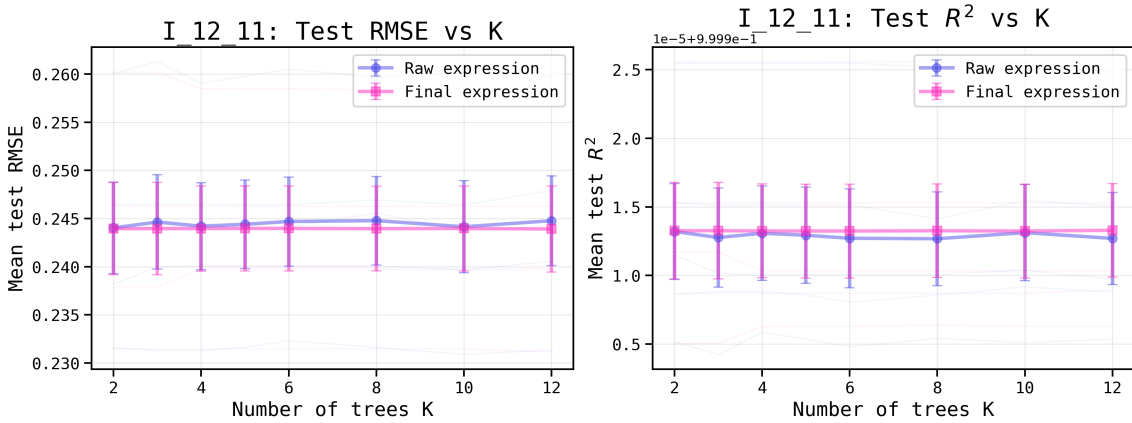


Figure 7: Average (over 5 independent repetitions) test RMSE and  $R^2$  of HierBOSSS across different values of  $K$  for learning I\_12.11 (FCE) under noise level  $\sigma = 0.25$ .

The runtime behavior is shown in Figure 8. As expected, the computational cost increases with  $K$ . This increase is moderate for smaller values of  $K$ , but becomes more pronounced for larger symbolic forest sizes. Thus, although larger  $K$  expands the symbolic expression search space, it also introduces additional computational overhead without commensurate gains in predictive accuracy for these Feynman equations.

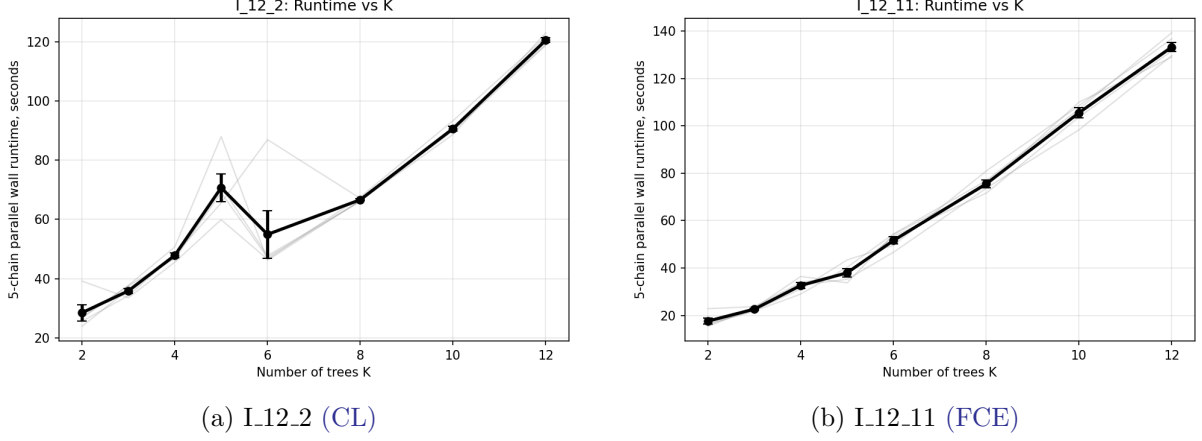


Figure 8: Runtime behavior of **HIERBOSSS** across different values of  $K$  under noise level  $\sigma = 0.25$  for I.12.2 (CL) and I.12.11 (FCE). Each panel reports the wall-clock runtime for 5 parallel chains across 5 independent repetitions, with the mean trend shown in black and repetition-level variability shown in gray.

Taken together, the preceding ablation study suggests a principled strategy for choosing  $K$ : select the smallest or a moderate value of  $K$  after which predictive accuracy stabilizes, while avoiding unnecessarily large forests that increase runtime. In this sense,  $K$  serves as a nominal expressivity parameter controlling the breadth of symbolic exploration, rather than the final number of symbolic components retained in the reported expression. The post-MCMC symbolic model refinement step of **HIERBOSSS** subsequently determines the effective forest size  $K_{\text{eff}}$  in a data-adaptive manner by removing redundant or collinear symbolic tree structures. Therefore, a moderate nominal  $K$  provides sufficient exploratory flexibility, while the *Final* expression remains parsimonious through the refinement step.



## H Posterior Diagnostics for the Bayesian SR Methods

We provide trace plots to examine the stochastic search and posterior convergence behavior of **HierBOSSS**, **BMS**, and **BSR**, while learning the Feynman equations in I.12.2 (CL)–II.36\_38 (FMMM). For each equation, we show one representative repetition at the corresponding fixed noise level ( $\sigma$ ), where **HierBOSSS** and **BMS** have been implemented with 5 parallel chains and **BSR** is executed for 5 restarts. All other experimental configurations follow from Sections D.1, B.7, and B.8. These traces visualize the stability of the posterior search and exploration of the Bayesian SR methods over the symbolic expression space. Since the plotted quantity differs across methods, the traces should be interpreted method-wise rather than as direct numerical comparisons across methods.

For **HierBOSSS**, the traces in Figure 9 show the evolution of the log joint marginal posterior value of  $\mathcal{T}$ , denoted by  $\log(\text{JMP})$ , across MCMC iterations. These traces indicate how the sampler explores the posterior distribution over symbolic forests and whether the chains move toward regions of high posterior support. Stable traces across chains suggest proper mixing and that the posterior search repeatedly identifies comparable high-scoring symbolic structures.

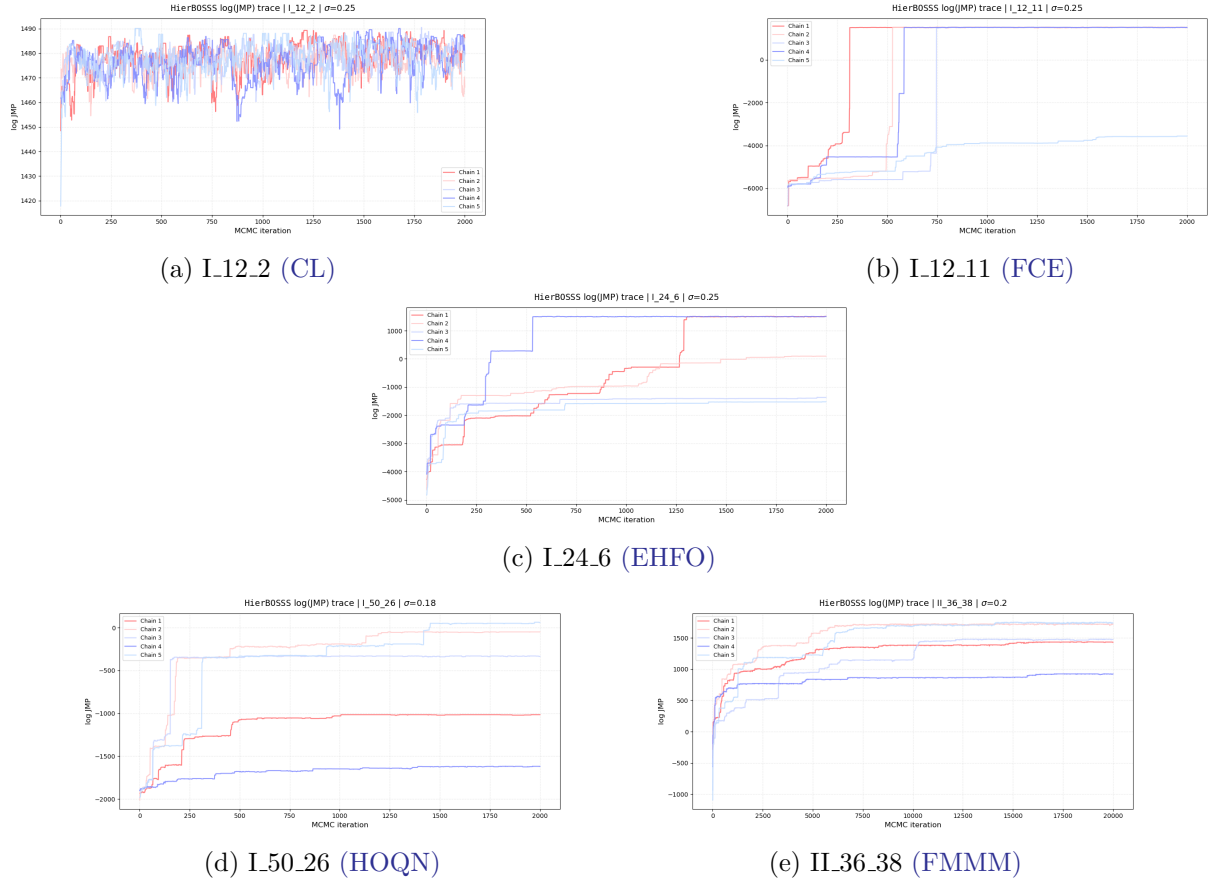
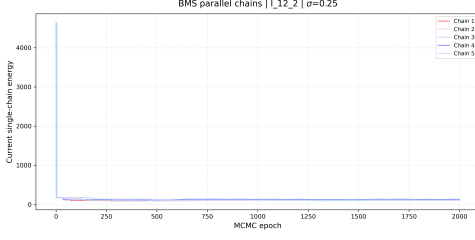


Figure 9: Trace plots for **HierBOSSS** across the Feynman equations in I.12.2 (CL)–II.36\_38 (FMMM).

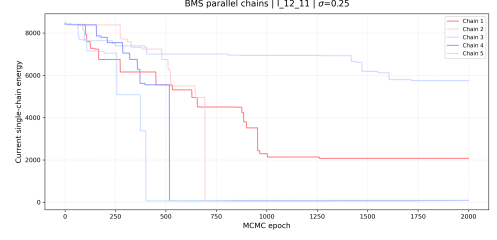
For **BMS**, the traces in Figure 10 report the current single-chain energy, or the description length [Guimerà et al., 2020], over MCMC epochs. This energy combines the data-fit term and the prior or symbolic complexity penalty. Lower values indicate models with better posterior

support. Therefore, downward movements correspond to improvements in fit, parsimony, or both, while flat regions indicate that the chain remains in similar symbolic states.

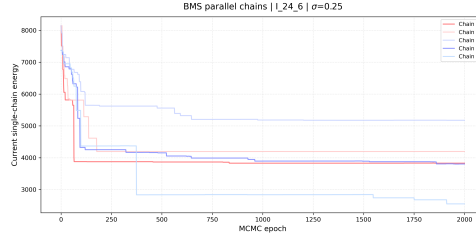
For BSR, the traces show the accepted-state log likelihood across proposal iterations [Jin et al., 2020]. Since the trace updates only when a proposed symbolic modification is accepted, flat segments correspond to rejected proposals, while jumps indicate accepted moves. Upward jumps typically reflect improved data fit, although occasional downward moves may occur due to the exploratory nature of the underlying MCMC routine.



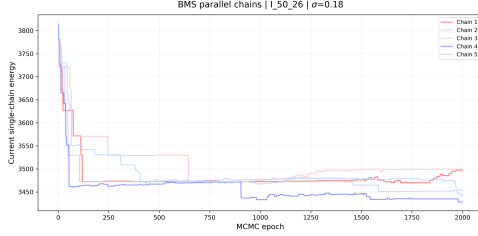
(a) I.12.2 (CL)



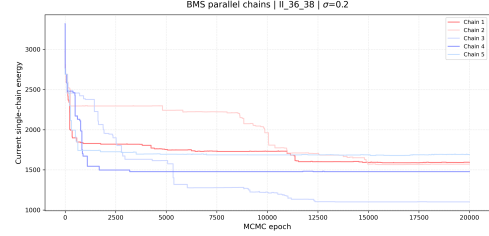
(b) I.12.11 (FCE)



(c) I.24.6 (EHFO)

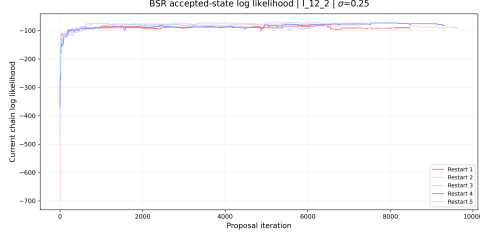


(d) I.50.26 (HOQN)

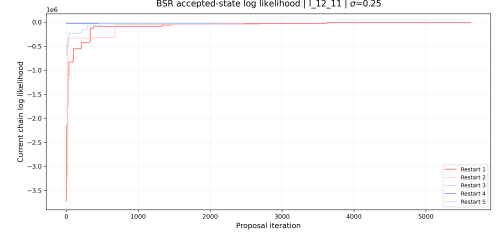


(e) IL36.38 (FMMM)

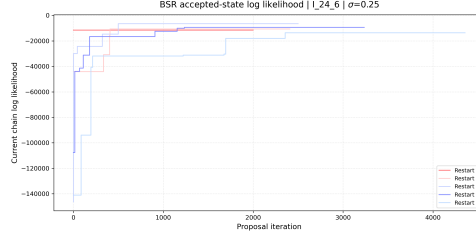
Figure 10: Trace plots for BMS across the Feynman equations in I.12.2 (CL)–IL36.38 (FMMM).



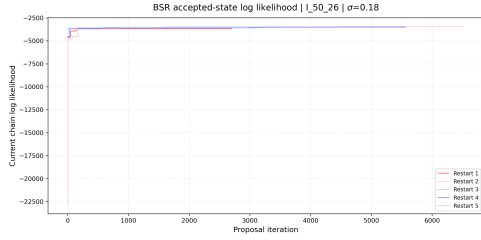
(a) I\_12\_2 (CL)



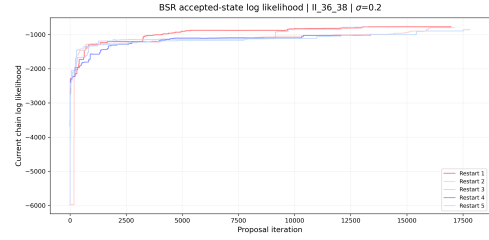
(b) I\_12\_11 (FCE)



(c) I\_24\_6 (EHFO)



(d) I\_50\_26 (HOQN)



(e) II\_36\_38 (FMMM)

Figure 11: Trace plots for BSR across the Feynman equations in I\_12\_2 (CL)–II\_36\_38 (FMMM).

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